



US012419180B2

(12) **United States Patent**
Kamada et al.

(10) **Patent No.:** **US 12,419,180 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **OPTICAL FUNCTIONAL DEVICE,
FUNCTIONAL PANEL, DISPLAY DEVICE,
INPUT/OUTPUT DEVICE, AND DATA
PROCESSING DEVICE**

(71) Applicant: **Semiconductor Energy Laboratory
Co., Ltd., Atsugi (JP)**

(72) Inventors: **Taisuke Kamada**, Niiza (JP); **Yasuhiro
Niikura**, Komae (JP); **Daisuke Kubota**,
Atsugi (JP); **Ryo Hatsumi**, Hadano
(JP); **Akio Yamashita**, Atsugi (JP);
Sachiko Kawakami, Atsugi (JP);
Satoshi Seo, Sagamiara (JP)

(73) Assignee: **Semiconductor Energy Laboratory
Co., Ltd., Atsugi (JP)**

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 459 days.

(21) Appl. No.: **17/925,628**

(22) PCT Filed: **May 17, 2021**

(86) PCT No.: **PCT/IB2021/054193**

§ 371 (c)(1),

(2) Date: **Nov. 16, 2022**

(87) PCT Pub. No.: **WO2021/240291**

PCT Pub. Date: **Dec. 2, 2021**

(65) **Prior Publication Data**

US 2023/0180491 A1 Jun. 8, 2023

(30) **Foreign Application Priority Data**

May 29, 2020 (JP) 2020-093856

(51) **Int. Cl.**

H10K 59/60 (2023.01)

G06F 9/30 (2018.01)

(Continued)

(52) **U.S. Cl.**

CPC **H10K 59/60** (2023.02); **G06F 9/3001**
(2013.01); **H10K 30/81** (2023.02); **H10K**
39/34 (2023.02);

(Continued)

(58) **Field of Classification Search**

CPC H10K 59/60; H10K 30/81; H10K 39/34;
H10K 50/125; H10K 59/8792;

(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,357,961 B2 1/2013 Hanari
9,406,898 B2 8/2016 Yamazaki et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 001669296 A 9/2005
CN 105144841 A 12/2015
(Continued)

OTHER PUBLICATIONS

International Search Report (Application No. PCT/IB2021/054193)
Dated Jul. 13, 2021.

(Continued)

Primary Examiner — S M Sohel Imtiaz

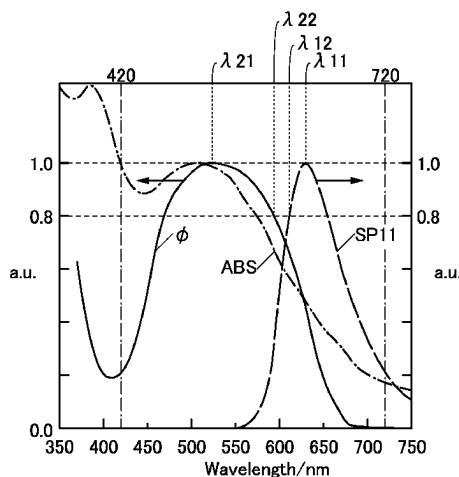
(74) Attorney, Agent, or Firm — Eric J. Robinson;
Robinson Intellectual Property Law Office, P.C.

(57)

ABSTRACT

A novel optical functional device that is highly convenient,
useful, or reliable is provided. The optical functional device
includes a light-emitting function, a photoelectric conver-
sion function, a first electrode, a second electrode, and an
optical functional layer. The light-emitting function converts
electrical energy into first light, the first light has a first
emission spectrum, and the first emission spectrum exhibits
a maximum peak at a first wavelength. At a second wave-
length, the intensity of the first emission spectrum is 80% of

(Continued)



the maximum peak. The photoelectric conversion function has a spectral sensitivity characteristic; at a third wavelength, the spectral sensitivity characteristic has a maximum sensitivity within a range of 420 to 720 nm inclusive; and at a fourth wavelength, the sensitivity of the spectral sensitivity characteristic is 80% of the maximum sensitivity. The third wavelength is positioned closer to the second wavelength than to the first wavelength, and the fourth wavelength is positioned closer to the first wavelength than to the third wavelength.

18 Claims, 23 Drawing Sheets

(51) Int. Cl.

H10K 30/81 (2023.01)

H10K 39/34 (2023.01)

H10K 50/125 (2023.01)

H10K 59/80 (2023.01)

(52) U.S. Cl.

CPC **H10K 50/125** (2023.02); **H10K 59/8792** (2023.02)

(58) Field of Classification Search

CPC H10K 59/38; G06F 9/3001; G06F 3/041;
G09F 9/00; G09G 3/20; H04N 25/76;
H05B 33/14; H10F 39/12

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,848,070 B2 12/2017 Yamazaki et al.
10,924,595 B2 2/2021 Yamazaki et al.

11,095,763 B2 8/2021 Yamazaki et al.
11,968,863 B2 4/2024 Ikeda et al.
2005/0253790 A1* 11/2005 Uchida H04N 1/028
345/76
2006/0246318 A1 11/2006 Nomura et al.
2009/0141004 A1* 6/2009 Yamazaki H10H 20/01
345/204
2010/0084642 A1 4/2010 Hanari
2014/0267955 A1 9/2014 Hibayashi et al.
2014/0306260 A1 10/2014 Yamazaki et al.
2014/0340363 A1 11/2014 Ikeda et al.
2017/0285404 A1* 10/2017 Kubota G02F 1/133553
2018/0102456 A1* 4/2018 Uemura H10F 30/20
2020/0152915 A1 5/2020 Mori et al.
2020/0395555 A1 12/2020 Kawano et al.
2020/0411608 A1* 12/2020 Tang G06F 3/0446
2021/0227061 A1 7/2021 Yamazaki et al.
2021/0343965 A1 11/2021 Seo
2024/0224676 A1 7/2024 Ikeda et al.

FOREIGN PATENT DOCUMENTS

JP 2004-045636 A 2/2004
JP 2010-091610 A 4/2010
JP 2015-005280 A 1/2015
JP 2015-109252 A 6/2015
KR 2015-0143638 A 12/2015
TW 201830736 8/2018
WO WO-2004/008736 1/2004
WO WO-2014/171336 10/2014
WO WO-2020/044170 3/2020

OTHER PUBLICATIONS

Written Opinion (Application No. PCT/IB2021/054193) Dated Jul. 13, 2021.
Taiwanese Office Action (Application No. 110118182) Dated Apr. 7, 2025.

* cited by examiner

FIG. 1A

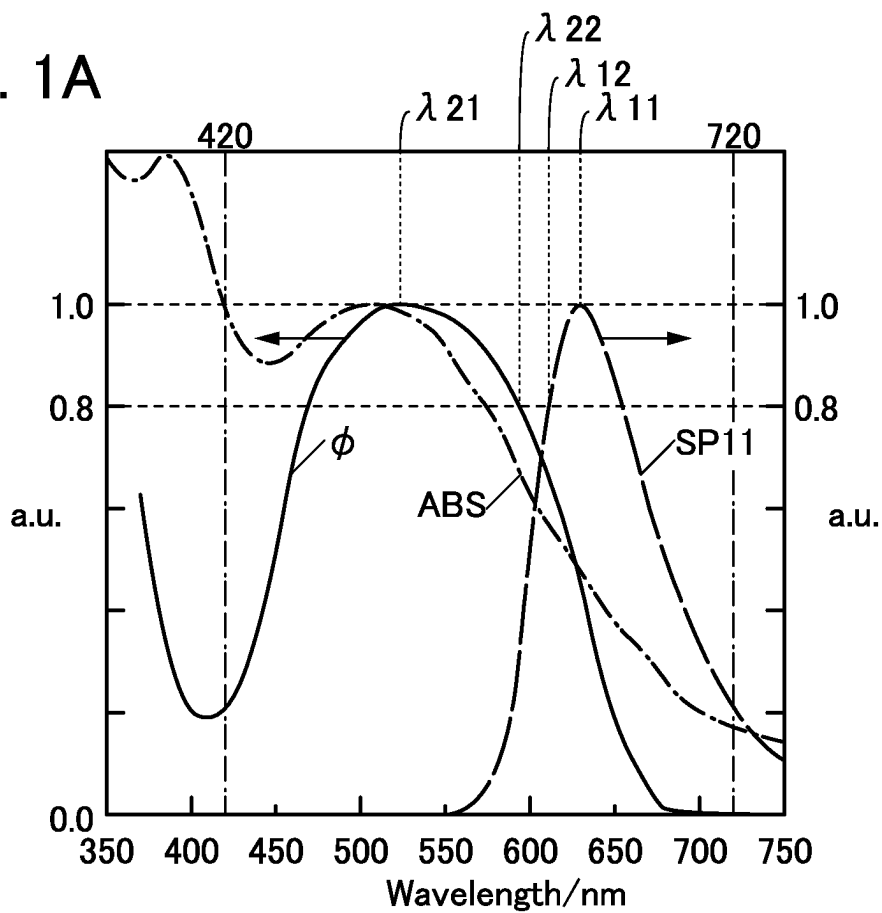


FIG. 1B

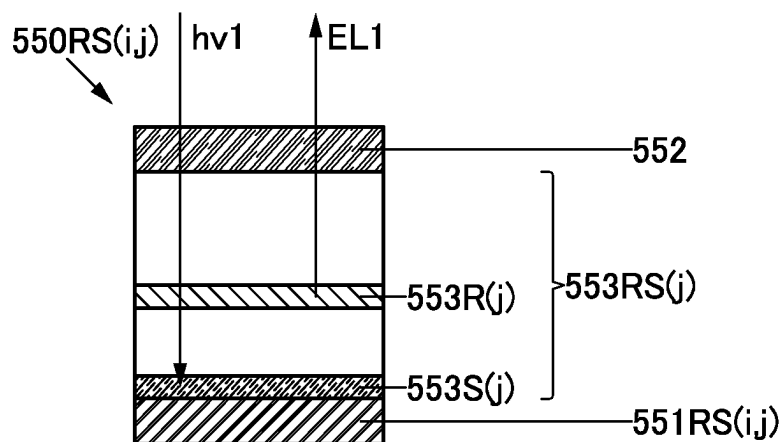


FIG. 2A

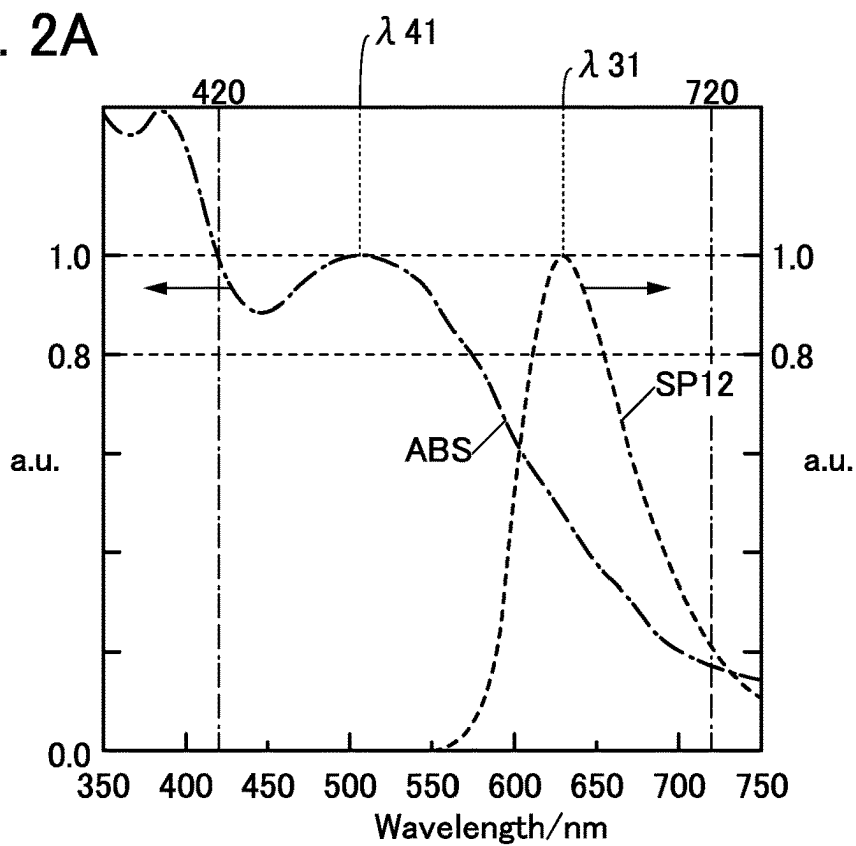


FIG. 2B

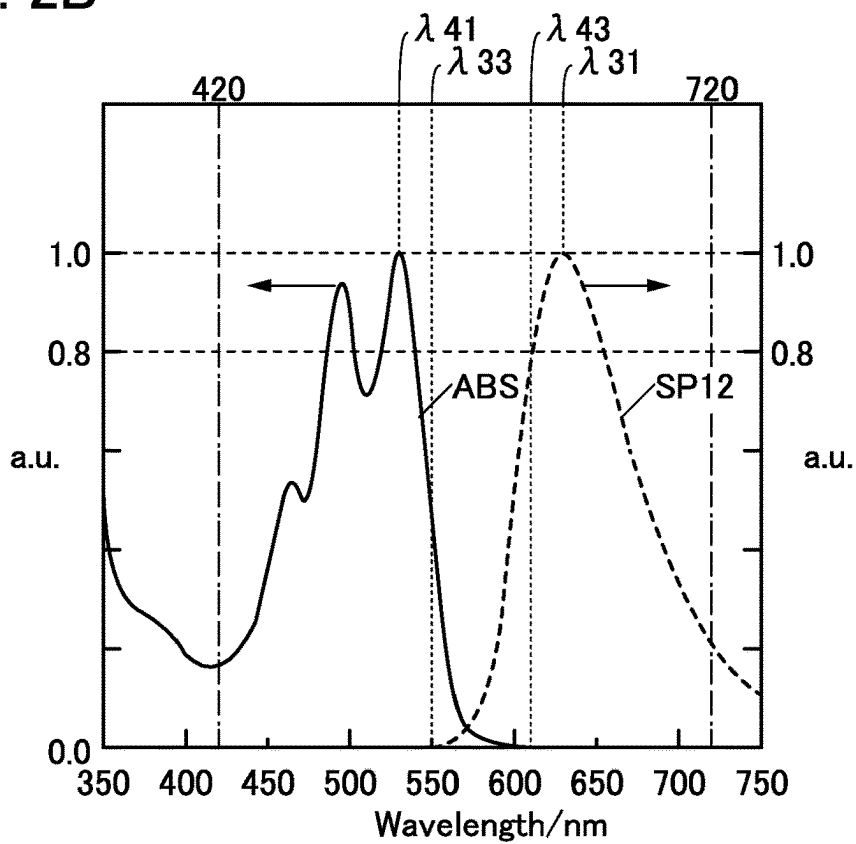


FIG. 3A

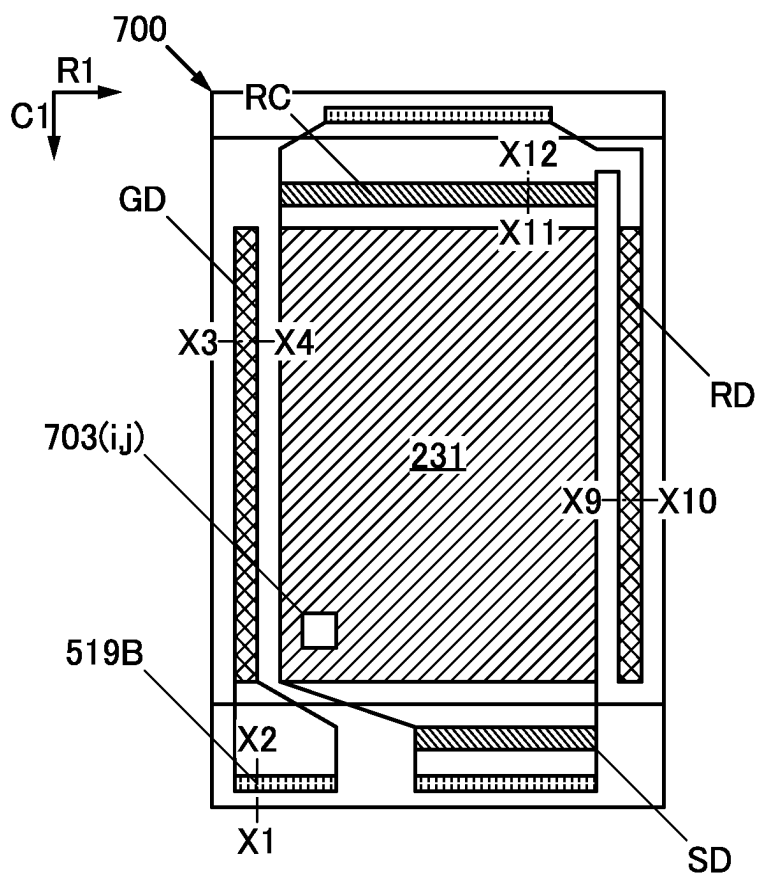


FIG. 3B

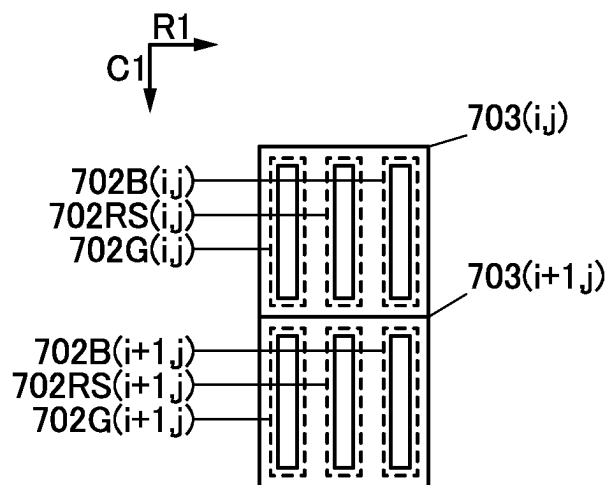


FIG. 3C

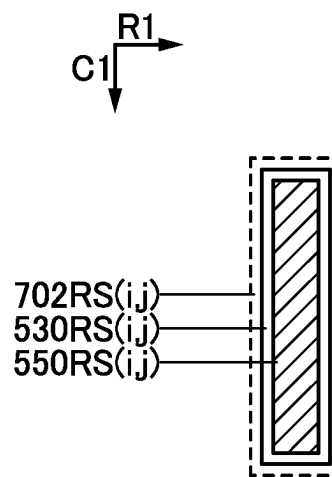


FIG. 4A

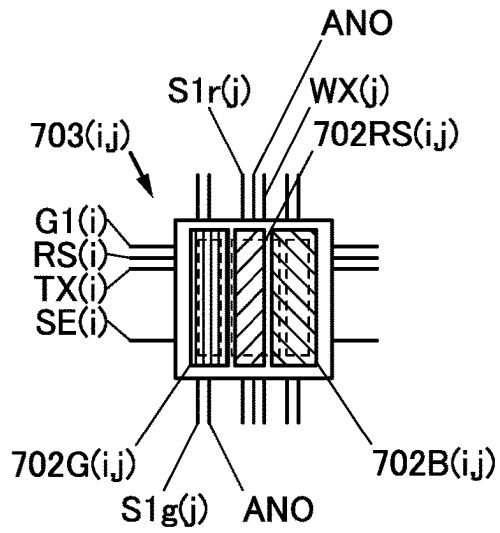


FIG. 4B

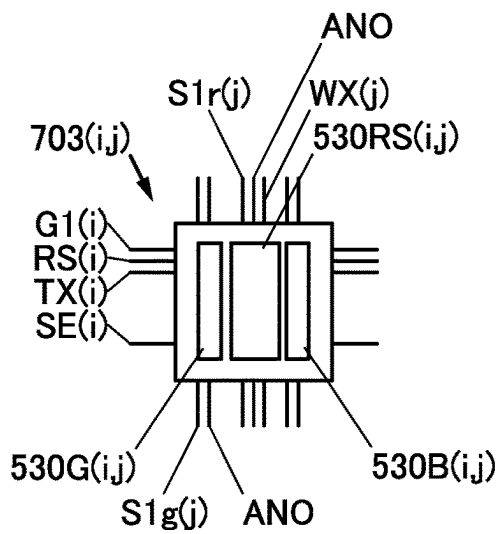


FIG. 4C

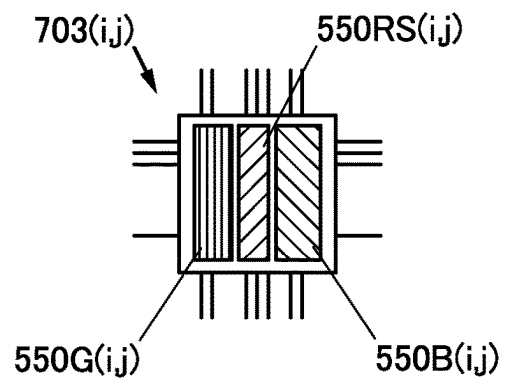


FIG. 5

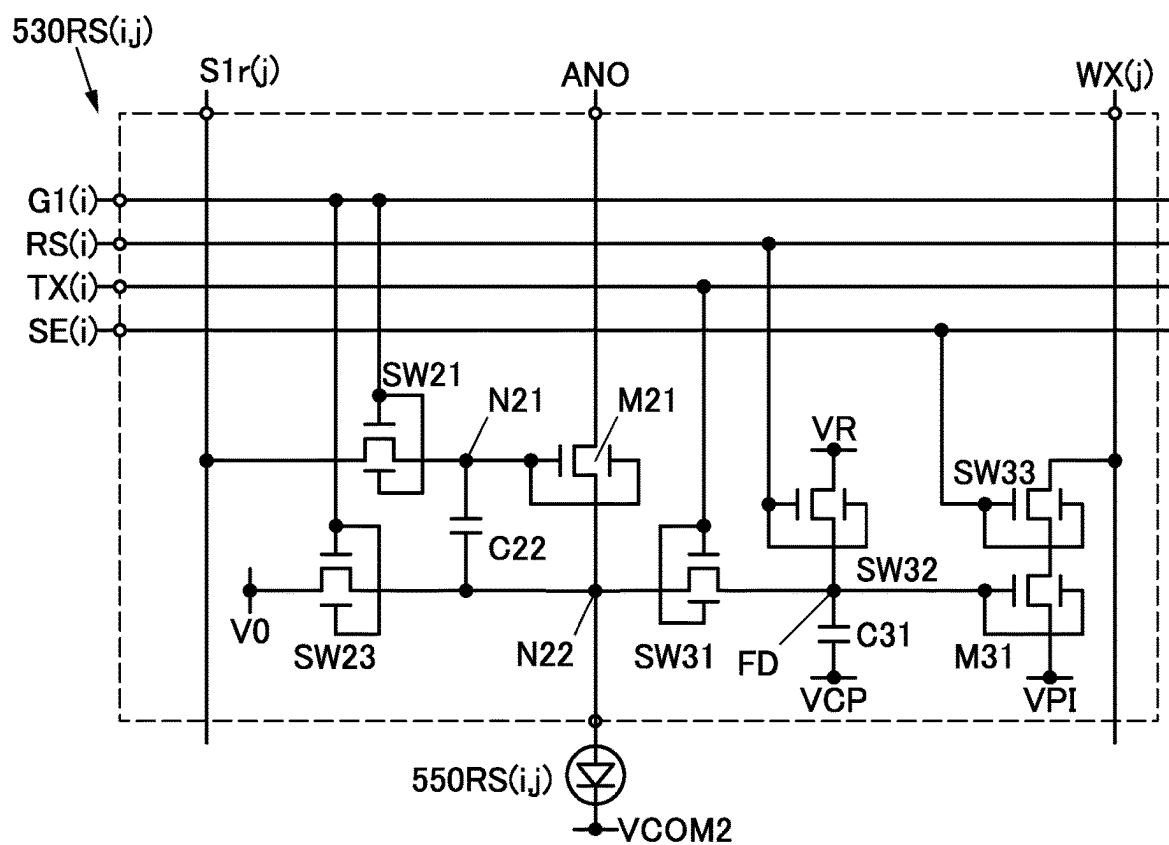


FIG. 6A

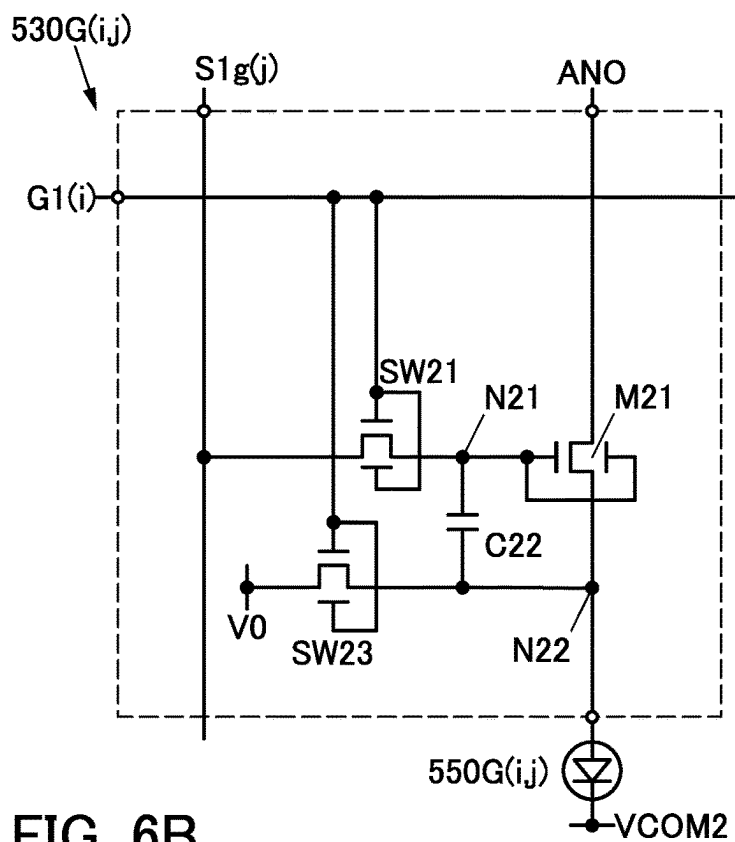


FIG. 6B

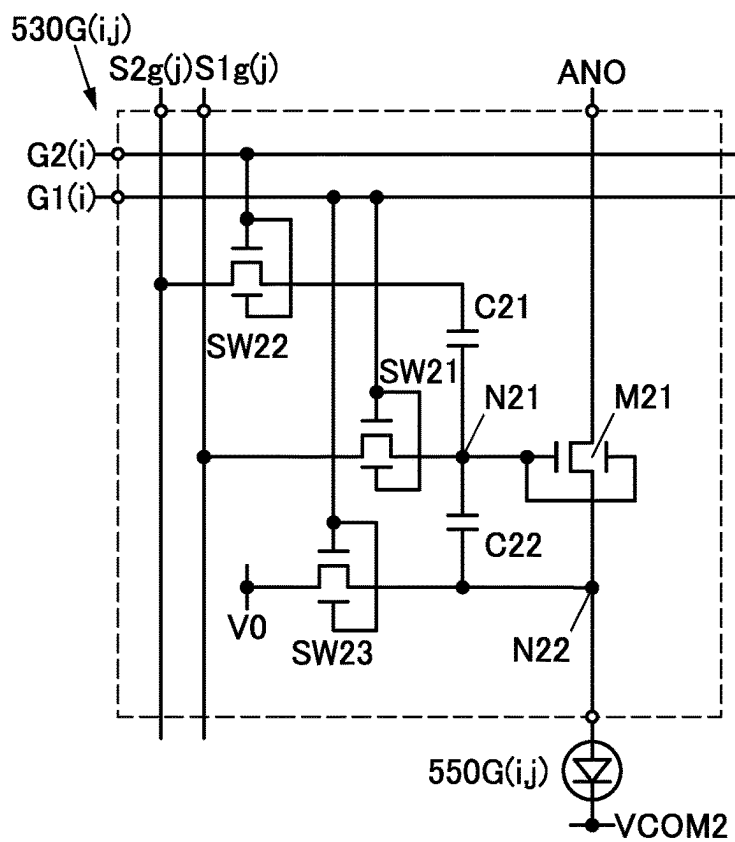


FIG. 7A

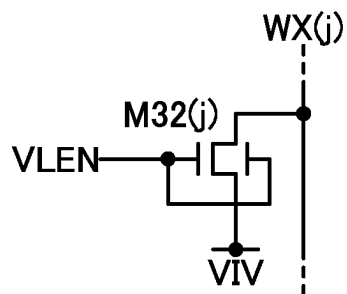


FIG. 7B

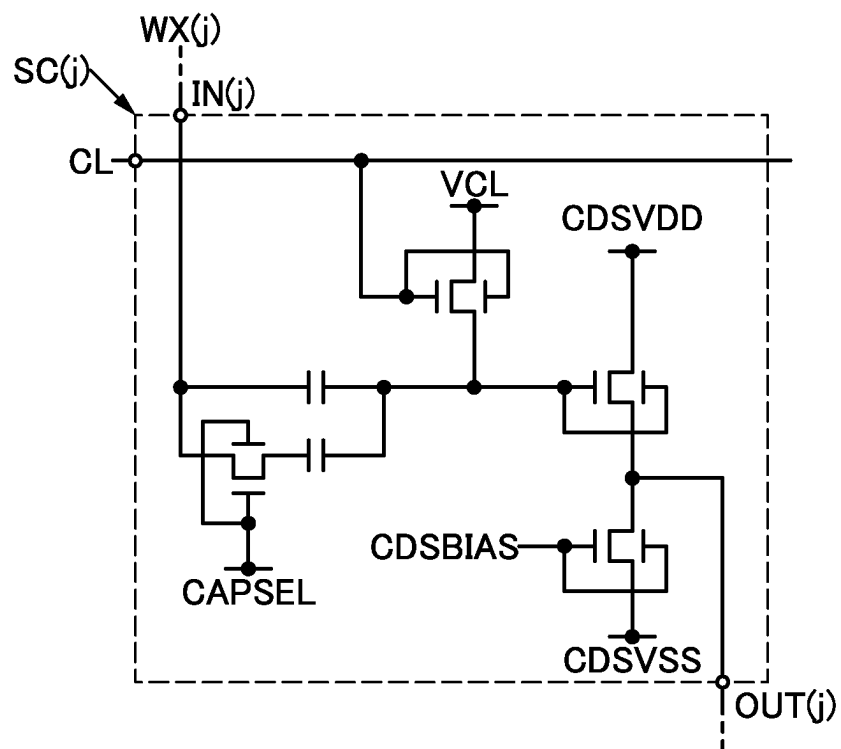


FIG. 8

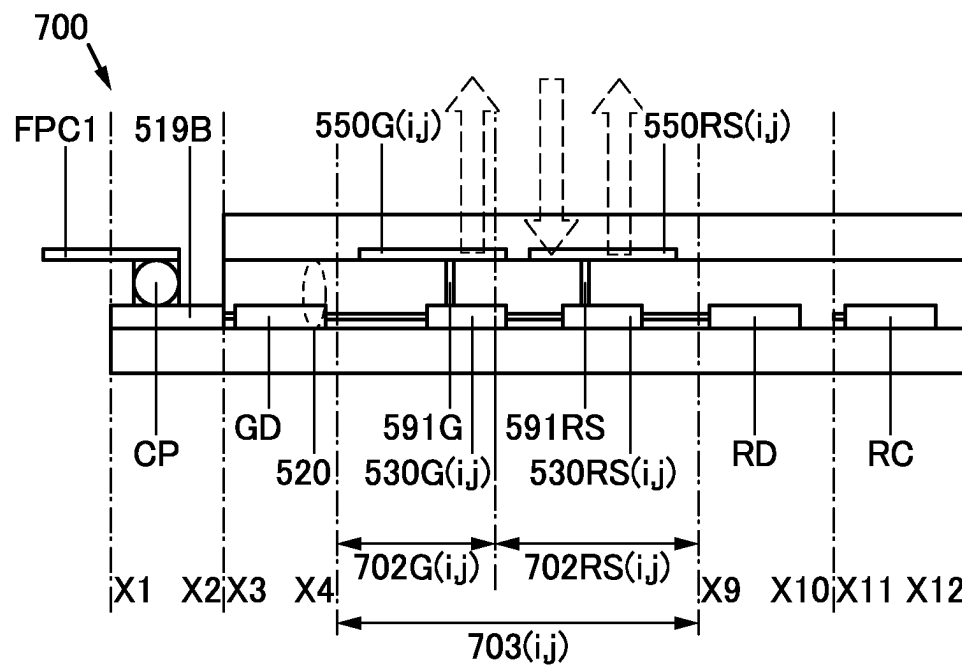


FIG. 9A

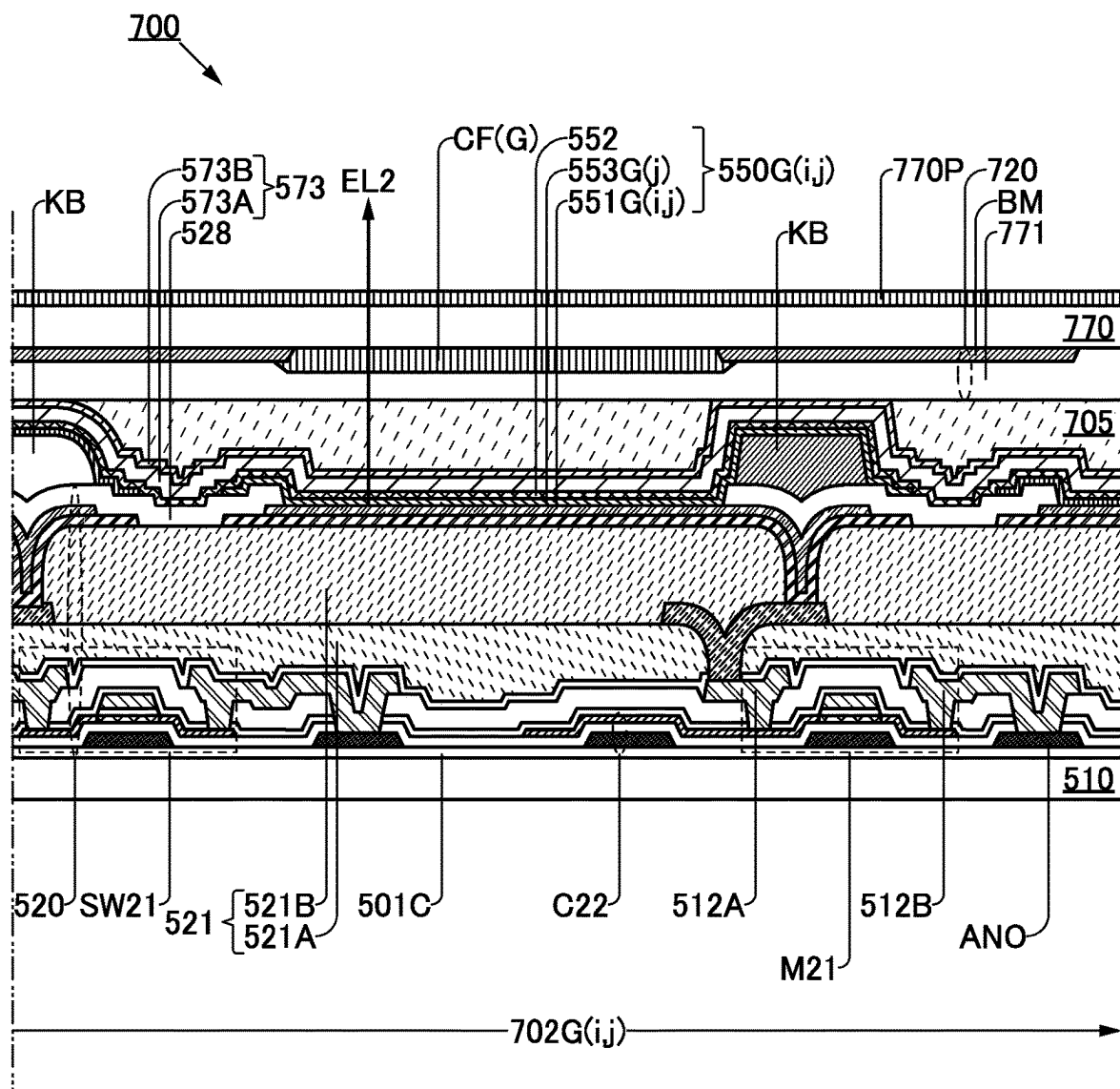


FIG. 9B

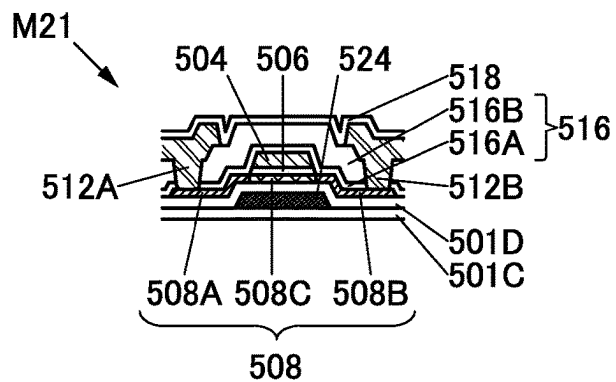


FIG. 10A

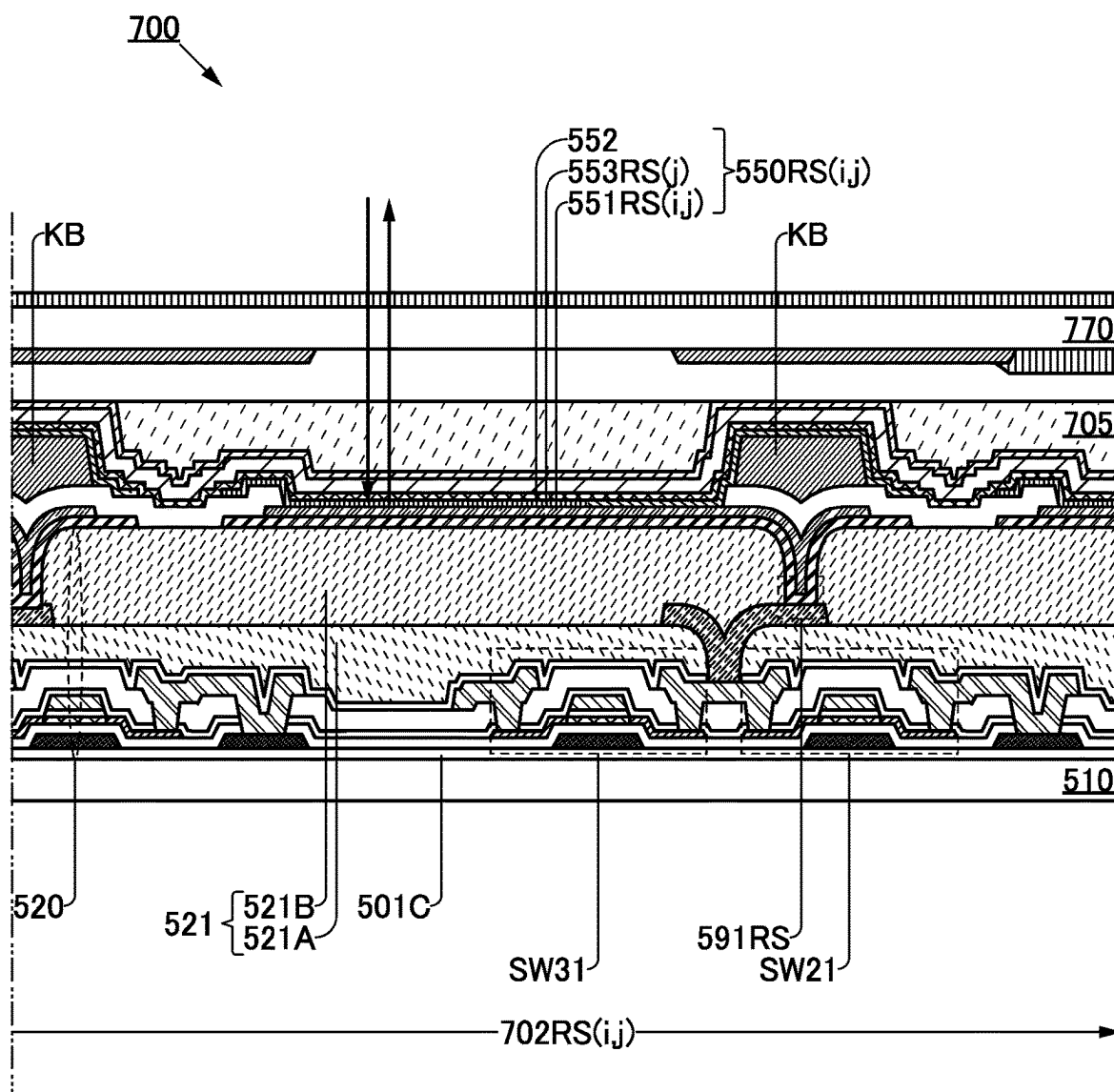


FIG. 10B

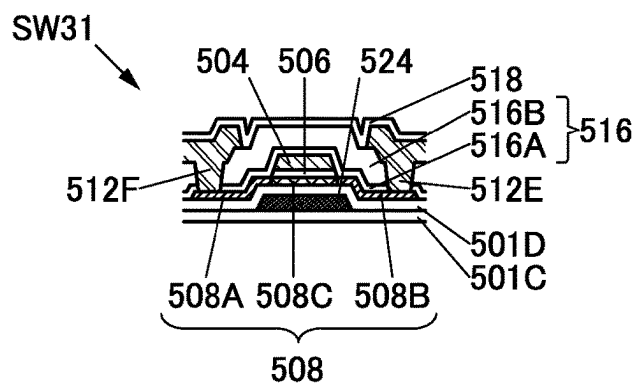


FIG. 11A

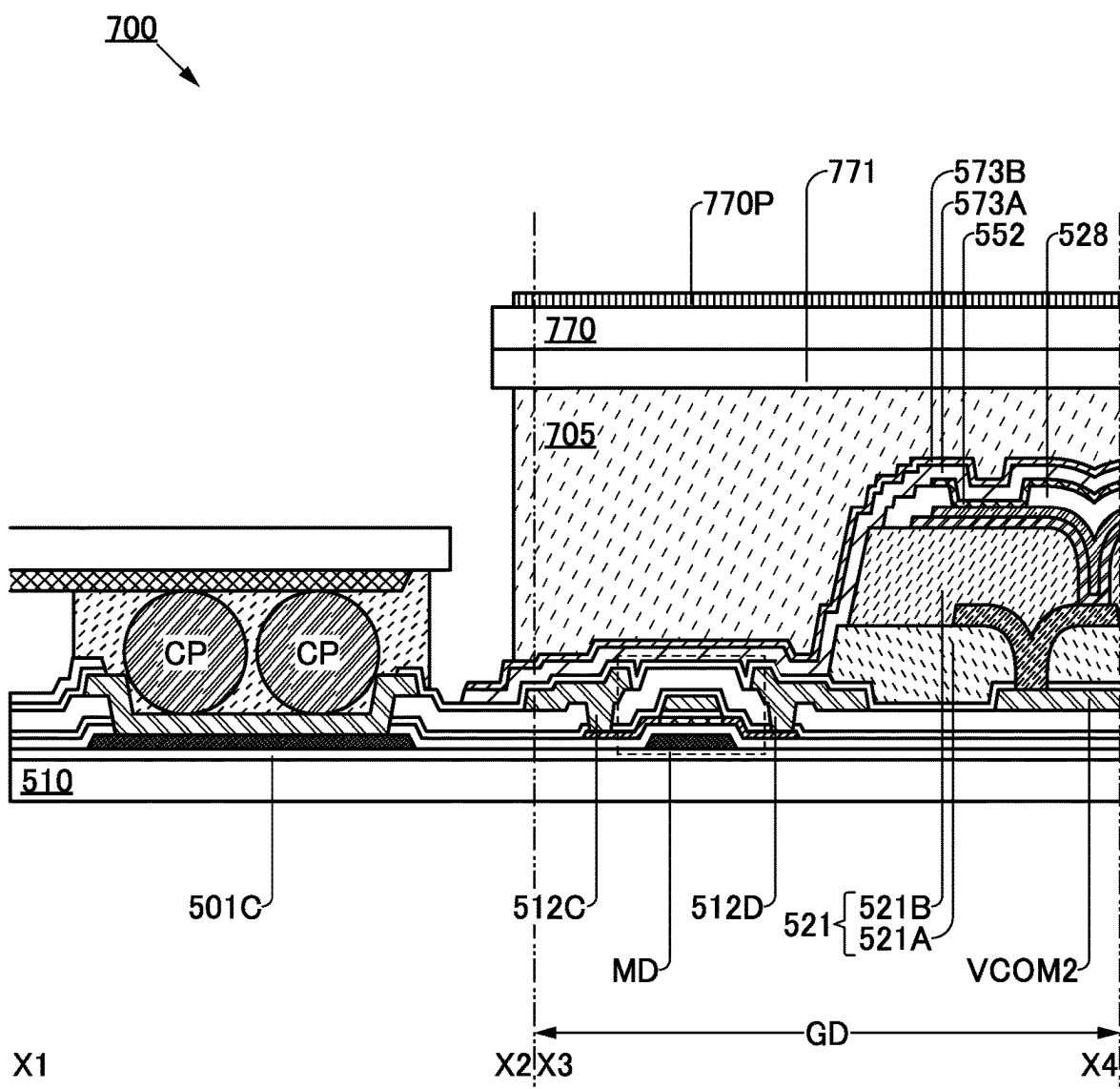


FIG. 11B

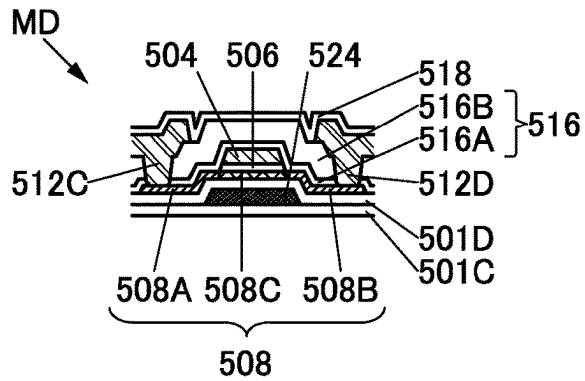


FIG. 12A

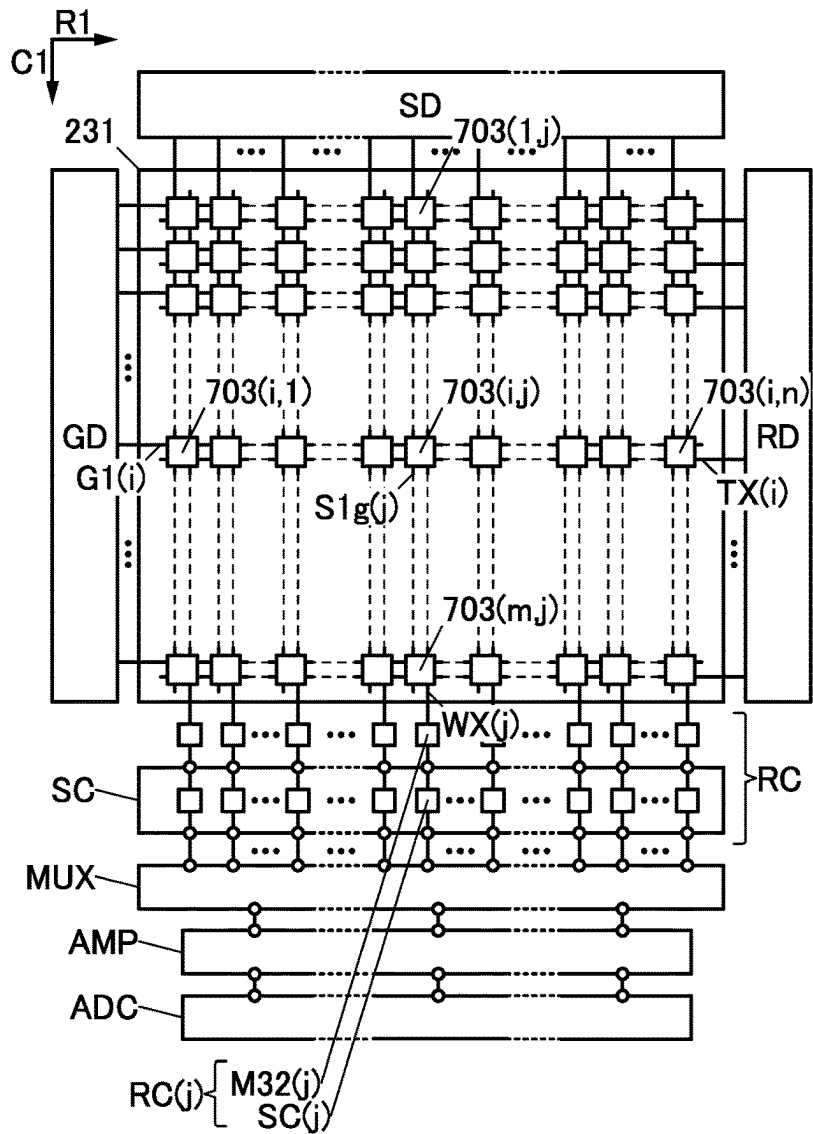


FIG. 12B

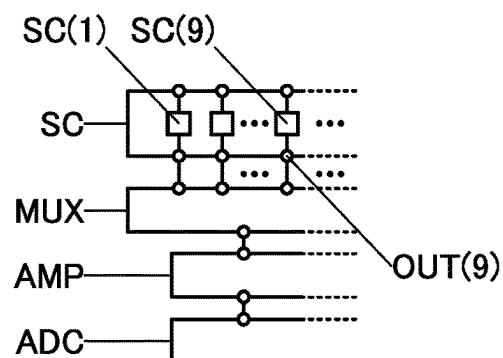


FIG. 13

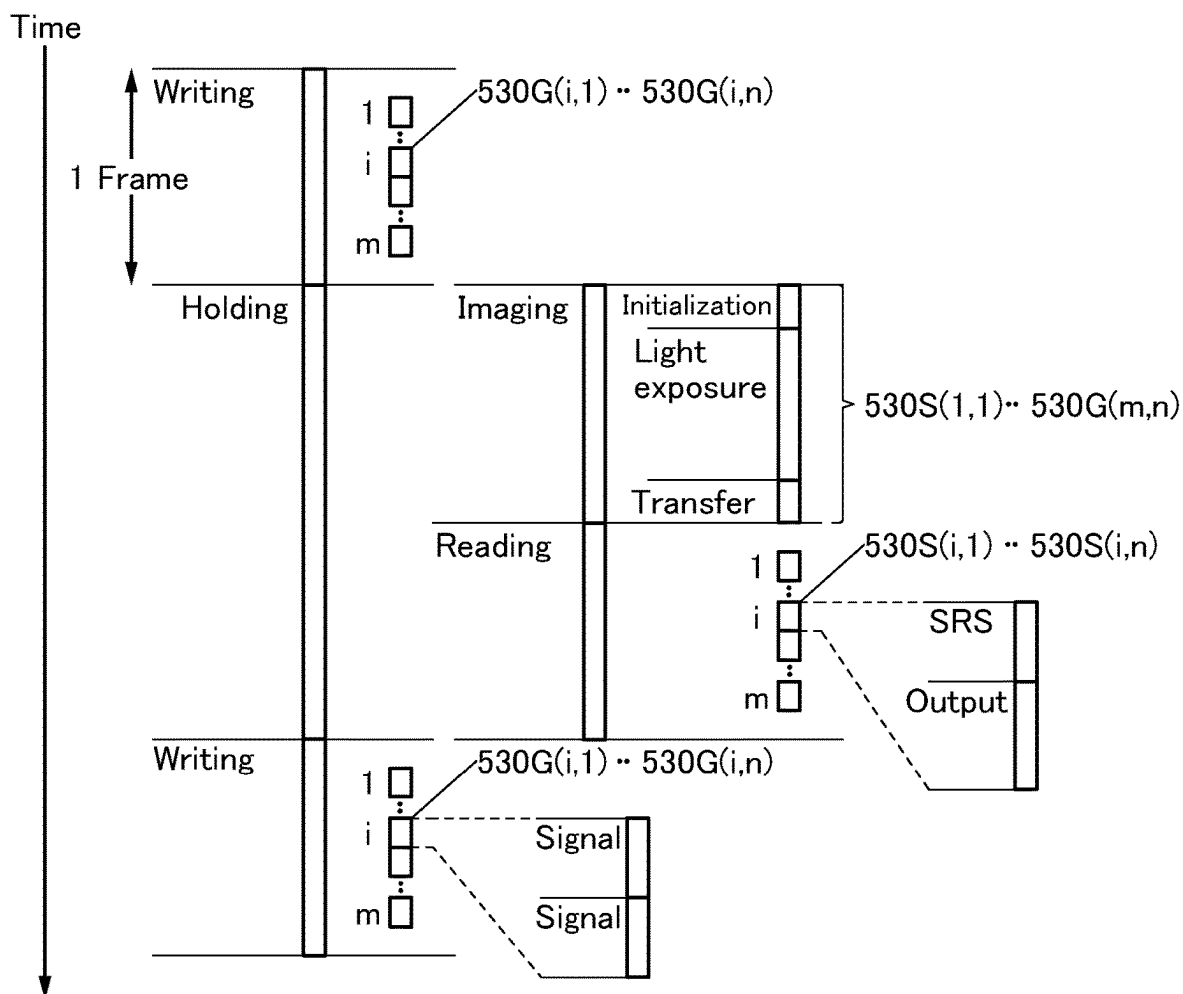


FIG. 14A

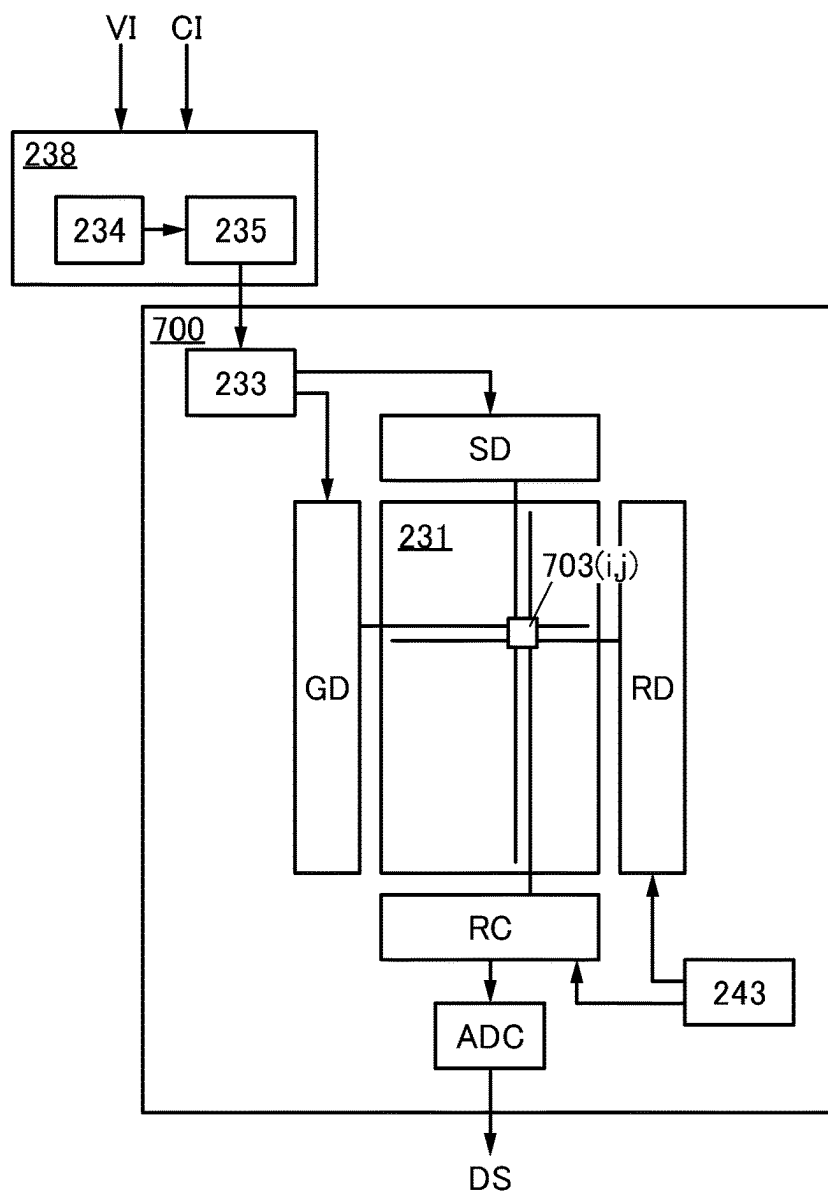


FIG. 14B

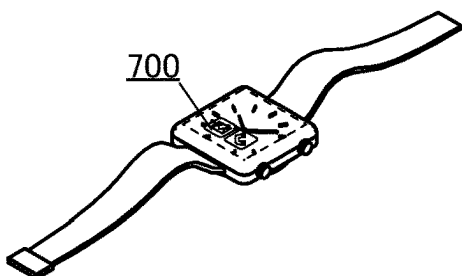


FIG. 14C

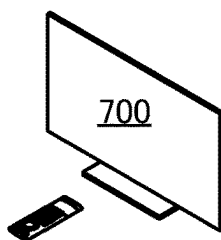


FIG. 14D

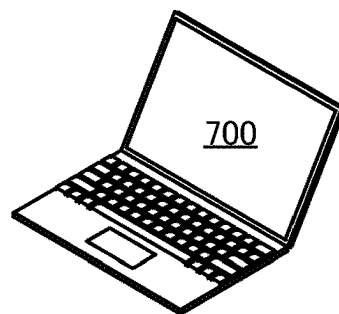


FIG. 15

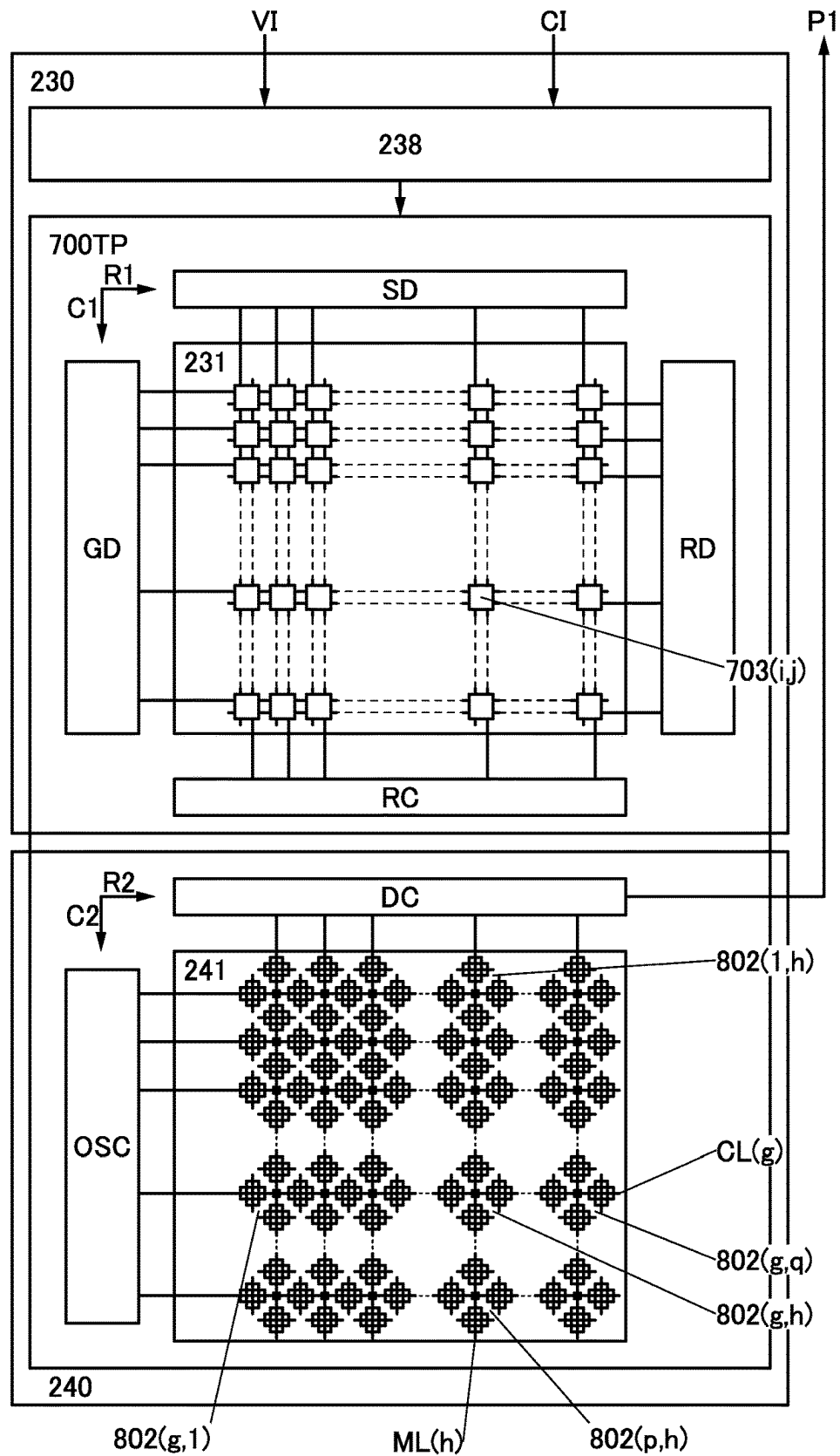


FIG. 16A

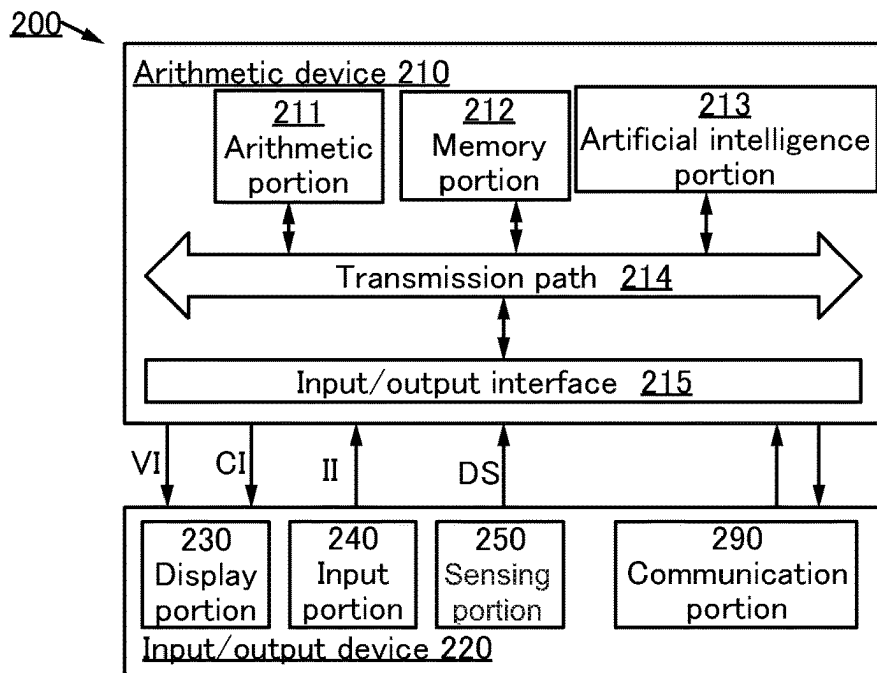


FIG. 16B

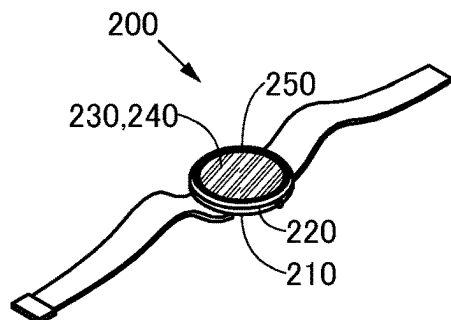


FIG. 16C

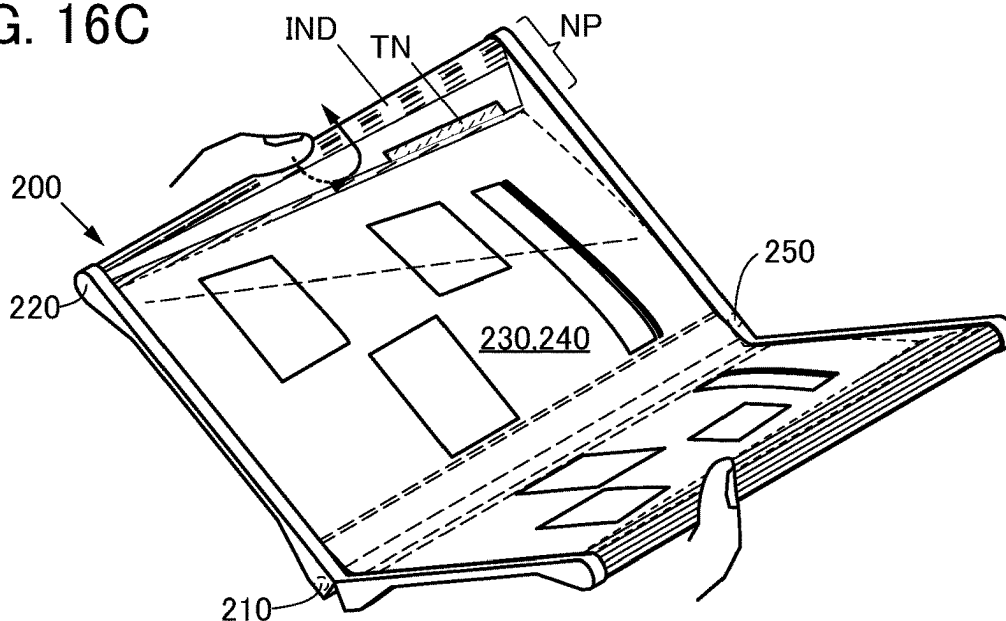


FIG. 17A

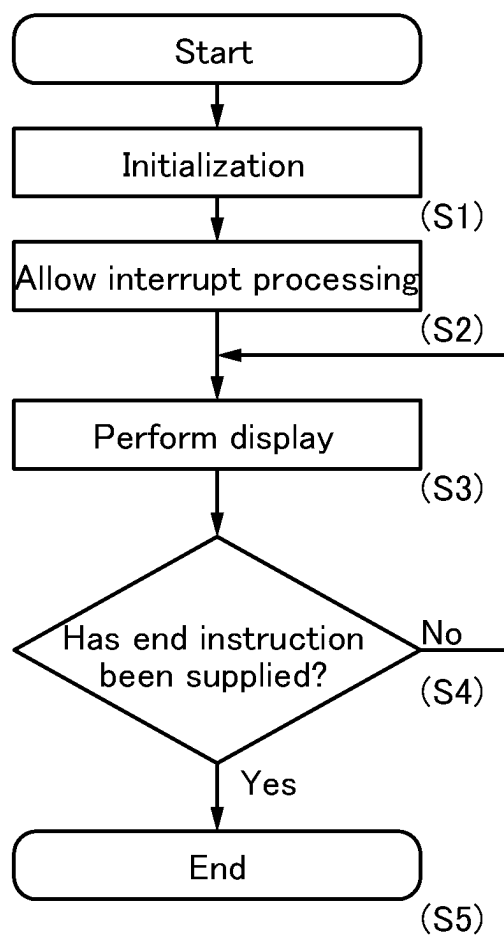


FIG. 17B

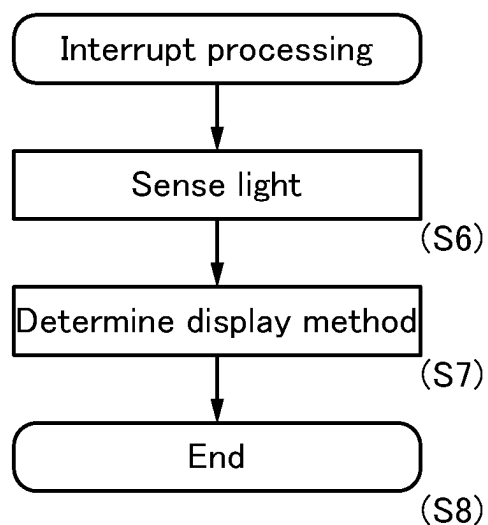


FIG. 18A

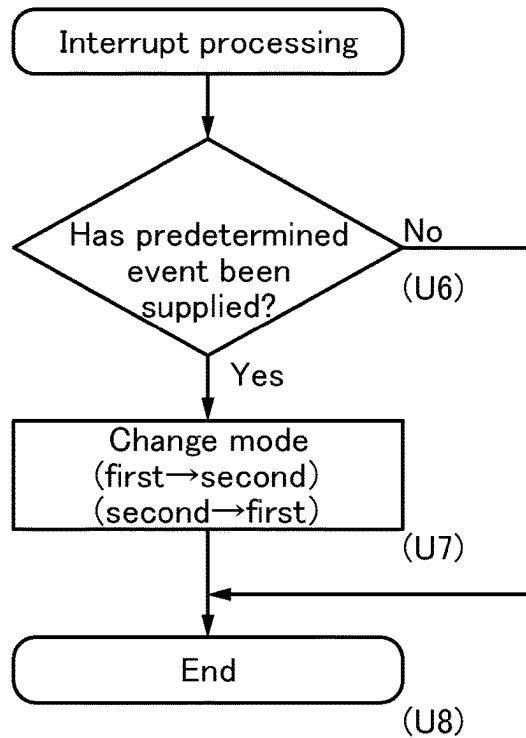


FIG. 18B

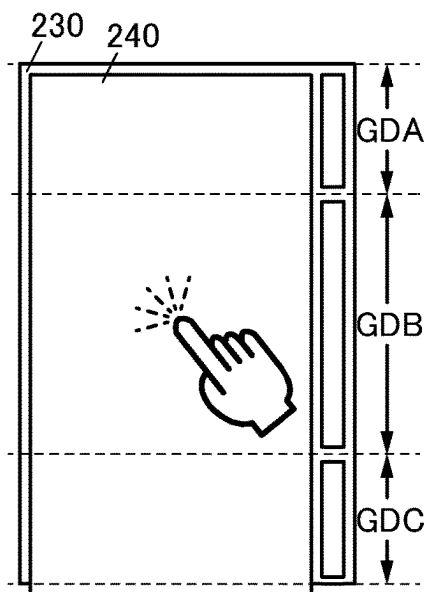


FIG. 18C

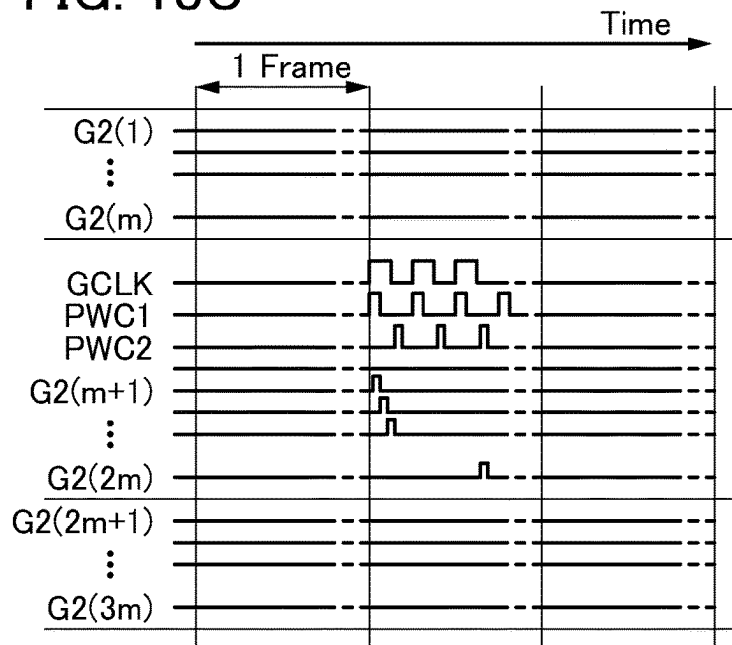


FIG. 19A

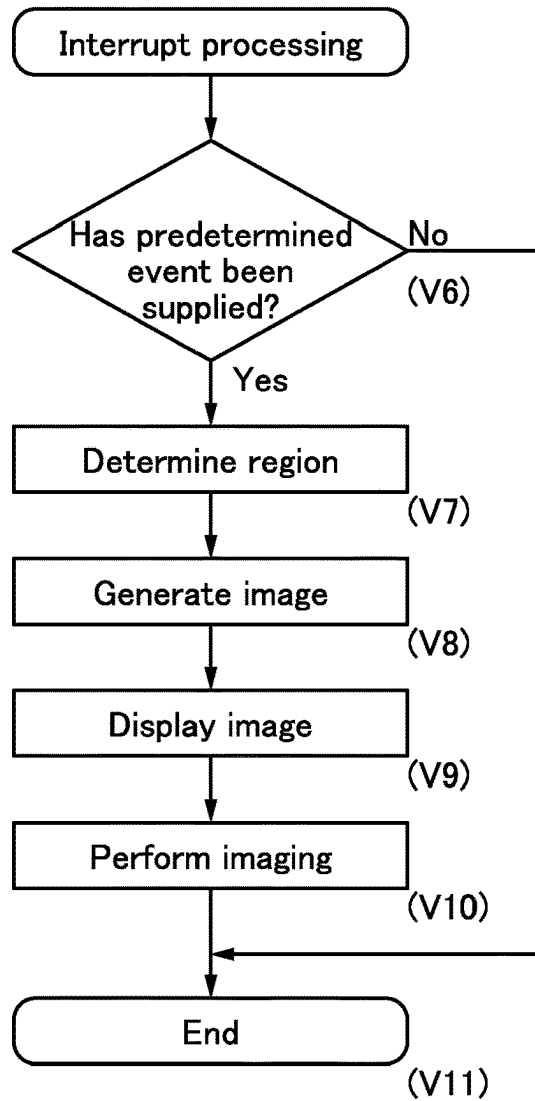


FIG. 19B

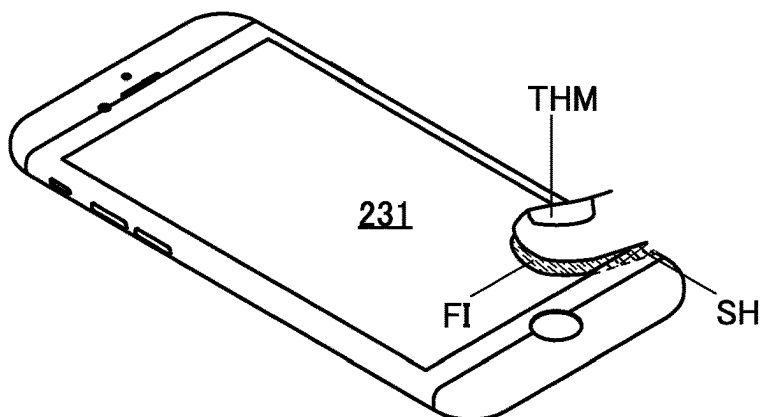


FIG. 19C



FIG. 20A

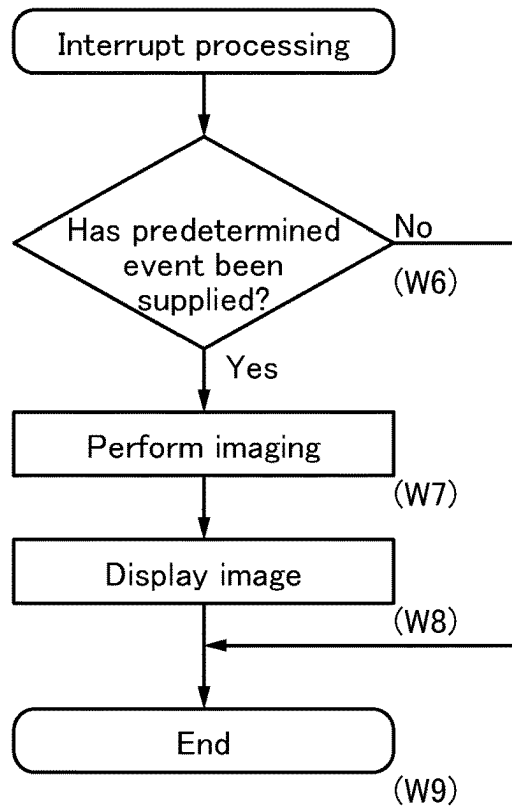


FIG. 20B

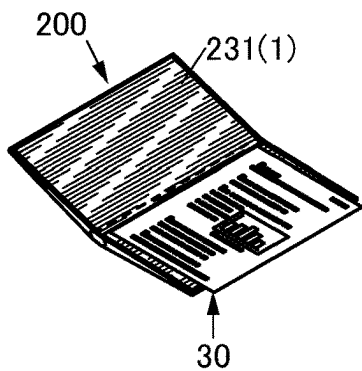


FIG. 20C

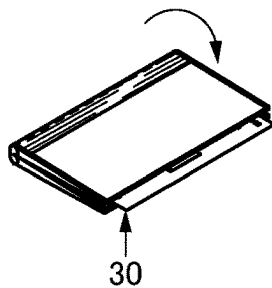


FIG. 20D

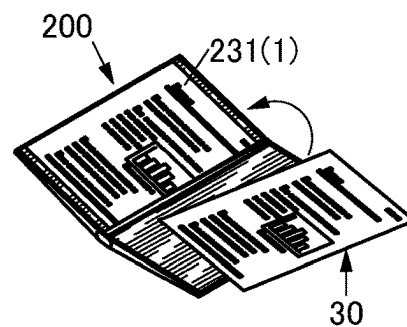


FIG. 21A

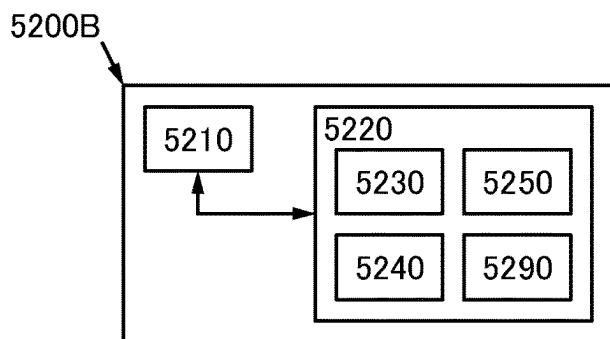


FIG. 21B

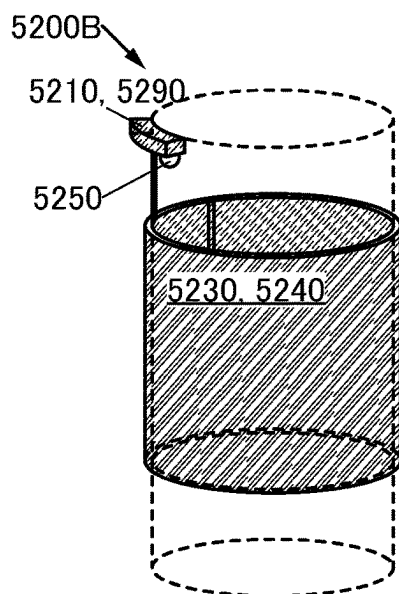


FIG. 21C

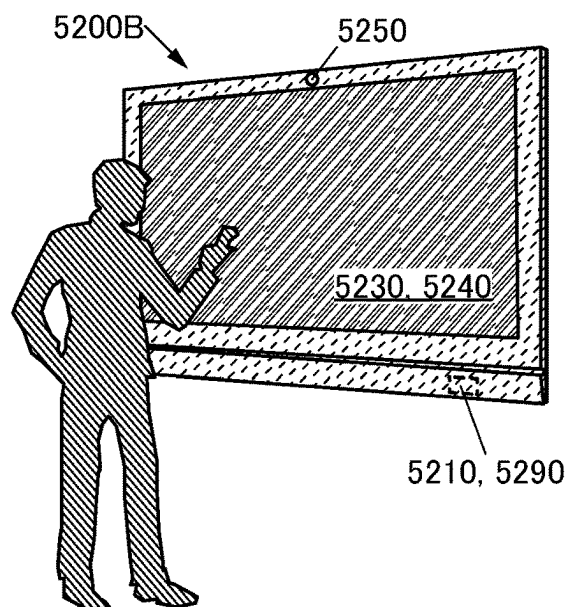


FIG. 21D

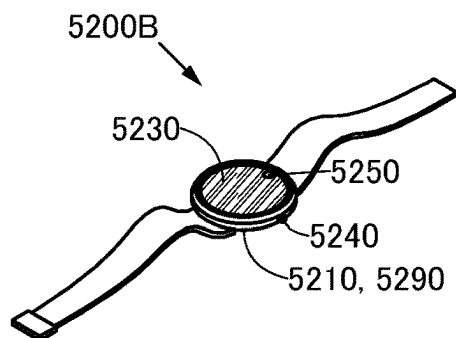


FIG. 21E

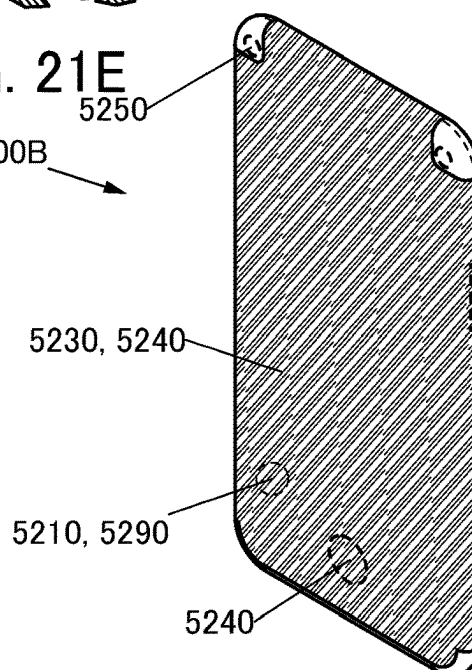


FIG. 22A

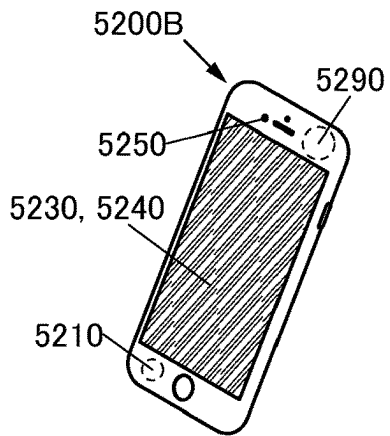


FIG. 22B

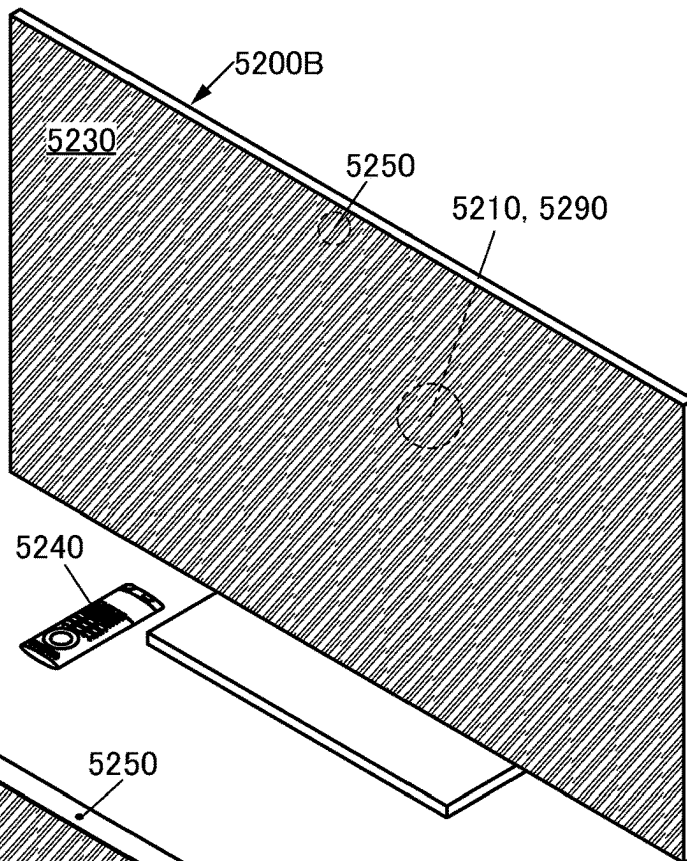


FIG. 22C

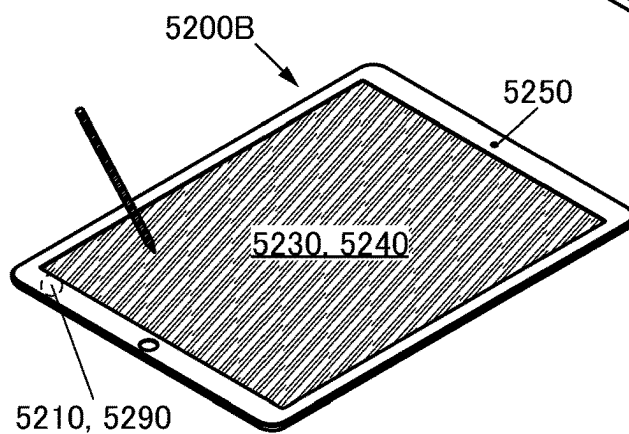


FIG. 22D

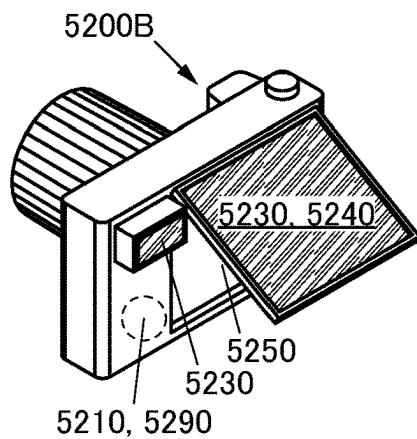


FIG. 22E

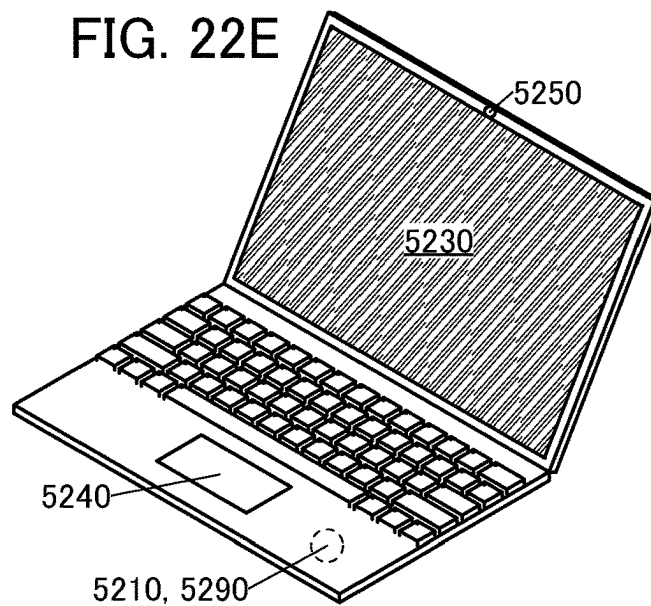


FIG. 23A

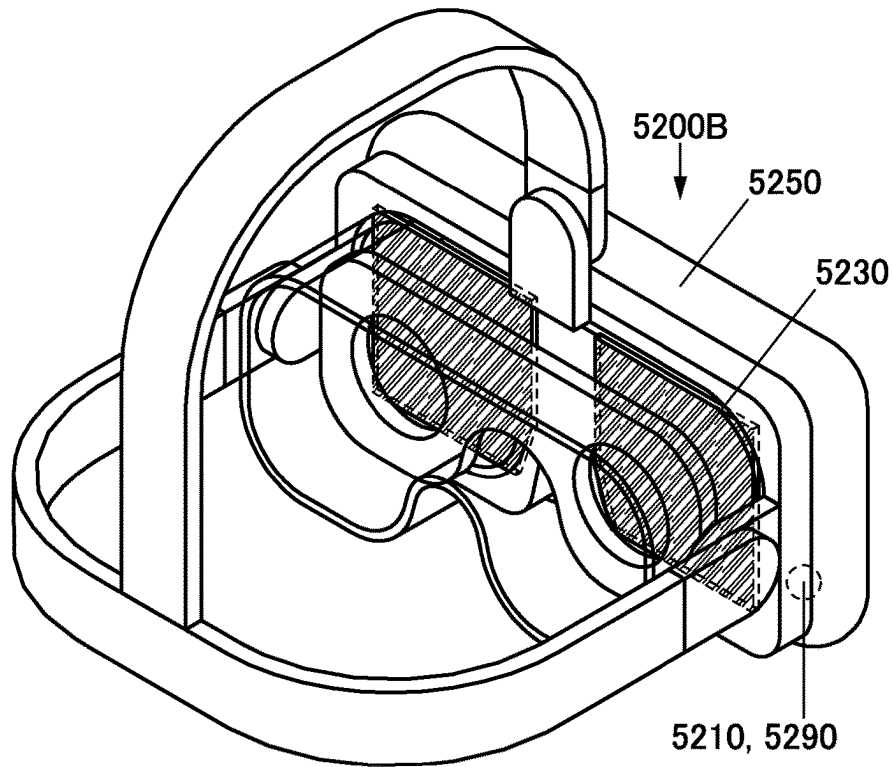
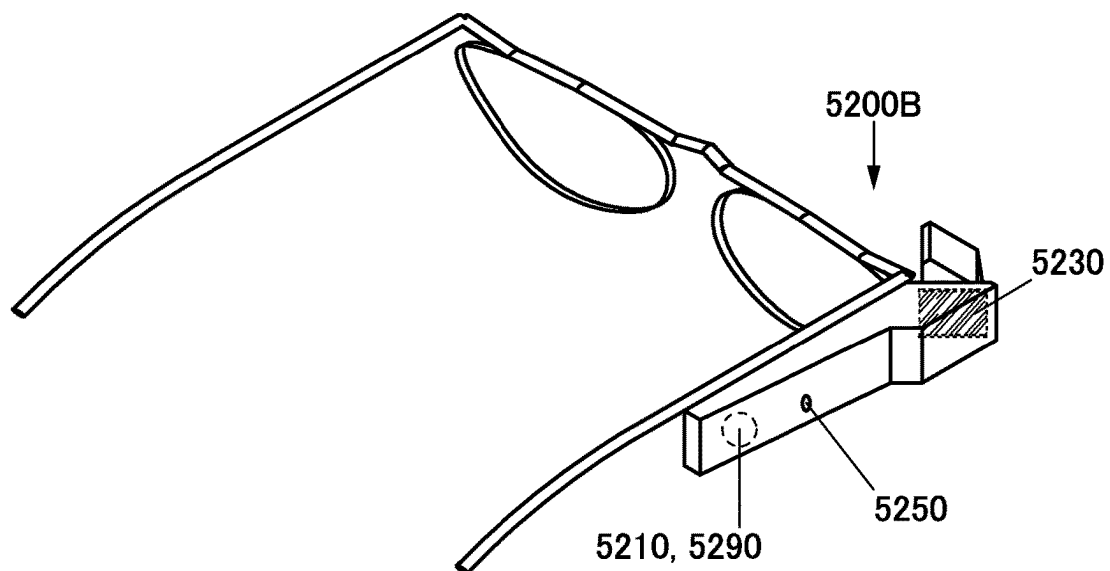


FIG. 23B



1

OPTICAL FUNCTIONAL DEVICE, FUNCTIONAL PANEL, DISPLAY DEVICE, INPUT/OUTPUT DEVICE, AND DATA PROCESSING DEVICE

TECHNICAL FIELD

One embodiment of the present invention relates to an optical functional device, a functional panel, a display device, an input/output device, a data processing device, or a semiconductor device.

Note that one embodiment of the present invention is not limited to the above technical field. The technical field of one embodiment of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. One embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter. Thus, more specifically, examples of the technical field of one embodiment of the present invention disclosed in this specification include a semiconductor device, a display device, a light-emitting apparatus, a power storage device, a memory device, a driving method thereof, and a manufacturing method thereof.

BACKGROUND ART

An imaging panel including a substrate with an insulating surface and a plurality of imaging pixels on the insulating surface is known (Patent Document 1). The imaging pixel includes a plurality of windows which are arranged in a matrix and transmit visible light, a photoelectric conversion element having a grid-like formation which extends between the plurality of windows and supplies a signal, and a sensor circuit supplied with the signal.

REFERENCE

Patent Document

[Patent Document 1] Japanese Published Patent Application No. 2015-005280

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

An object of one embodiment of the present invention is to provide a novel optical functional device that is highly convenient, useful, or reliable. Another object is to provide a novel functional panel that is highly convenient, useful, or reliable. Another object is to provide a novel display device that is highly convenient, useful, or reliable. Another object is to provide a novel input/output device that is highly convenient, useful, or reliable. Another object is to provide a novel data processing device that is highly convenient, useful, or reliable. Another object is to provide a novel method for driving a data processing device that is highly convenient, useful, or reliable. Another object is to provide a novel optical functional device, a novel functional panel, a novel display device, a novel input/output device, a novel data processing device, a novel method for driving a data processing device, or a novel semiconductor device.

Note that the description of these objects does not preclude the existence of other objects. One embodiment of the present invention does not have to achieve all these objects. Other objects will be apparent from the description of the specification, the drawings, the claims, and the like, and

2

other objects can be derived from the description of the specification, the drawings, the claims, and the like.

Means for Solving the Problems

(1) One embodiment of the present invention is an optical functional device including a light-emitting function, a photoelectric conversion function, a first electrode, a second electrode, and an optical functional layer.

The light-emitting function converts electrical energy into first light; the first light has a first emission spectrum; the first emission spectrum exhibits a maximum peak at a first wavelength; the maximum peak exhibits a first value; and at a second wavelength, the intensity of the first emission spectrum is 80% of the first value.

The photoelectric conversion function has a spectral sensitivity characteristic; at a third wavelength, the spectral sensitivity characteristic has a maximum sensitivity within a range of 420 to 720 nm inclusive; and at a fourth wavelength, the sensitivity of the spectral sensitivity characteristic is 80% of the maximum sensitivity.

The third wavelength is positioned closer to the second wavelength than to the first wavelength, and the fourth wavelength is positioned closer to the first wavelength than to the third wavelength.

The optical functional layer includes a region interposed between the first electrode and the second electrode, and the optical functional layer includes a first layer and a second layer. The first layer emits light including the first light, and the second layer includes a region overlapping with the first layer. The second layer includes a light-absorbing material, the light-absorbing material has a first absorption spectrum, and the first absorption spectrum has a region overlapping with the spectral sensitivity characteristic.

(2) Another embodiment of the present invention is the above optical functional device in which the third wavelength is shorter than the first wavelength.

Thus, the overlap between a first spectrum and the spectral sensitivity characteristic can be reduced. The light-absorbing material can be inhibited from absorbing the light. The first light can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic can be performed. Thus, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

(3) Another embodiment of the present invention is the above optical functional device including a light-emitting function, a photoelectric conversion function, a first electrode, a second electrode, and an optical functional layer.

The light-emitting function converts electrical energy into first light, the first light has a first emission spectrum, and the first emission spectrum exhibits a maximum peak at a first wavelength.

The photoelectric conversion function has a spectral sensitivity characteristic, at a third wavelength, the spectral sensitivity characteristic has a maximum sensitivity within a range of 420 to 720 nm inclusive, and the third wavelength is positioned in a range of wavelength longer than or equal to 420 nm and shorter than the first wavelength.

The optical functional layer includes a region interposed between the first electrode and the second electrode, and the optical functional layer includes a first layer and a second layer. The first layer emits light including the first light. The second layer includes a light-absorbing material; the light-absorbing material has a first absorption spectrum; and the first absorption spectrum has a region overlapping with the spectral sensitivity characteristic.

3

Accordingly, the spectral sensitivity characteristic can exhibit high sensitivity to light with a wavelength shorter than the wavelength of the maximum peak of the first spectrum. The first light can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

(4) Another embodiment of the present invention is the above optical functional device in which the second wavelength is shorter than the first wavelength, the fourth wavelength is longer than the third wavelength, and the fourth wavelength is shorter than the second wavelength.

Accordingly, the spectral sensitivity characteristic can exhibit high sensitivity to light with a wavelength shorter than the wavelength of the maximum peak of the first spectrum. The overlap between the first spectrum and the spectral sensitivity characteristic can be reduced. The first light can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

(5) Another embodiment of the present invention is the above optical functional device including a function of emitting red light and a function of photoelectric conversion of green light.

Thus, the optical functional device can be used for a biosensor, for example. A change in bloodstream can be observed. The spectral sensitivity characteristic can exhibit high sensitivity to green light. Red light can be emitted efficiently. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

(6) Another embodiment of the present invention is the above optical functional device including a first electrode, a second electrode, and an optical functional layer.

The optical functional layer includes a region interposed between the first electrode and the second electrode, and the optical functional layer includes a first layer and a second layer.

The first layer includes a light-emitting material, the light-emitting material has a function of emitting second light, the second light has a second spectrum, and the second spectrum exhibits a maximum peak at a fifth wavelength.

The second layer includes a light-absorbing material, and the light-absorbing material has a first absorption spectrum. At a sixth wavelength, the first absorption spectrum exhibits maximum absorption within a range of 420 to 720 nm inclusive. The sixth wavelength is positioned in a range of wavelength longer than or equal to 420 nm and shorter than the fifth wavelength.

(7) Another embodiment of the present invention is the above optical functional device in which the first absorption includes an absorption edge at a seventh wavelength and the seventh wavelength is shorter than the fifth wavelength.

(8) Another embodiment of the present invention is the above optical functional device in which the second spectrum includes an emission edge at an eighth wavelength and the eighth wavelength is shorter than the fifth wavelength and longer than the sixth wavelength.

Accordingly, a phenomenon in which the light-absorbing material absorbs light emitted by the light-emitting material can be inhibited. The first light can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensi-

4

tivity characteristic can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

(9) Another embodiment of the present invention is the above functional panel including a first pixel, a first conductive film, a second conductive film, a third conductive film, a fourth conductive film, a fifth conductive film, a sixth conductive film, and a seventh conductive film.

The first pixel includes the above optical functional device and a first pixel circuit.

The first pixel circuit is electrically connected to the optical functional device. The first pixel circuit is electrically connected to the first conductive film, the second conductive film, the third conductive film, the fourth conductive film, the fifth conductive film, the sixth conductive film, and the seventh conductive film.

(10) Another embodiment of the present invention is the above functional panel including a pixel set.

The pixel set includes the first pixel and a second pixel, and the second pixel includes a light-emitting device and a second pixel circuit.

The light-emitting device is electrically connected to the second pixel circuit, the light-emitting device emits third light, and the optical functional device is capable of photoelectric conversion of the third light.

(11) One embodiment of the present invention is the above functional panel including a functional layer.

The functional layer includes the first pixel circuit, and the first pixel circuit includes a first transistor and a second transistor. The functional layer includes a driver circuit, and the driver circuit includes a third transistor.

The first transistor includes a semiconductor film, the second transistor includes a semiconductor film that can be formed in the step of forming the semiconductor film, and the third transistor also includes a semiconductor film that can be formed in the step of forming the semiconductor film.

Thus, the first pixel circuit can be formed in the functional layer. In the step of forming the semiconductor film of a transistor included in the first pixel circuit, the semiconductor film of the transistor included in the driver circuit can be formed, for example. The manufacturing process of the functional panel can be simplified. As a result, a novel functional panel that is highly convenient, useful, or reliable can be provided.

(12) Another embodiment of the present invention is the above functional panel including a region.

The region includes a group of pixel sets and a different group of pixel sets.

The group of pixel sets is arranged in a row direction, and the group of pixel sets includes the pixel set. The group of pixel sets is electrically connected to the first conductive film, and the group of pixel sets is electrically connected to the fifth conductive film.

The different group of pixel sets is arranged in a column direction intersecting with the row direction, and the different group of pixel sets includes the pixel set. The different group of pixels is electrically connected to the third conductive film, and the different group of pixels is electrically connected to the seventh conductive film.

Thus, imaging data can be obtained from a plurality of pixels. In addition, image data can be supplied to a plurality of pixels. Thus, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

(13) Another embodiment of the present invention is the above functional panel including a group of sampling

5

circuits, a multiplexer, an amplifier circuit, and an analog-digital converter circuit.

The group of sampling circuits includes the sampling circuit; the multiplexer selects one from the group of sampling circuits and obtains an imaging signal; the multiplexer has a function of supplying the imaging signal to the amplifier circuit; and the amplifier circuit has a function of supplying the imaging signal to the analog-digital converter circuit.

Thus, imaging data can be obtained by selecting a predetermined pixel from a plurality of pixels arranged in the row direction. The number of imaging signals obtained at the same time can be limited to a predetermined number. It is possible to use the analog-digital converter circuit ADC in which the number of input channels is smaller than the number of pixels arranged in the row direction. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

(14) One embodiment of the present invention is a display device including a control portion and the above functional panel.

The control portion is supplied with image data and control data; the control portion generates data on the basis of the image data; the control portion generates a control signal on the basis of the control data; and the control portion supplies the data and the control signal.

The functional panel is supplied with the data and the control signal, and the pixel set performs display on the basis of the data.

Thus, the image data can be displayed using the optical functional device. Thus, a novel display device that is highly convenient, useful, or reliable can be provided.

(15) One embodiment of the present invention is an input/output device including an input portion and a display portion.

The display portion includes the above functional panel, and the input portion includes a sensing region.

The input portion senses an object approaching the sensing region, and the sensing region includes a region overlapping with the pixel.

Thus, the object approaching the region overlapping with the display portion can be sensed while image data is displayed using the display portion. A finger or the like approaching the display portion can be used as a pointer to input position data. Position data can be associated with image data displayed on the display portion. Thus, a novel input/output device that is highly convenient, useful, or reliable can be provided.

(16) One embodiment of the present invention is a data processing device including an arithmetic device and an input/output device.

The arithmetic device is supplied with input data or sensing data, the arithmetic device generates control data and image data on the basis of the input data or the sensing data, and the arithmetic device supplies the control data and the image data.

The input/output device supplies the input data and the sensing data, the input/output device is supplied with the control data and the image data, the input/output device includes a display portion, an input portion, and a sensing portion, and the display portion includes the above optical functional device.

The display portion displays the image data on the basis of the control data, the input portion generates the input data, and the sensing portion generates the sensing data.

Thus, the control data can be generated on the basis of the input data or the sensing data. The image data can be

6

displayed on the basis of the input data or the sensing data. Thus, a novel data processing device that is highly convenient, useful, or reliable can be provided.

(17) One embodiment of the present invention is a data processing device including the above optical functional device and one or more of a keyboard, a hardware button, a pointing device, a touch sensor, an illuminance sensor, an imaging device, an audio input device, an eye-gaze input device, and an attitude detection device.

Thus, the arithmetic device can generate the image data or the control data on the basis of the data supplied using a variety of input devices. Thus, a novel data processing device that is highly convenient, useful, or reliable can be provided.

In a block diagram shown in the drawings attached to this specification, components are classified according to their functions and shown as independent blocks; however, it is difficult to separate actual components completely according to their functions, and it is possible for one component to relate to a plurality of functions.

In this specification, the names of a source and a drain of a transistor interchange with each other depending on the polarity of the transistor and the levels of potentials applied to the terminals. In general, in an n-channel transistor, a terminal to which a lower potential is applied is called a source, and a terminal to which a higher potential is applied is called a drain. In a p-channel transistor, a terminal to which a lower potential is applied is called a drain, and a terminal to which a higher potential is applied is called a source. In this specification, for the sake of convenience, the connection relationship of a transistor is sometimes described assuming that the source and the drain are fixed; in reality, the source and the drain interchange with each other according to the above relationship of the potentials.

In this specification, a source of a transistor means a source region that is part of a semiconductor film functioning as an active layer or a source electrode connected to the semiconductor film. Similarly, a drain of a transistor means a drain region that is part of the semiconductor film or a drain electrode connected to the semiconductor film. Moreover, a gate means a gate electrode.

In this specification, a state in which transistors are connected in series means, for example, a state in which only one of a source and a drain of a first transistor is connected to only one of a source and a drain of a second transistor. In addition, a state in which transistors are connected in parallel means a state in which one of a source and a drain of a first transistor is connected to one of a source and a drain of a second transistor and the other of the source and the drain of the first transistor is connected to the other of the source and the drain of the second transistor.

In this specification, connection means electrical connection and corresponds to a state in which a current, a voltage, or a potential can be supplied or transmitted. Accordingly, a state of being connected does not necessarily mean a state of being directly connected and also includes, in its category, a state of being indirectly connected through a circuit element such as a wiring, a resistor, a diode, or a transistor that allows a current, a voltage, or a potential to be supplied or transmitted.

In this specification, even when independent components are connected to each other in a circuit diagram, there is actually a case where one conductive film has functions of a plurality of components, such as a case where part of a wiring functions as an electrode, for example. Connection in

this specification also includes such a case where one conductive film has functions of a plurality of components, in its category.

Furthermore, in this specification, one of a first electrode and a second electrode of a transistor refers to a source electrode and the other refers to a drain electrode.

Effect of the Invention

According to one embodiment of the present invention, a novel optical functional device that is highly convenient, useful, or reliable can be provided. A novel functional panel that is highly convenient, useful, or reliable can be provided. A novel display device that is highly convenient, useful, or reliable can be provided. A novel input/output device that is highly convenient, useful, or reliable can be provided. A novel data processing device that is highly convenient, useful, or reliable can be provided. A novel method for driving a data processing device that is highly convenient, useful, or reliable can be provided. A novel functional panel, a novel display device, a novel input/output device, a novel data processing device, a novel method for driving a data processing device, or a novel semiconductor device can be provided.

Note that the description of these effects does not preclude the existence of other effects. One embodiment of the present invention does not have to have all of these effects. Other effects will be apparent from the description of the specification, the drawings, the claims, and the like and other effects can be derived from the description of the specification, the drawings, the claims, and the like.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A and FIG. 1B are diagrams illustrating a function and a structure of an optical functional device of an embodiment.

FIG. 2A and FIG. 2B are diagrams illustrating a function of an optical functional device of an embodiment.

FIG. 3A to FIG. 3C are diagrams illustrating a structure of a functional panel of an embodiment.

FIG. 4A to FIG. 4C are diagrams illustrating a structure of a functional panel of an embodiment.

FIG. 5 is a circuit diagram illustrating a structure of a functional panel of an embodiment.

FIG. 6A and FIG. 6B are circuit diagrams each illustrating a structure of a functional panel of an embodiment.

FIG. 7A and FIG. 7B are circuit diagrams illustrating structures of a functional panel of an embodiment.

FIG. 8 is a cross-sectional view illustrating a structure of a functional panel of an embodiment.

FIG. 9A and FIG. 9B are cross-sectional views illustrating a structure of a functional panel of an embodiment.

FIG. 10A and FIG. 10B are cross-sectional views illustrating a structure of a functional panel of an embodiment.

FIG. 11A and FIG. 11B are cross-sectional views illustrating a structure of a functional panel of an embodiment.

FIG. 12A and FIG. 12B are diagrams illustrating a structure of a functional panel of an embodiment.

FIG. 13 is a diagram illustrating an operation of a functional panel of an embodiment.

FIG. 14A to FIG. 14D are diagrams illustrating structures of a display device of an embodiment.

FIG. 15 is a block diagram illustrating a structure of an input/output device of an embodiment.

FIG. 16A to FIG. 16C are diagrams illustrating a structure of a data processing device of an embodiment.

FIG. 17A and FIG. 17B are flow charts illustrating a method for driving a data processing device of an embodiment.

FIG. 18A to FIG. 18C are diagrams illustrating a method for driving a data processing device of an embodiment.

FIG. 19A to FIG. 19C are diagrams illustrating a method for driving a data processing device of an embodiment.

FIG. 20A to FIG. 20D are diagrams illustrating a method for driving a data processing device of an embodiment.

FIG. 21A to FIG. 21E are diagrams illustrating structures of a data processing device of an embodiment.

FIG. 22A to FIG. 22E are diagrams illustrating structures of a data processing device of an embodiment.

FIG. 23A and FIG. 23B are diagrams illustrating structures of a data processing device of an embodiment.

MODE FOR CARRYING OUT THE INVENTION

An optical functional device of one embodiment of the present invention includes a light-emitting function, a photoelectric conversion function, a first electrode, a second electrode, and an optical functional layer. The light-emitting function converts electrical energy into first light; the first light has a first emission spectrum; the first emission spectrum exhibits a maximum peak at a first wavelength; the maximum peak exhibits a first value; and at a second wavelength, the intensity of the first emission spectrum is 80% of the first value. The photoelectric conversion function has a spectral sensitivity characteristic; at a third wavelength, the spectral sensitivity characteristic has a maximum sensitivity within a range of 420 to 720 nm inclusive; and at a fourth wavelength, the sensitivity of the spectral sensitivity characteristic is 80% of the maximum sensitivity. The third wavelength is positioned closer to the second wavelength than to the first wavelength; and the fourth wavelength is positioned closer to the first wavelength than to the third wavelength. The optical functional layer includes a region interposed between the first electrode and the second electrode; the optical functional layer includes a first layer and a second layer; the first layer emits light including the first light; the second layer includes a region overlapping with the first layer; and the second layer includes a light-absorbing material. The light-absorbing material has a first absorption spectrum, and the first absorption spectrum has a region overlapping with the spectral sensitivity characteristic.

Thus, the overlap between the first spectrum and the spectral sensitivity characteristic can be reduced. The first light can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

Embodiments are described in detail with reference to the drawings. Note that the present invention is not limited to the following description, and it will be readily appreciated by those skilled in the art that modes and details of the present invention can be modified in various ways without departing from the spirit and scope of the present invention. Thus, the present invention should not be construed as being limited to the description in the following embodiments. Note that in structures of the invention described below, the same portions or portions having similar functions are denoted by the same reference numerals in different drawings, and a description thereof is not repeated.

Embodiment 1

In this embodiment, functions and structures of an optical functional device of one embodiment of the present invention will be described with reference to FIG. 1 and FIG. 2.

FIG. 1A is a diagram illustrating a function of the optical functional device of one embodiment of the present invention. FIG. 1B is a cross-sectional view illustrating a structure of the optical functional device of one embodiment of the present invention.

FIG. 2A and FIG. 2B are diagrams illustrating a function of the optical functional device of one embodiment of the present invention.

Note that in this specification, an integer variable of 1 or more is sometimes used in reference numerals. For example, (p) where p is an integer variable of 1 or more is sometimes used in part of a reference numeral that specifies any of p components at a maximum. As another example, (m, n) where m and n are each an integer variable of 1 or more is sometimes used in part of a reference numeral that specifies any of m×n components at a maximum.

<Structure Example 1 of Optical Functional Device>

An optical functional device 550RS(i, j) described in this embodiment includes a light-emitting function, a photoelectric conversion function, an electrode 551RS(i, j), an electrode 552, and an optical functional layer 553RS(j) (see FIG. 1A and FIG. 1B).

<<Example of Light-Emitting Function>>

The optical functional device 550RS(i, j) has the light-emitting function of converting electrical energy into light EL1. The light EL1 exhibits an emission spectrum SP11. The emission spectrum SP11 has a maximum peak at a wavelength λ_{11} , and the maximum peak exhibits a maximum value I_{max} . At a wavelength λ_{12} , the intensity of the emission spectrum SP11 is 80% of the maximum value I_{max} . Note that the emission spectrum SP11 normalized by the maximum value I_{max} is shown (see FIG. 1A).

<<Example 1 of Photoelectric Conversion Function>>

The optical functional device 550RS(i, j) has a photoelectric conversion function, and the photoelectric conversion function has a spectral sensitivity characteristic ϕ . At a wavelength λ_{21} , the spectral sensitivity characteristic ϕ has a maximum sensitivity V_{max} within the range of 420 to 720 nm inclusive. At a wavelength λ_{22} , the sensitivity of the spectral sensitivity characteristic ϕ is 80% of the maximum sensitivity V_{max} . Note that the spectral sensitivity characteristic ϕ normalized by the maximum value V_{max} is shown (see FIG. 1A). In addition, light incident on the optical functional device 550RS(i, j) is indicated by light $h\nu_1$ (see FIG. 1B).

The wavelength λ_{21} is positioned closer to the wavelength λ_{12} than to the wavelength λ_{11} , and the wavelength λ_{22} is positioned closer to the wavelength λ_{11} than to the wavelength λ_{21} . Although the case where the wavelength λ_{21} is shorter than the wavelength λ_{11} is shown, the present invention is not limited to this. For example, the wavelength λ_{21} may be longer than the wavelength λ_{11} . Specifically, the wavelength λ_{11} may be positioned in the wavelength region of green light and the wavelength λ_{21} may be positioned in the wavelength region of red light or the wavelength region of near-infrared light.

<<Structure Example 1 of Optical Functional Layer 553RS(j)>>

The optical functional layer 553RS(j) includes a region interposed between the electrode 551RS(i, j) and the electrode 552, and the optical functional layer 553RS(j) includes a layer 553R(j) and a layer 553S(j) (see FIG. 1B).

The layer 553R(j) emits light including the light EL1. A light-emitting organic material can be used for the layer 553R(j), for example. Specifically, a fluorescent material or a phosphorescent material can be used for the layer 553R(j). Alternatively, a light-emitting organic material dispersed in

a material having a carrier-transport property can be used. Specifically, an electron-transport material and a hole-transport material can be used for the layer 553R(j). For example, a known material that can be used for an organic EL element can be used for the layer 553R(j).

The layer 553S(j) includes a region overlapping with the layer 553R(j) and the layer 553S(j) includes a light-absorbing material. The light-absorbing material has an absorption spectrum ABS, and the absorption spectrum ABS has a region overlapping with the spectral sensitivity characteristic ϕ . Specifically, the absorption spectrum ABS that is normalized by the maximum absorption within the range of 420 to 720 nm inclusive and the spectral sensitivity characteristic ϕ that is normalized by the maximum value V_{max} within the range of 420 to 720 nm inclusive overlap with each other in this range. Preferably, the region occupying 50% or more of the area of the normalized absorption spectrum ABS and the normalized spectral sensitivity characteristic overlap with each other in this range. Further preferably, the region occupying 65% or more of the area of the normalized absorption spectrum ABS and the normalized spectral sensitivity characteristic overlap with each other in this range.

For example, a known material that can be used for an organic solar cell can be used for the layer 553S(j). Specifically, an electron-accepting material and an electron-donating material can be used for the layer 553S(j). As the electron-accepting material, a fullerene derivative or a non-fullerene electron acceptor can be used, for example. Specifically, a perylene derivative, a compound having a dicyanomethyleneindanone group, or the like can be used as a non-fullerene electron acceptor. As the electron-donating material, a phthalocyanine compound, a tetracene derivative, a quinacridone derivative, a rubrene derivative, or the like can be used.

<<Example 2 of Photoelectric Conversion Function>>

In the optical functional device 550RS(i, j) described in this embodiment, the wavelength λ_{21} is shorter than the wavelength λ_{11} .

Thus, the overlap between the emission spectrum SP11 and the spectral sensitivity characteristic ϕ can be reduced. The light-absorbing material can be inhibited from absorbing the light EL. The light EL1 can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic ϕ can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

<Structure Example 2 of Optical Functional Device>

The optical functional device 550RS(i, j) described in this embodiment includes the light-emitting function, the photoelectric conversion function, the electrode 551RS(i, j), the electrode 552, and the optical functional layer 553RS(j) (see FIG. 1A and FIG. 1B).

<<Example of Light-Emitting Function>>

The optical functional device 550RS(i, j) has the light-emitting function of converting electrical energy into the light EL1. The light EL1 has the emission spectrum SP11. The emission spectrum SP11 has the maximum peak at the wavelength λ_{11} .

<<Example 3 of Photoelectric Conversion Function>>

The optical functional device 550RS(i, j) has the photoelectric conversion function, and the photoelectric conversion function has the spectral sensitivity characteristic ϕ . At the wavelength λ_{21} , the spectral sensitivity characteristic ϕ has the maximum sensitivity V_{max} within the range of 420 to 720 nm inclusive.

11

The wavelength λ_{21} is positioned in a range of wavelength longer than or equal to 420 nm and shorter than the wavelength λ_{11} .

<<Structure Example 2 of Optical Functional Layer 553RS(j)>>

The optical functional layer 553RS(j) includes a region interposed between the electrode 551RS(i, j) and the electrode 552, and the optical functional layer 553RS(j) includes the layer 553R(j) and the layer 553S(j) (see FIG. 1B).

The layer 553R(j) emits light including the light EL1.

The layer 553S(j) includes a light-absorbing material. The light-absorbing material has an absorption spectrum ABS, and the absorption spectrum ABS has a region overlapping with the spectral sensitivity characteristic ϕ . Specifically, the absorption spectrum ABS that is normalized by the maximum absorption within the range of 420 to 720 nm inclusive and the spectral sensitivity characteristic ϕ that is normalized by the maximum value Vmax within the range of 420 to 720 nm inclusive overlap with each other in this range. Preferably, the region occupying 50% or more of the area of the normalized absorption spectrum ABS and the normalized spectral sensitivity characteristic ϕ overlap with each other in this range. Further preferably, the region occupying 65% or more of the area of the normalized absorption spectrum ABS and the normalized spectral sensitivity characteristic overlap with each other in this range.

Accordingly, the spectral sensitivity characteristic can exhibit high sensitivity to light with a wavelength shorter than the wavelength of the maximum peak of the emission spectrum SP11. The light EL1 can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

<Structure Example 3 of Optical Functional Device>

In the optical functional device 550RS(i, j) described in this embodiment, the wavelength λ_{12} is shorter than the wavelength λ_{11} , the wavelength λ_{22} is longer than the wavelength λ_{21} , and the wavelength λ_{22} is shorter than the wavelength λ_{12} .

Accordingly, the spectral sensitivity characteristic can exhibit high sensitivity to light with a wavelength shorter than the wavelength of the maximum peak of the emission spectrum SP11. The overlap between the emission spectrum SP11 and the spectral sensitivity characteristic ϕ can be reduced. The light EL1 can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic ϕ can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

<Structure Example 4 of Optical Functional Device>

The optical functional device 550RS(i, j) described in this embodiment has a function of emitting red light and a function of photoelectric conversion of green light.

Thus, the optical functional device can be used for a biosensor, for example. A change in bloodstream can be observed. The spectral sensitivity characteristic can exhibit high sensitivity to green light. Red light can be emitted efficiently. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

<Structure Example 5 of Optical Functional Device>

The optical functional device 550RS(i, j) described in this embodiment includes the electrode 551RS(i, j), the electrode 552, and the optical functional layer 553RS(j) (see FIG. 1B).

12

<<Structure Example 3 of Optical Functional Layer 553RS(j)>>

The optical functional layer 553RS(j) includes a region interposed between the electrode 551RS(i, j) and the electrode 552, and the optical functional layer 553RS(j) includes the layer 553R(j) and the layer 553S(j).

<<Example 1 of Layer 553R(j)>>

The layer 553R(j) includes a light-emitting material, and the light-emitting material has a function of emitting light (photoluminescence) PL. The light (photoluminescence) PL has a spectrum SP12, and the spectrum SP12 exhibits a maximum peak at a wavelength λ_{31} .

<<Example 1 of Layer 553S(j)>>

The layer 553S(j) includes a light-absorbing material, and the light-absorbing material has the absorption spectrum ABS. At a wavelength λ_{41} , the absorption spectrum ABS exhibits the maximum absorption within the range of 420 to 720 nm inclusive. The wavelength λ_{41} is positioned in a range of wavelength longer than or equal to 420 nm and shorter than the wavelength λ_{31} .

<<Example 2 of Layer 553S(j)>>

The layer 553S(j) includes a light-absorbing material, and the light-absorbing material has the absorption spectrum ABS. The absorption spectrum ABS has an absorption edge at a wavelength λ_{43} , and the wavelength λ_{43} is shorter than the wavelength λ_{31} .

<<Example 2 of Layer 553R(j)>>

The layer 553R(j) includes a light-emitting material and the light-emitting material has a function of emitting light (photoluminescence) PL. The light (photoluminescence) PL has the spectrum SP12, the spectrum SP12 has an emission edge at a wavelength λ_{33} , and the wavelength λ_{33} is shorter than the wavelength λ_{31} and longer than the wavelength λ_{41} .

Accordingly, a phenomenon in which the light-absorbing material absorbs light emitted by the light-emitting material can be inhibited. The first light EL1 can be emitted efficiently. Photoelectric conversion on the basis of the spectral sensitivity characteristic ϕ can be performed. As a result, a novel optical functional device that is highly convenient, useful, or reliable can be provided.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 2

In this embodiment, structures of a functional panel of one embodiment of the present invention will be described with reference to FIG. 3 to FIG. 7.

FIG. 3A is a top view illustrating a structure of the functional panel of one embodiment of the present invention, and FIG. 3B is a diagram illustrating part of FIG. 3A. FIG. 3C is a diagram illustrating part of FIG. 3B.

FIG. 4A is a diagram illustrating part of FIG. 3A, which illustrates a structure of a pixel that can be used for the functional panel of one embodiment of the present invention. FIG. 4B is a diagram illustrating part of FIG. 4A, and FIG. 4C is a diagram illustrating another part of FIG. 4A.

FIG. 5 is a circuit diagram illustrating part of FIG. 4B, which illustrates a structure of a pixel circuit that can be used for the functional panel of one embodiment of the present invention.

FIG. 6A is a circuit diagram illustrating part of FIG. 4B, which illustrates a structure of a pixel that can be used for the functional panel of one embodiment of the present invention. FIG. 6B is a circuit diagram illustrating a structure different from that in FIG. 6A.

FIG. 7A is a circuit diagram illustrating part of an amplifier circuit that can be used in the functional panel of

one embodiment of the present invention, and FIG. 7B is a circuit diagram of a sampling circuit that can be used in the functional panel of one embodiment of the present invention.

<Structure Example 1 of Functional Panel 700>

A functional panel 700 described in this embodiment includes a pixel 702RS(i, j), a conductive film G1(i), a conductive film S1r(j), a conductive film ANO, a conductive film RS(i), a conductive film TX(i), a conductive film SE(i), and a conductive film WX(j) (see FIG. 3A and FIG. 4A). The functional panel 700 includes a conductive film VCOM2.

For example, the conductive film G1(i) is supplied with a first selection signal, and the conductive film S1r(j) is supplied with an image signal. For example, the conductive film RS(i) is supplied with a second selection signal, the conductive film TX(i) is supplied with a third selection signal, and the conductive film SE(i) is supplied with a fourth selection signal.

<<Structure Example 1 of Pixel 702RS(i, j)>>

The pixel 702RS(i, j) includes the optical functional device 550RS(i, j) and a pixel circuit 530RS(i, j) (see FIG. 3C). For example, the structure described in Embodiment 1 can be used for the optical functional device 550RS(i, j).

<<Structure Example of Pixel Circuit 530RS(i, j)>>

The pixel circuit 530RS(i, j) is electrically connected to the optical functional device 550RS(i, j) (see FIG. 5).

The pixel circuit 530RS(i, j) is electrically connected to the conductive film G1(i), the conductive film S1r(j), the conductive film ANO, the conductive film RS(i), the conductive film TX(i), the conductive film SE(i), and the conductive film WX(j).

The pixel circuit 530RS(i, j) includes a switch SW21, a transistor M21, and a node N21. In addition, the pixel circuit 530RS(i, j) includes a node N22, a capacitor C22, and a switch SW23.

The transistor M21 includes a gate electrode electrically connected to the node N21, a first electrode electrically connected to the optical functional device 550RS(i, j), and a second electrode electrically connected to a conductive film ANO.

The switch SW21 includes a first terminal electrically connected to the node N21 and a second terminal electrically connected to a conductive film S1r(j), and has a function of controlling the conduction state or the non-conduction state on the basis of a potential of a conductive film G1(i).

Thus, the supplied image signal can be held in the node N21. Alternatively, the optical functional device 550RS(i, j) can be made to emit light with brightness in accordance with the potential of the node N21. The image data can be displayed. As a result, a novel display device that is highly convenient, useful, or reliable can be provided.

The pixel circuit 530RS(i, j) includes a switch SW31, a switch SW32, a switch SW33, a transistor M31, a capacitor C31, and a node FD (see FIG. 5).

The switch SW31 includes a first terminal electrically connected to the optical functional device 550RS(i, j) and a second terminal electrically connected to the node FD, and has a function of controlling the conduction state or the non-conduction state on the basis of a potential of a conductive film TX(i).

The switch SW32 includes a first terminal electrically connected to the node FD and a second terminal electrically connected to a conductive film VR, and has a function of controlling the conduction state or the non-conduction state on the basis of a potential of a conductive film RS(i).

The capacitor C31 includes a conductive film electrically connected to the node FD and a conductive film electrically connected to a conductive film VCP.

The transistor M31 includes a gate electrode electrically connected to the node FD and a first electrode electrically connected to a conductive film VPI.

The switch SW33 includes a first terminal electrically connected to a second electrode of the transistor M31 and a second terminal electrically connected to a conductive film WX(j), and has a function of controlling the conduction state or the non-conduction state on the basis of a potential of a conductive film SE(i).

Thus, an imaging signal generated by the optical functional device 550RS(i, j) can be transferred to the node FD using the switch SW31. The imaging signal generated by the optical functional device 550RS(i, j) can be stored in the node FD using the switch SW31. Electrical continuity between the pixel circuit 530RS(i, j) and the optical functional device 550RS(i, j) can be broken by the switch SW31. A correlated double sampling method can be used. Noise included in the imaging signal can be reduced. Thus, a novel functional panel that is highly convenient or reliable can be provided.

<Structure Example 2 of Functional Panel 700>

The functional panel 700 described in this embodiment includes a pixel set 703(i, j), and the pixel set 703(i, j) includes the pixel 702RS(i, j) and a pixel 702G(i, j) (see FIG. 3B and FIG. 4A).

The pixel 702G(i, j) includes a light-emitting device 550G(i, j) and a pixel circuit 530G(i, j).

The light-emitting device 550G(i, j) is electrically connected to the pixel circuit 530G(i, j) and the light-emitting device 550G(i, j) emits light EL2.

The optical functional device 550RS(i, j) is capable of photoelectric conversion of the light EL2. In other words, the optical functional device 550RS(i, j) has a spectral sensitivity characteristic ϕ , and the spectral sensitivity overlaps with the emission spectrum of the light EL2 emitted by the light-emitting device 550G(i, j).

Thus, capturing using the optical functional device 550RS(i, j) can be performed while the light-emitting device 550G(i, j) is used for lighting. As a result, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example 1 of Pixel Circuit 530G(i, j)>>

The pixel circuit 530G(i, j) is supplied with the first selection signal, and the pixel circuit 530G(i, j) obtains an image signal on the basis of the first selection signal. For example, the first selection signal can be supplied using the conductive film G1(i) (see FIG. 4A). The image signal can be supplied using the conductive film S1g(j). Note that the operation of supplying the first selection signal and making the pixel circuit 530G(i, j) obtain the image signal can be referred to as "writing" (see FIG. 13).

<<Structure Example 2 of Pixel Circuit 530G(i, j)>>

The pixel circuit 530G(i, j) includes the switch SW21, the switch SW23, the transistor M21, the capacitor C22, and the node N21 (see FIG. 6A). The pixel circuit 530G(i, j) includes the node N22.

The transistor M21 includes a gate electrode electrically connected to the node N21, a first electrode electrically connected to the light-emitting device 550G(i, j), and a second electrode electrically connected to a conductive film ANO.

The switch SW21 includes a first terminal electrically connected to the node N21 and a second terminal electrically connected to a conductive film S1g(j), and has a function of

controlling the conduction state or the non-conduction state on the basis of a potential of a conductive film G1(i).

<<Structure Example 3 of Pixel Circuit 530G(i, j)>>

The pixel circuit 530G(i, j) includes the switch SW21, a switch SW22, the transistor M21, the capacitor C22, and the node N21, for example (see FIG. 6B). In addition, the pixel circuit 530G(i, j) includes the node N22, the capacitor C21, and the switch SW23.

The switch SW22 includes a first terminal electrically connected to a conductive film S2g(j), and has a function of controlling the conduction state or the non-conduction state on the basis of a potential of a conductive film G2(i) (see FIG. 6B).

The capacitor C21 includes a conductive film electrically connected to the node N21 and a conductive film electrically connected to a second electrode of the switch SW22.

Thus, an image signal can be stored in the node N21. A potential of the node N21 can be changed using the switch SW22. The intensity of light emitted from the light-emitting device 550G(i, j) can be controlled with the potential of the node N21. Thus, a novel functional panel that is highly convenient or reliable can be provided.

<<Structure Example of Light-Emitting Device 550G(i, j)>>

The light-emitting device 550G(i, j) is electrically connected to the pixel circuit 530G(i, j) (see FIG. 6A). The light-emitting device 550G(i, j) includes the electrode 551G(i, j) electrically connected to the pixel circuit 530G(i, j), the electrode 552 electrically connected to the conductive film VCOM2, and the layer 553G(j) (see FIG. 6A, FIG. 6B, and FIG. 9A). The layer 553G(j) emits light including the light EL2. A light-emitting organic material can be used for the layer 553G(j), for example. Note that the light-emitting device 550G(i, j) has a function of operating on the basis of the potential of the node N21.

For example, an organic electroluminescence element, an inorganic electroluminescence element, a light-emitting diode, a mini LED, a micro LED, a QDLED (Quantum Dot LED), or the like can be used as the light-emitting device 550G(i, j).

<<Structure Example of Pixel 703(i, j)>>

A plurality of pixels can be used in the pixel 703(i, j). For example, a plurality of pixels capable of displaying colors with different hues can be used. Note that a plurality of pixels can be referred to as subpixels. A set of subpixels can be rephrased as a pixel.

This enables additive mixture or subtractive mixture of colors displayed by the plurality of pixels. It is possible to display a color of a hue that an individual pixel cannot display.

Specifically, a pixel 702B(i, j) displaying blue, the pixel 702G(i, j) displaying green, and a pixel 702RS(i, j) displaying red can be used in the pixel 703(i, j). The pixel 702B(i, j), the pixel 702G(i, j), and the pixel 702RS(i, j) can each be referred to as a subpixel (see FIG. 3B).

A pixel displaying white or the like can be used in addition to the above set in the pixel 703(i, j), for example. A pixel displaying cyan, a pixel displaying magenta, and a pixel displaying yellow can be used in the pixel 703(i, j).

A pixel emitting infrared rays can be used in addition to the above set in the pixel 703(i, j), for example. Specifically, a pixel that emits light including light with a wavelength of greater than or equal to 650 nm and less than or equal to 1000 nm can be used in the pixel 703(i, j).

<Structure Example 3 of Functional Panel 700>

The functional panel described in this embodiment includes a driver circuit GD, a driver circuit SD, and a driver circuit RD (see FIG. 3A).

<<Structure Example of Driver Circuit GD>>

The driver circuit GD has a function of supplying the first selection signal. For example, the driver circuit GD is electrically connected to the conductive film G1(i) and supplies the first selection signal. In addition, the driver circuit GD is electrically connected to the conductive film G2(i) and can also supply another selection signal.

<<Structure Example of Driver Circuit SD>>

The driver circuit SD has a function of supplying an image signal and a control signal, and the control signal includes a first level and a second level. For example, the driver circuit SD is electrically connected to the conductive film S1g(j) and supplies the image signal. In addition, the driver circuit SD is electrically connected to the conductive film S2g(j) and can also supply the control signal.

<<Structure Example of Driver Circuit RD>>

The driver circuit RD has a function of supplying the second selection signal to the fourth selection signal. For example, the driver circuit RD is electrically connected to the conductive film RS(i), supplies the second selection signal, is electrically connected to the conductive film TX(i), supplies the third selection signal, is electrically connected to the conductive film SE(i), and supplies the fourth selection signal.

<Structure Example 4 of Functional Panel 700>

The functional panel described in this embodiment includes a reading circuit RC (see FIG. 3A). The functional panel includes a conductive film VLEN and a conductive film VIV. Note that the reading circuit RC includes a reading circuit RC(j). The functional panel of one embodiment of the present invention includes a conductive film CAPSEL, a conductive film CDSBIAS, a conductive film CDSVDD, a conductive film CDSVSS, and a conductive film VCL.

<<Structure example of reading circuit RC(j)>>

The reading circuit RC(j) includes an amplifier circuit and a sampling circuit SC(j) (see FIG. 7A and FIG. 7B).

<<Structure Example of Amplifier Circuit>>

The amplifier circuit includes a transistor M32(j) (see FIG. 7A). The transistor M32(j) includes a gate electrode electrically connected to the conductive film VLEN, a first electrode electrically connected to the conductive film WX(j), and a second electrode electrically connected to the conductive film VIV.

Note that the conductive film WX(j) connects the transistor M31 and the transistor M32 when the switch SW33 is in a conduction state (see FIG. 5 and FIG. 7A). Thus, a source follower circuit can be configured with the transistor M31 and the transistor M32. The potential of the conductive film WX(j) can be changed on the basis of the potential of the node FD.

<<Structure Example of Sampling Circuit SC(j)>>

The sampling circuit SC(j) includes a first terminal IN(j), a second terminal, and a third terminal OUT(j) (see FIG. 7B).

The first terminal IN(j) is electrically connected to the conductive film WX(j), the second terminal is electrically connected to a conductive film CL, and the third terminal OUT(j) has a function of supplying a signal that changes on the basis of a potential of the first terminal IN(j).

Accordingly, an imaging signal can be obtained from the pixel circuit 530RS(i, j). A correlated double sampling method can be employed, for example. A differential signal of the pixel circuit 530RS(i, j) can be obtained by the corresponding conductive film WX(j). Noise can be

17

reduced. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 3

In this embodiment, structures of a functional panel of one embodiment of the present invention will be described with reference to FIG. 8 to FIG. 11.

FIG. 8 is a diagram illustrating a structure of the functional panel of one embodiment of the present invention. FIG. 8 is a cross-sectional view taken along cutting lines X1-X2, X3-X4, X9-X10, and X11-X12 in FIG. 3A and in the pixel set 703(i, j).

FIG. 9A is a diagram illustrating a structure of the functional panel of one embodiment of the present invention. FIG. 9A is a cross-sectional view of the pixel 702G(i, j) shown in FIG. 3B. FIG. 9B is a cross-sectional view illustrating part of FIG. 9A.

FIG. 10A is a diagram illustrating a structure of the functional panel of one embodiment of the present invention. FIG. 10A is a cross-sectional view of the pixel 702RS(i, j) shown in FIG. 3B. FIG. 10B is a cross-sectional view illustrating part of FIG. 10A.

FIG. 11A is a diagram illustrating a structure of the functional panel of one embodiment of the present invention. FIG. 11A is a cross-sectional view taken along cutting lines X1-X2 and X3-X4 in FIG. 3A, and FIG. 11B is a diagram illustrating part of FIG. 11A.

<Structure Example 1 of Functional Panel 700>

The functional panel 700 described in this embodiment includes a functional layer 520 (see FIG. 8).

<<Structure Example 1 of Functional Layer 520>>

The functional layer 520 includes the pixel circuit 530RS(i, j), and the pixel circuit 530RS(i, j) includes the transistor M21 and the transistor M31 (see FIG. 5, FIG. 8, and FIG. 10). That is, the functional layer 520 includes, for example, the transistor M21 and the transistor M31 of the pixel circuit 530RS(i, j).

The functional layer 520 includes an opening portion 591RS and an opening portion 591G. The pixel circuit 530RS(i, j) is electrically connected to the optical functional device 550RS(i, j) through the opening portion 591RS (see FIG. 8 and FIG. 10A). The pixel circuit 530G(i, j) is electrically connected to the light-emitting device 550G(i, j) through the opening portion 591G (see FIG. 8).

The functional layer 520 includes the driver circuit GD and the driver circuit GD includes the transistor MD (see FIG. 8 and FIG. 11).

The transistor M21 includes a semiconductor film, and the transistor M31 includes a semiconductor film that can be formed in the step of forming the semiconductor film included in the transistor M21.

The transistor MD includes a semiconductor film, and the transistor MD includes a semiconductor film that can be formed in the step of forming the semiconductor film included in the transistor M21.

Thus, the pixel circuit 530RS(i, j) can be formed in the functional layer 520. The semiconductor film of the transistor included in the driver circuit GD can be formed in the step of forming the semiconductor film of the transistor included in the pixel circuit 530RS(i, j), for example. The manufacturing process of the functional panel can be simplified. As a result, a novel functional panel that is highly convenient, useful, or reliable can be provided.

18

<<Structure Example 2 of Functional Layer 520>>

The functional layer 520 includes a driver circuit GD (see FIG. 3A and FIG. 8). The functional layer 520 includes, for example, a transistor MD used in the driver circuit GD (see FIG. 8 and FIG. 11A).

The functional layer 520 includes a driver circuit RD and a reading circuit RC (see FIG. 8).

Thus, the semiconductor film used in the driver circuit GD can be formed in the step of forming the semiconductor film used in the pixel circuit 530RS(i, j), for example. Semiconductor films used in the driver circuit RD and the reading circuit RC can be formed in the step of forming the semiconductor film used in the pixel circuit 530RS(i, j). The manufacturing process of the functional panel can be simplified. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example of Transistor>>

A bottom-gate transistor, a top-gate transistor, or the like can be used in the functional layer 520. Specifically, a transistor can be used as a switch.

The transistor includes a semiconductor film 508, a conductive film 504, a conductive film 512A, and a conductive film 512B (see FIG. 9B). Alternatively, the transistor includes a conductive film 512C and a conductive film 512D (see FIG. 11B). Alternatively, the transistor includes a conductive film 512E and a conductive film 512F (see FIG. 10B).

The semiconductor film 508 includes a region 508A electrically connected to the conductive film 512A and a region 508B electrically connected to the conductive film 512B. The semiconductor film 508 includes a region 508C between the region 508A and the region 508B.

The conductive film 504 includes a region overlapping with the region 508C, and the conductive film 504 has a function of a gate electrode.

An insulating film 506 includes a region interposed between the semiconductor film 508 and the conductive film 504. The insulating film 506 has a function of a gate insulating film.

The conductive film 512A has one of a function of a source electrode and a function of a drain electrode, and the conductive film 512B has the other of the function of the source electrode and the function of the drain electrode.

A conductive film 524 can be used for the transistor. The semiconductor film 508 is interposed between a region of the conductive film 524 and the conductive film 504. The conductive film 524 has a function of a second gate electrode.

Note that the semiconductor film used in the transistor of the driver circuit can be formed in the step of forming the semiconductor film used in the transistor of the pixel circuit. A semiconductor film having the same composition as the semiconductor film used in the transistor of the pixel circuit can be used in the driver circuit, for example.

<<Structure Example 1 of Semiconductor Film 508>>

A semiconductor containing a Group 14 element can be used for the semiconductor film 508, for example. Specifically, a semiconductor containing silicon can be used for the semiconductor film 508.

[Hydrogenated Amorphous Silicon]

For example, hydrogenated amorphous silicon can be used for the semiconductor film 508. Alternatively, microcrystalline silicon or the like can be used for the semiconductor film 508. Thus, a functional panel having less display unevenness than a functional panel using polysilicon for the semiconductor film 508, for example, can be provided. The size of the functional panel can be easily increased.

[Polysilicon]

For example, polysilicon can be used for the semiconductor film 508. In this case, the field-effect mobility of the transistor can be higher than that of a transistor using hydrogenated amorphous silicon for the semiconductor film 508, for example. The driving capability can be higher than that of a transistor using hydrogenated amorphous silicon for the semiconductor film 508, for example. The aperture ratio of the pixel can be higher than that in the case of using a transistor that uses hydrogenated amorphous silicon for the semiconductor film 508, for example.

The reliability of the transistor can be higher than that of a transistor using hydrogenated amorphous silicon for the semiconductor film 508, for example.

The temperature required for fabrication of the transistor can be lower than that required for a transistor using single crystal silicon, for example.

The semiconductor film used in the transistor of the driver circuit can be formed in the same step as the semiconductor film used in the transistor of the pixel circuit. The driver circuit can be formed over the same substrate where the pixel circuit is formed. The number of components included in an electronic device can be reduced.

[Single Crystal Silicon]

For example, single crystal silicon can be used for the semiconductor film 508. In this case, a functional panel with higher resolution than a functional panel using hydrogenated amorphous silicon for the semiconductor film 508, for example, can be provided. A functional panel having less display unevenness than a functional panel using polysilicon for the semiconductor film 508, for example, can be provided. Smart glasses or a head-mounted display can be provided, for example.

<<Structure Example 2 of Semiconductor Film 508>>

For example, a metal oxide can be used for the semiconductor film 508. In this case, the pixel circuit can hold an image signal for a longer time than a pixel circuit utilizing a transistor using amorphous silicon for a semiconductor film. Specifically, a selection signal can be supplied at a frequency of lower than 30 Hz, preferably lower than 1 Hz, and further preferably less than once per minute with the suppressed occurrence of flickers. Consequently, fatigue accumulation in a user of a data processing device can be reduced. Moreover, power consumption for driving can be reduced.

The pixel circuit can hold an imaging signal for a longer time than a pixel circuit utilizing a transistor using amorphous silicon for a semiconductor film. Specifically, the first selection signal can be supplied at a frequency lower than 30 Hz, preferably lower than 1 Hz, and further preferably less than once per minute. Accordingly, an image can be taken by a global shutter method. An image of a moving object can be taken while distortion is inhibited.

A transistor using an oxide semiconductor can be used, for example. Specifically, an oxide semiconductor containing indium, an oxide semiconductor containing indium, gallium, and zinc, or an oxide semiconductor containing indium, gallium, zinc, and tin can be used for the semiconductor film.

A transistor having a lower leakage current in an off state than a transistor using amorphous silicon for a semiconductor film can be used, for example. Specifically, a transistor using an oxide semiconductor for a semiconductor film can be used as a switch or the like. In that case, a potential of a floating node can be held for a longer time than in a circuit in which a transistor using amorphous silicon is used as a switch.

A 25-nm-thick film containing indium, gallium, and zinc can be used as the semiconductor film 508, for example.

A conductive film in which a 10-nm-thick film containing tantalum and nitrogen and a 300-nm-thick film containing copper are stacked can be used as the conductive film 504, for example. Note that the film containing tantalum and nitrogen is interposed between a region of the film containing copper and the insulating film 506.

A stacked film in which a 400-nm-thick film containing silicon and nitrogen and a 200-nm-thick film containing silicon, oxygen, and nitrogen are stacked can be used as the insulating film 506, for example. Note that the film containing silicon, oxygen, and nitrogen is interposed between a region of the film containing silicon and nitrogen and the semiconductor film 508.

A conductive film in which a 50-nm-thick film containing tungsten, a 400-nm-thick film containing aluminum, and a 100-nm-thick film containing titanium are stacked in this order can be used as the conductive film 512A or the conductive film 512B, for example. Note that the film containing tungsten includes a region in contact with the semiconductor film 508.

A manufacturing line for a bottom-gate transistor using amorphous silicon for a semiconductor can be easily remodeled into a manufacturing line for a bottom-gate transistor using an oxide semiconductor for a semiconductor, for example. Furthermore, a manufacturing line for a top-gate transistor using polysilicon for a semiconductor can be easily remodeled into a manufacturing line for a top-gate transistor using an oxide semiconductor for a semiconductor, for example. In either remodeling, an existing manufacturing line can be effectively utilized.

Accordingly, flickering of a display can be inhibited. Power consumption can be reduced. A moving image with quick movements can be smoothly displayed. A photograph and the like can be displayed with a wide range of grayscale. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example 3 of Semiconductor Film 508>>

For example, a compound semiconductor can be used for the semiconductor of the transistor. Specifically, a semiconductor containing gallium arsenide can be used.

For example, an organic semiconductor can be used for the semiconductor of the transistor. Specifically, an organic semiconductor containing any of polyacenes or graphene can be used for the semiconductor film.

<<Structure Example of Capacitor>>

A capacitor includes one conductive film, a different conductive film, and an insulating film. The insulating film includes a region interposed between the one conductive film and the different conductive film.

For example, a conductive film used as the source electrode or the drain electrode of the transistor, a conductive film used as the gate electrode, and an insulating film used as the gate insulating film can be used for the capacitor.

<<Structure Example 3 of Functional Layer 520>>

The functional layer 520 includes an insulating film 521, an insulating film 518, an insulating film 516, the insulating film 506, an insulating film 501C, and the like (see FIG. 9A and FIG. 9B). The insulating film 521 includes an insulating film 521A and an insulating film 521B. The insulating film 516 includes an insulating film 516A and an insulating film 516B.

The insulating film 521 includes a region interposed between the pixel circuit 530G(i, j) and the light-emitting device 550G(i, j).

21

The insulating film **518** includes a region interposed between the insulating film **521** and the insulating film **501C**.

The insulating film **516** includes a region interposed between the insulating film **518** and the insulating film **501C**.

The insulating film **506** includes a region interposed between the insulating film **516** and the insulating film **501C**.

[Insulating Film **521**]

An insulating inorganic material, an insulating organic material, or an insulating composite material containing an inorganic material and an organic material, for example, can be used for the insulating film **521**.

Specifically, an inorganic oxide film, an inorganic nitride film, an inorganic oxynitride film, or the like, or a stacked-layer material in which a plurality of films selected from these films are stacked can be used as the insulating film **521**.

For example, a film including a silicon oxide film, a silicon nitride film, a silicon oxynitride film, an aluminum oxide film, or the like, or a film including a stacked-layer material in which a plurality of films selected from these films are stacked can be used as the insulating film **521**. Note that the silicon nitride film is a dense film and has an excellent function of inhibiting diffusion of impurities.

For example, for the insulating film **521**, polyester, polyolefin, polyamide, polyimide, polycarbonate, polysiloxane, an acrylic resin, or the like, or a stacked-layer material, a composite material, or the like of a plurality of resins selected from these resins can be used. Note that polyimide is excellent in thermal stability, insulating property, toughness, low dielectric constant, low coefficient of thermal expansion, chemical resistance, and other properties compared with other organic materials. Accordingly, in particular, polyimide can be suitably used for the insulating film **521** or the like.

The insulating film **521** may be formed using a photo-sensitive material. Specifically, a film formed using photo-sensitive polyimide, a photosensitive acrylic resin, or the like can be used as the insulating film **521**.

Thus, the insulating film **521** can eliminate a level difference due to various components overlapping with the insulating film **521**, for example.

[Insulating Film **518**]

The material that can be used for the insulating film **521**, for example, can be used for the insulating film **518**.

For example, a material having a function of inhibiting diffusion of oxygen, hydrogen, water, an alkali metal, an alkaline earth metal, and the like can be used for the insulating film **518**. Specifically, a nitride insulating film can be used as the insulating film **518**. For example, silicon nitride, silicon nitride oxide, aluminum nitride, aluminum nitride oxide, or the like can be used for the insulating film **518**. Thus, diffusion of impurities into the semiconductor film of the transistor can be inhibited.

[Insulating Film **516**]

The material that can be used for the insulating film **521**, for example, can be used for the insulating film **516**.

Specifically, a film formed by a fabrication method different from that of the insulating film **518** can be used as the insulating film **516**.

[Insulating Film **506**]

The material that can be used for the insulating film **521**, for example, can be used for the insulating film **506**.

Specifically, a film including a silicon oxide film, a silicon oxynitride film, a silicon nitride oxide film, a silicon nitride film, an aluminum oxide film, a hafnium oxide film, an

22

yttrium oxide film, a zirconium oxide film, a gallium oxide film, a tantalum oxide film, a magnesium oxide film, a lanthanum oxide film, a cerium oxide film, or a neodymium oxide film can be used as the insulating film **506**.

[Insulating Film **501D**]

An insulating film **501D** includes a region interposed between the insulating film **501C** and the insulating film **516**.

The material that can be used for the insulating film **506**, for example, can be used for the insulating film **501D**.

[Insulating Film **501C**]

The material that can be used for the insulating film **521**, for example, can be used for the insulating film **501C**. Specifically, a material containing silicon and oxygen can be used for the insulating film **501C**. Thus, diffusion of impurities into the pixel circuit, the light-emitting device **550G**(i, j), the optical functional device **550RS**(i, j), or the like can be inhibited.

<<Structure Example 4 of Functional Layer **520**>>

The functional layer **520** includes a conductive film, a wiring, and a terminal. A material having conductivity can be used for the wiring, an electrode, the terminal, the conductive film, and the like.

[Wiring and the Like]

For example, an inorganic conductive material, an organic conductive material, a metal, a conductive ceramic, or the like can be used for the wiring and the like.

Specifically, a metal element selected from aluminum, gold, platinum, silver, copper, chromium, tantalum, titanium, molybdenum, tungsten, nickel, iron, cobalt, palladium, and manganese, or the like can be used for the wiring and the like. Alternatively, an alloy containing the above-described metal element, or the like can be used for the wiring and the like. In particular, an alloy of copper and manganese is suitable for microfabrication using a wet etching method.

Specifically, a two-layer structure in which a titanium film is stacked over an aluminum film, a two-layer structure in which a titanium film is stacked over a titanium nitride film, a two-layer structure in which a tungsten film is stacked over a titanium nitride film, a two-layer structure in which a tungsten film is stacked over a tantalum nitride film or a tungsten nitride film, a three-layer structure of a titanium film, an aluminum film stacked over the titanium film, and a titanium film further formed thereover, or the like can be used for the wiring and the like.

Specifically, a conductive oxide such as indium oxide, indium tin oxide, indium zinc oxide, zinc oxide, or zinc oxide to which gallium is added can be used for the wiring and the like.

Specifically, a film containing graphene or graphite can be used for the wiring and the like.

For example, a film containing graphene oxide is formed and the film containing graphene oxide is reduced, so that a film containing graphene can be formed. As a reducing method, a method with application of heat, a method using a reducing agent, or the like can be given.

For example, a film including a metal nanowire can be used for the wiring and the like. Specifically, a nanowire containing silver can be used.

Specifically, a conductive polymer can be used for the wiring and the like.

Note that a terminal **519B** can be electrically connected to a flexible printed circuit **FPC1** using a conductive material, for example (see FIG. 8). Specifically, the terminal **519B** can be electrically connected to the flexible printed circuit **FPC1** using a conductive material **CP**.

<Structure Example 2 of Functional Panel 700>

The functional panel 700 includes a base material 510, the base material 770, and a sealant 705 (see FIG. 9A). In addition, the functional panel 700 includes a structure body KB.

<<Base Material 510 and Base Material 770>>

A material having a light-transmitting property can be used for the base material 510 or the base material 770.

For example, a flexible material can be used for the base material 510 or the base material 770. Thus, a flexible functional panel can be provided.

For example, a material with a thickness less than or equal to 0.7 mm and greater than or equal to 0.1 mm can be used. Specifically, a material polished to a thickness of approximately 0.1 mm can be used. Thus, the weight can be reduced.

A glass substrate of the 6th generation (1500 mm×1850 mm), the 7th generation (1870 mm×2200 mm), the 8th generation (2200 mm×2400 mm), the 9th generation (2400 mm×2800 mm), the 10th generation (2950 mm×3400 mm), or the like can be used as the base material 510 or the base material 770. Thus, a large-sized display device can be fabricated.

For the base material 510 or the base material 770, an organic material, an inorganic material, a composite material of an organic material and an inorganic material, or the like can be used.

For example, an inorganic material such as glass, ceramic, or a metal can be used. Specifically, non-alkali glass, soda-lime glass, potash glass, crystal glass, aluminosilicate glass, tempered glass, chemically tempered glass, quartz, sapphire, or the like can be used for the base material 510 or the base material 770. Aluminosilicate glass, tempered glass, chemically tempered glass, sapphire, or the like can be suitably used for the base material 510 or the base material 770 that is provided on the side close to a user of the functional panel. Thus, the functional panel can be prevented from being broken or damaged by the use thereof.

Specifically, an inorganic oxide film, an inorganic nitride film, an inorganic oxynitride film, or the like can be used. For example, a silicon oxide film, a silicon nitride film, a silicon oxynitride film, an aluminum oxide film, or the like can be used. Stainless steel, aluminum, or the like can be used for the base material 510 or the base material 770.

For example, a single crystal semiconductor substrate or a polycrystalline semiconductor substrate of silicon or silicon carbide, a compound semiconductor substrate of silicon germanium or the like, an SOI substrate, or the like can be used as the base material 510 or the base material 770. Thus, a semiconductor element can be formed on the base material 510 or the base material 770.

For example, an organic material such as a resin, a resin film, or plastic can be used for the base material 510 or the base material 770. Specifically, a material containing polyester, polyolefin, polyamide (e.g., nylon or aramid), polyimide, polycarbonate, polyurethane, an acrylic resin, an epoxy resin, or a resin having a siloxane bond, such as silicone, can be used for the base material 510 or the base material 770. For example, a resin film, a resin plate, a stacked-layer material, or the like containing any of these materials can be used. Thus, the weight can be reduced. The frequency of occurrence of breakage or the like due to dropping can be reduced, for example.

Specifically, polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polyethersulfone (PES), a

cycloolefin polymer (COP), a cycloolefin copolymer (COC), or the like can be used for the base material 510 or the base material 770.

For example, a composite material formed by attaching a metal plate, a thin glass plate, or a film of an inorganic material or the like and a resin film or the like to each other can be used for the base material 510 or the base material 770. For example, a composite material formed by dispersing a fibrous or particulate metal, glass, an inorganic material, or the like into a resin can be used for the base material 510 or the base material 770. For example, a composite material formed by dispersing a fibrous or particulate resin, an organic material, or the like into an inorganic material can be used for the base material 510 or the base material 770.

Furthermore, a single-layer material or a material in which a plurality of layers are stacked can be used for the base material 510 or the base material 770. For example, a material in which insulating films and the like are stacked can be used. Specifically, a material in which one or a plurality of films selected from a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, and the like are stacked can be used. Thus, diffusion of impurities contained in the base materials can be prevented, for example. Diffusion of impurities contained in glass or a resin can be prevented. Diffusion of impurities that pass through a resin can be prevented.

Furthermore, paper, wood, or the like can be used for the base material 510 or the base material 770.

For example, a material having heat resistance high enough to withstand heat treatment in the fabricating process can be used for the base material 510 or the base material 770. Specifically, a material having heat resistance to heat applied in the formation process of directly forming the transistor, the capacitor, or the like can be used for the base material 510 or the base material 770.

For example, it is possible to employ a method in which an insulating film, a transistor, a capacitor, or the like is formed on a process substrate having heat resistance to heat applied in the fabricating process, and the formed insulating film, transistor, capacitor, or the like is transferred to the base material 510 or the base material 770. Accordingly, an insulating film, a transistor, a capacitor, or the like can be formed on a flexible substrate, for example.

<<Sealant 705>>

The sealant 705 includes a region interposed between the functional layer 520 and the base material 770 and has a function of bonding the functional layer 520 and the base material 770 together (see FIG. 9A).

An inorganic material, an organic material, a composite material of an inorganic material and an organic material, or the like can be used for the sealant 705.

For example, an organic material such as a thermally fusible resin or a curable resin can be used for the sealant 705.

For example, an organic material such as a reactive curable adhesive, a photocurable adhesive, a thermosetting adhesive, and/or an anaerobic adhesive can be used for the sealant 705.

Specifically, an adhesive containing an epoxy resin, an acrylic resin, a silicone resin, a phenol resin, a polyimide resin, an imide resin, a PVC (polyvinyl chloride) resin, a PVB (polyvinyl butyral) resin, an EVA (ethylene vinyl acetate) resin, or the like can be used for the sealant 705.

<<Structure Body KB>>

The structure body KB includes a region interposed between the functional layer 520 and the base material 770. The structure body KB has a function of providing a

25

predetermined space between the functional layer **520** and the base material **770**. Note that a colored material can be used for the structure body KB. Thus, stray light can be absorbed.

<Structure Example 3 of Functional Panel **700**>

The functional panel **700** includes the insulating film **528** and an insulating film **573** (see FIG. 9A).

<<Insulating Film **528**>>

The insulating film **528** includes a region interposed between the functional layer **520** and the base material **770**, and the insulating film **528** has an opening portion in a region overlapping with the light-emitting device **550G(i, j)** and the optical functional device **550RS(i, j)** (see FIG. 9A).

The material that can be used for the insulating film **521**, for example, can be used for the insulating film **528**. Specifically, a silicon oxide film, a film containing an acrylic resin, a film containing polyimide, or the like can be used as the insulating film **528**.

<<Insulating Film **573**>>

The light-emitting device **550G(i, j)** and the optical functional device **550RS(i, j)** are interposed between a region of the insulating film **573** and the functional layer **520** (see FIG. 9A).

For example, a single film or a stacked film in which a plurality of films are stacked can be used as the insulating film **573**. Specifically, a stacked film, in which an insulating film **573A** which can be formed by a method that hardly damages the light-emitting device **550G(i, j)** and the optical functional device **550RS(i, j)** and a dense insulating film **573B** with a few defects are stacked, can be used as the insulating film **573**.

Thus, diffusion of impurities into the light-emitting device **550G(i, j)** and the optical functional device **550RS(i, j)** can be inhibited. The reliability of the light-emitting device **550G(i, j)** and the optical functional device **550RS(i, j)** can be increased.

<Structure Example 4 of Functional Panel **700**>

The functional panel **700** includes a functional layer **720** (see FIG. 9A).

<<Functional Layer **720**>>

The functional layer **720** includes a light-blocking film BM, a coloring film CF(G), and an insulating film **771**. A color conversion layer can also be used.

<<Light-Blocking Film BM>>

The light-blocking film BM has an opening portion in a region overlapping with the pixel **702G(i, j)**. The light-blocking film BM has an opening portion in a region overlapping with the pixel **702RS(i, j)** (see FIG. 9A).

A material of a dark color can be used for the light-blocking film BM, for example. Thus, the display contrast can be increased.

<<Coloring Film CF(G)>>

The coloring film CF(G) includes a region interposed between the base material **770** and the light-emitting device **550G(i, j)**. A material that selectively transmits light of a predetermined color, for example, can be used for the coloring film CF(G). Specifically, a material that transmits red light, green light, or blue light can be used for the coloring film CF(G).

<<Structure Example of Insulating Film **771**>>

The insulating film **771** includes a region interposed between the base material **770** and the light-emitting device **550G(i, j)**.

The insulating film **771** includes a region where the light-blocking film BM, the coloring film CF(G), or the color conversion layer is interposed between the insulating film **771** and the base material **770**. Thus, unevenness due to

26

the thickness of the light-blocking film BM, the coloring film CF(G), or the color conversion layer can be reduced.

<Structure Example 5 of Functional Panel **700**>

The functional panel **700** includes a functional film **770P** or the like (see FIG. 9A).

<<Functional Film **770P** and the Like>>

The functional film **770P** includes a region overlapping with the light-emitting device **550G(i, j)**.

An anti-reflection film, a polarizing film, a retardation film, a light diffusion film, a condensing film, or the like can be used as the functional film **770P**, for example.

For example, an anti-reflection film with a thickness less than or equal to 1 μm can be used as the functional film **770P**. Specifically, a stacked film in which three or more layers, preferably five or more layers, and further preferably 15 or more layers of dielectrics are stacked can be used as the functional film **770P**. This allows the reflectance to be as low as 0.5% or less, preferably 0.08% or less.

For example, a circularly polarizing film can be used as the functional film **770P**.

Furthermore, an antistatic film inhibiting the attachment of a dust, a water repellent film inhibiting the attachment of a stain, an oil repellent film inhibiting the attachment of a stain, a non-glare film (anti-glare film), a hard coat film inhibiting generation of a scratch in use, a self-healing film that self-heals from generated scratches, or the like can be used as the functional film **770P**.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 4

In this embodiment, structures of the functional panel of one embodiment of the present invention will be described with reference to FIG. 12 to FIG. 13.

FIG. 12A is a block diagram illustrating a structure of the functional panel of one embodiment of the present invention, and FIG. 12B is a block diagram illustrating part of FIG. 12A.

FIG. 13 is a diagram illustrating an operation of the functional panel of one embodiment of the present invention.

<Structure Example 1 of Functional Panel **700**>

The functional panel **700** described in this embodiment includes a region **231** (see FIG. 12).

<<Structure Example 1 of Region **231**>>

The region **231** includes a group of pixel sets **703(i, 1)** to **703(i, n)** and a different group of pixel sets **703(1, j)** to **703(m, j)**. The region **231** also includes the conductive film **G1(i)**, the conductive film **TX(i)**, the conductive film **S1g(j)**, and the conductive film **WX(j)**.

The group of pixel sets **703(i, 1)** to **703(1, n)** is arranged in the row direction (the direction indicated by an arrow **R1** in the drawing), and the group of pixel sets **703(i, 1)** to **703(1, n)** includes the pixel **703(i, j)**.

The group of pixel sets **703(i, 1)** to **703(i, n)** is electrically connected to the conductive film **G1(i)**, and the group of pixel sets **703(i, 1)** to **703(i, n)** is electrically connected to the conductive film **TX(i)**.

The different group of pixel sets **703(1, j)** to **703(m, j)** is arranged in the column direction intersecting the row direction (the direction indicated by an arrow **C1** in the drawing), and the different group of pixel sets **703(1, j)** to **703(m, j)** includes the pixel **703(i, j)**.

The different group of pixel sets **703(1, j)** to **703(m, j)** is electrically connected to the conductive film **S1g(j)**, and the

different group of pixel sets **703**(1, *j*) to **703**(*m*, *j*) is electrically connected to the conductive film **WX**(*j*).

Thus, image data can be supplied to a plurality of pixels. Imaging data can be obtained from a plurality of pixels. As a result, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example 2 of Region **231**>>

The region **231** includes 500 or more groups of pixel sets per inch. Furthermore, the region **231** includes 1000 or more groups of pixel sets per inch, preferably 5000 or more groups of pixel sets per inch, further preferably 10000 or more groups of pixel sets per inch. This can reduce a screen-door effect, for example. Note that the group of pixel sets include the pixel set **703**(*i*, *j*).

<<Structure Example 3 of Region **231**>>

The region **231** includes a plurality of pixels in a matrix. For example, the region **231** includes 7600 or more pixels in the row direction and the region **231** includes 4300 or more pixels in the column direction. Specifically, 7680 pixels are provided in the row direction and 4320 pixels are provided in the column direction.

Thus, a high-resolution image can be displayed. Thus, a novel functional panel that is highly convenient or reliable can be provided.

<<Structure Example 4 of Region **231**>>

The diagonal size of the region **231** is greater than or equal to 40 inches, preferably greater than or equal to 60 inches, further preferably greater than or equal to 80 inches. The diagonal size of the region **231** is preferably, for example, less than or equal to 150 inches, in which case the weight can be light.

This enables display of a realistic image. As a result, a novel functional panel that is highly convenient or reliable can be provided.

<Structure Example 2 of Functional Panel **700**>

The functional panel **700** of one embodiment of the present invention includes a group of sampling circuits **SC**, a multiplexer **MUX**, an amplifier circuit **AMP**, and an analog-digital converter circuit **ADC** (see FIG. **12A**). Note that the group of sampling circuits **SC** includes the sampling circuit **SC**(*j*).

Accordingly, the sampling circuit **SC**(*j*) can be provided for each conductive film **WX**(*j*). A differential signal of the pixel circuit **530RS**(*i*, *j*) can be obtained by the corresponding conductive film **WX**(*j*). The operating frequency of the sampling circuit **SC**(*j*) can be low. Noise can be reduced. As a result, a novel functional panel that is highly convenient or reliable can be provided.

<<Structure Example of Multiplexer **MUX**>>

The multiplexer **MUX** has a function of selecting one sampling circuit from a group of sampling circuits to obtain an imaging signal. For example, the multiplexer **MUX** selects the sampling circuit **SC**(*j*) to obtain the imaging signal.

Specifically, the multiplexer **MUX** is electrically connected to a sampling circuit **SC**(1) to a sampling circuit **SC**(9), and selects one sampling circuit **SC** to obtain an imaging signal (see FIG. **12B**). For example, the multiplexer **MUX** is electrically connected to a third terminal **OUT**(9) of the sampling circuit **SC**(9).

The multiplexer **MUX** is electrically connected to the amplifier circuit **AMP** and has a function of supplying the obtained imaging signal.

Thus, a predetermined pixel can be selected from a plurality of pixels arranged in the row direction. Alternatively, imaging data can be obtained from a predetermined pixel. Alternatively, the number of imaging signals obtained

concurrently can be reduced with use of a plurality of multiplexers. Alternatively, an analog-digital converter circuit **ADC** in which the number of input channels is smaller than the number of pixels arranged in the row direction can be used. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example of Amplifier Circuit **AMP**>>

The amplifier circuit **AMP** can amplify the imaging signal and supply the amplified signal to the analog-digital converter circuit **ADC**.

Note that the functional layer **520** includes the multiplexer **MUX** and the amplifier circuit **AMP**.

Accordingly, for example, in the step of forming the semiconductor film used in the pixel circuit **530G**(*i*, *j*), semiconductor films used in the multiplexer **MUX** and the amplifier circuit **AMP** can be formed. The manufacturing process of the functional panel can be simplified. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example of Analog-Digital Converter Circuit **ADC**>>

The analog-digital converter circuit **ADC** has a function of converting an analog imaging signal to a digital signal. This can suppress deterioration of an imaging signal due to transmission.

<Structure Example 3 of Functional Panel **700**>

The functional panel **700** of one embodiment of the present invention includes the driver circuit **GD**, the driver circuit **RD**, and one pixel set **703**(0). The driver circuit **GD** has a function of supplying the first selection signal, and the driver circuit **RD** has a function of supplying the third selection signal and the fourth selection signal.

<<Structure Example 1 of Pixel **703**(*i*, *j*)>>

The pixel set **703**(*i*, *j*) is supplied with the third selection signal and the fourth selection signal in a period during which the first selection signal is not supplied (see FIG. **13**). In addition, the pixel circuit **530RS**(*i*, *j*) obtains an imaging signal on the basis of the third selection signal, and supplies the imaging signal on the basis of the fourth selection signal.

Note that for example, the first selection signal can be supplied with use of the conductive film **G1**(*i*), the third selection signal can be supplied with use of the conductive film **TX**(*i*), and the fourth selection signal can be supplied with use of the conductive film **SE**(*i*) (see FIG. **5**).

The operation of supplying the third selection signal and making the pixel circuit **530RS**(*i*, *j*) obtain an imaging signal can be referred to as "imaging" (see FIG. **13**). The operation of reading an imaging signal from the pixel circuit **530RS**(*i*, *j*) can be referred to as "reading". The operation of supplying a predetermined voltage to the optical functional device **550RS**(*i*, *j*) can be referred to as "initialization"; the operation of exposing the optical functional device **550RS**(*i*, *j*) to light in a predetermined period after the initialization as "light exposure"; and the operation of reflecting a voltage that has been changed along with the light exposure on the pixel circuit **530RS**(*i*, *j*) as "transfer". Moreover, in the figure, **SRS** corresponds to the operation of supplying a reference signal used in a correlated double sampling method, and "output" corresponds to the operation of supplying an imaging signal.

For example, image data of one frame can be written in 16.7 msec. Specifically, the operation can be performed at a frame rate of 60 Hz. Note that an image signal can be written to the pixel circuit **530G**(*i*, *j*) in 15.2 μsec.

For example, image data of one frame can be held in a period corresponding to 16 frames. Imaging data of one frame can be imaged and read in a period corresponding to 16 frames.

Specifically, it is possible to perform the initialization in 15 μ sec, the light exposure in a period from 1 msec to 5 msec, and the transfer in 150 μ sec. Moreover, the reading can be performed in 250 msec.

Accordingly, imaging can be performed in a period during which the first selection signal is not supplied. Noise in imaging can be inhibited. An imaging signal can be read in the period during which the first selection signal is not supplied. Noise in reading can be inhibited. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

<<Structure Example 2 of Pixel 703(i, j)>>

The pixel 703(i, j) is supplied with the third selection signal in a period during which the pixel 703(i, j) holds one image signal. For example, in a period during which the pixel circuit 530G(i, j) holds one image signal, the pixel 703(i, j) can emit light using the light-emitting device 550G(i, j) on the basis of the image signal (see FIG. 13). The pixel circuit 530RS(i, j) is supplied with the third selection signal after the pixel circuit 530G(i, j) obtains one image signal on the basis of the first selection signal by the time when the pixel circuit 530G(i, j) is supplied with the first selection signal again.

Accordingly, the intensity of light emitted from the light-emitting device 550G(i, j) can be controlled using the image signal. Light having a controlled intensity can be emitted to an object. The object can be imaged using the optical functional device 550RS(i, j). The object can be imaged using the optical functional device 550RS(i, j) while the intensity of emitted light is controlled. The influence of a change from one image signal to another image signal held in the pixel circuit 530G(i, j) on an imaging signal can be eliminated. Thus, a novel functional panel that is highly convenient, useful, or reliable can be provided.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 5

In this embodiment, structures of display devices of one embodiment of the present invention are described with reference to drawings.

FIG. 14A is a block diagram of the display device of one embodiment of the present invention, and FIG. 14B to FIG. 14D are perspective views each illustrating the appearance of the display device of one embodiment of the present invention.

<<Structure Example of Display Device>>

The display device described in this embodiment includes a control portion 238 and the functional panel 700 (see FIG. 14A).

<<Structure Example 1 of Control Portion 238>>

The control portion 238 is supplied with image data VI and control data CI. For example, a clock signal, a timing signal, or the like can be used as the control data CI.

The control portion 238 generates data on the basis of the image data VI and the control portion 238 generates a control signal on the basis of the control data CI. Furthermore, the control portion 238 supplies the data and the control signal.

The data includes a grayscale of 8 bits or more, preferably 12 bits or more, for example. In addition, a clock signal, a

start pulse, or the like of a shift register used for a driver circuit can be used as the control signal, for example.

<<Structure Example 2 of Control Portion 238>>

For example, a decompression circuit 234 and an image processing circuit 235 can be used in the control portion 238. <<Decompression Circuit 234>>

The decompression circuit 234 has a function of decompressing the image data VI supplied in a compressed state. The decompression circuit 234 includes a memory portion. The memory portion has a function of storing decompressed image data, for example.

<<Image Processing Circuit 235>>

The image processing circuit 235 includes a memory region, for example. The memory region has a function of storing data included in the image data VI, for example.

The image processing circuit 235 has a function of generating the data by correcting the image data VI on the basis of a predetermined characteristic curve and a function of supplying the data, for example.

<<Structure Example 1 of Functional Panel 700>>

The functional panel 700 is supplied with the data and the control signal. For example, the functional panel 700 described in any one of Embodiment 2 to Embodiment 4 can be used.

<<Structure Example of Pixel 703(i, j)>>

The pixel 703(i, j) performs display on the basis of the data.

Thus, the image data VI can be displayed using the optical functional device 550RS(i, j). As a result, a novel display device that is highly convenient, useful, or reliable can be provided. For example, an information terminal (see FIG. 14B), a video display system (see FIG. 14C), a computer (see FIG. 14D), or the like can be provided.

<<Structure Example 2 of Functional Panel 700>>

The functional panel 700 includes a driver circuit and a control circuit, for example.

<<Driver Circuit>>

The driver circuit operates on the basis of the control signal. Using the control signal enables a synchronized operation of a plurality of driver circuits (see FIG. 14A).

For example, the driver circuit GD can be used in the functional panel 700. The driver circuit GD is supplied with the control signal and has a function of supplying the first selection signal.

For example, a driver circuit SD can be used in the functional panel 700. The driver circuit SD is supplied with the control signal and the data and can supply an image signal.

For example, the driver circuit RD can be used in the functional panel 700. The driver circuit RD is supplied with the control signal and can supply the second selection signal to the fourth selection signal.

For example, the reading circuit RC can be used in the functional panel 700. The reading circuit RC is supplied with the control signal, and can read an imaging signal by a correlated double sampling method, for example.

<<Control Circuit>>

The control circuit has a function of generating and supplying the control signal. For example, a clock signal or a timing signal can be used as the control signal.

Specifically, the control circuit formed over a rigid substrate can be used in the functional panel. The control circuit formed over the rigid substrate and the control portion 238 can be electrically connected to each other using a flexible printed circuit.

31

A timing controller **233** can be used as a control circuit, for example. With the use of a control circuit **243**, operation of the driver circuit RD can be synchronized with operation of the reading circuit RC.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 6

In this embodiment, a structure of an input/output device of one embodiment of the present invention will be described with reference to a drawing.

FIG. **15** is a block diagram illustrating the structure of the input/output device of one embodiment of the present invention.

<<Structure Example 1 of Input/Output Device>

The input/output device described in this embodiment includes an input portion **240** and a display portion **230** (see FIG. **15**).

<<Structure Example 1 of Display Portion **230**>>

The display portion **230** includes the functional panel **700**. For example, the functional panel **700** described in any one of Embodiment 2 to Embodiment 4 can be used for the display portion **230**. Note that a structure including the input portion **240** and the display portion **230** can be referred to as a functional panel **700TP**.

<<Structure Example 1 of Input Portion **240**>>

The input portion **240** includes a sensing region **241**. The input portion **240** senses an object approaching the sensing region **241**.

The sensing region **241** includes a region overlapping with the pixel **703(i,j)**.

Thus, the object approaching the region overlapping with a display portion **230** can be sensed while image data is displayed on the display portion **230**. A finger or the like approaching the display portion **230** can be used as a pointer to input position data. The position data can be associated with the image data displayed on the display portion **230**. Thus, a novel input/output device that is highly convenient, useful, or reliable can be provided.

<<Structure Example 1 of Sensing Region **241**>>

The sensing region **241** includes one or a plurality of sensors, for example.

The sensing region **241** includes a group of sensors **802(g, 1)** to **802(g, q)** and a different group of sensors **802(1, h)** to **802(p, h)**. Note that g is an integer greater than or equal to 1 and less than or equal to p, h is an integer greater than or equal to 1 and less than or equal to q, and p and q are each an integer greater than or equal to 1.

The group of sensors **802(g, 1)** to **802(g, q)** includes a sensor **802(g, h)**, is provided in a row direction (the direction indicated by an arrow R2 in the drawing), and is electrically connected to a conductive film CL(g). Note that the direction indicated by the arrow R2 may be the same as or different from the direction indicated by the arrow R1.

The different group of sensors **802(1, h)** to **802(p, h)** includes the sensor **802(g, h)**, is provided in a column direction (the direction indicated by an arrow C2 in the drawing) intersecting the row direction, and is electrically connected to a wiring ML(h).

<<Sensor>>

The sensor has a function of sensing an approaching pointer. For example, a finger or a stylus pen can be used as the pointer. For example, a piece of metal or a coil can be used for the stylus pen.

32

Specifically, a capacitive proximity sensor, an electromagnetic inductive proximity sensor, an optical proximity sensor, a resistive proximity sensor, or the like can be used as the sensor.

A plurality of types of sensors can be used in combination. For example, a sensor that senses a finger and a sensor that senses a stylus pen can be used in combination.

This allows determination of the kind of a pointer. Different instructions can be associated with sensing data depending on the kind of a pointer that has been determined. Specifically, in the case where it is determined that a finger is used as a pointer, sensing data can be associated with a gesture. In the case where it is determined that a stylus pen is used as a pointer, sensing data can be associated with drawing processing.

Specifically, a finger can be sensed using a capacitive, pressure-sensitive, or optical proximity sensor. A stylus pen can be sensed using an electromagnetic inductive or optical proximity sensor.

<<Structure Example 2 of Input Portion **240**>>

The input portion **240** includes an oscillation circuit OSC and a sensing circuit DC (see FIG. **15**).

The oscillation circuit OSC supplies a search signal to the sensor **802(g, h)**. For example, a rectangular wave, a sawtooth wave, a triangular wave, or a sine wave can be used as the search signal.

The sensor **802(g, h)** generates and supplies a sensing signal that changes in accordance with the search signal and the distance to a pointer approaching the sensor **802(g, h)**.

The sensing circuit DC supplies input data in response to the sensing signal.

Accordingly, the distance from an approaching pointer to the sensing region **241** can be sensed. The position in the sensing region **241** where the pointer comes the closest can be sensed.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 7

In this embodiment, structures of a data processing device of one embodiment of the present invention will be described with reference to drawings.

FIG. **16A** is a block diagram illustrating the structure of the data processing device of one embodiment of the present invention. FIG. **16B** and FIG. **16C** are projection views each illustrating examples of the appearance of the data processing device.

FIG. **17A** is a flow chart showing main processing of the program of one embodiment of the present invention, and FIG. **17B** is a flow chart showing interrupt processing.

FIG. **18A** is a flow chart showing interrupt processing of the program of one embodiment of the present invention. FIG. **18B** is a schematic view illustrating handling of the data processing device of one embodiment of the present invention, and FIG. **18C** is a timing chart showing the operation of the data processing device of one embodiment of the present invention.

FIG. **19A** is a flow chart showing interrupt processing different from the interrupt processing shown in FIG. **17B**. FIG. **19B** is a schematic view illustrating operation of the program shown in FIG. **19A**, and FIG. **19C** is a schematic view of a taken image of a fingerprint.

FIG. **20A** is a flow chart showing interrupt processing different from the interrupt processing shown in FIG. **17B**. FIG. **20B** to FIG. **20D** are schematic views illustrating operation of the program shown in FIG. **20A**.

<<Structure Example 1 of Data Processing Device>

The data processing device described in this embodiment includes the arithmetic device **210** and an input/output device **220** (see FIG. 16A). Note that the input/output device **220** is electrically connected to the arithmetic device **210**. A data processing device **200** can also include a housing (see FIG. 16B and FIG. 16C).

<<Structure Example 1 of Arithmetic Device 210>>

The arithmetic device **210** is supplied with input data II or sensing data DS. The arithmetic device **210** generates the control data CI and the image data VI on the basis of the input data II or the sensing data DS, and supplies the control data CI and the image data VI.

The arithmetic device **210** includes an arithmetic portion **211** and a memory portion **212**. The arithmetic device **210** also includes a transmission path **214** and an input/output interface **215**.

The transmission path **214** is electrically connected to the arithmetic portion **211**, the memory portion **212**, and the input/output interface **215**.

<<Arithmetic Portion 211>>

The arithmetic portion **211** has a function of executing a program, for example.

<<Memory Portion 212>>

The memory portion **212** has a function of storing, for example, the program executed by the arithmetic portion **211**, initial data, setting data, an image, or the like.

Specifically, a hard disk, a flash memory, a memory using a transistor including an oxide semiconductor, or the like can be used.

<<Input/Output Interface 215 and Transmission Path 214>>

The input/output interface **215** includes a terminal or a wiring and has a function of supplying data and being supplied with data. The input/output interface **215** can be electrically connected to the transmission path **214**, for example. The input/output interface **215** can also be electrically connected to the input/output device **220**.

The transmission path **214** includes a wiring and has a function of supplying data and being supplied with data. The transmission path **214** can be electrically connected to the input/output interface **215**, for example. The transmission path **214** can also be electrically connected to the arithmetic portion **211**, the memory portion **212**, or the input/output interface **215**.

<<Structure Example of Input/Output Device 220>>

The input/output device **220** supplies the input data II and the sensing data DS. The input/output device **220** is supplied with the control data CI and the image data VI (see FIG. 16A).

As the input data II, for example, a scan code of a keyboard, position data, data on button handling, sound data, or image data can be used. As the sensing data DS, for example, illuminance data, attitude data, acceleration data, direction data, pressure data, temperature data, humidity data, or the like of the environment where the data processing device **200** is used, or the like can be used.

As the control data CI, for example, a signal controlling the luminance of display of the image data VI, a signal controlling the color saturation, or a signal controlling the hue can be used. A signal that changes display of part of the image data VI can be used as the control data CI.

The input/output device **220** includes the display portion **230**, the input portion **240**, and a sensing portion **250**. For example, the input/output device described in Embodiment 4 can be used as the input/output device **220**. The input/output device **220** can include a communication portion **290**.

<<Structure Example of Display Portion 230>>

The display portion **230** displays the image data VI on the basis of the control data CI. For example, the display device described in Embodiment 3 can be used for the display portion **230**.

<<Structure Example of Input Portion 240>>

The input portion **240** generates the input data II. For example, the input portion **240** has a function of supplying position data P1.

For example, a human interface or the like can be used as the input portion **240** (see FIG. 16A). Specifically, a keyboard, a mouse, a touch sensor, a microphone, a camera, or the like can be used as the input portion **240**.

A touch sensor including a region overlapping with the display portion **230** can be used. Note that an input/output device including the display portion **230** and a touch sensor including a region overlapping with the display portion **230** can be referred to as a touch panel or a touch screen.

A user can make various gestures (tap, drag, swipe, pinch in, and the like) using his/her finger touching the touch panel as a pointer, for example.

The arithmetic device **210**, for example, analyzes data on the position, path, or the like of a finger in contact with the touch panel and can determine that a predetermined gesture is supplied when the analysis results meet predetermined conditions. Thus, the user can supply a predetermined operation instruction associated with a predetermined gesture in advance by using the gesture.

For instance, the user can supply a "scroll instruction" for changing the display position of image data by using a gesture of moving a finger in contact with the touch panel along the touch panel.

The user can supply a "dragging instruction" for pulling out and displaying a navigation panel NP at an edge portion of the region **231** by using a gesture of moving a finger in contact with the edge portion of the region **231** (see FIG. 16C). Moreover, the user can supply a "leafing through instruction" for displaying index images IND, some parts of other pages, or thumbnail images TN of other pages in a predetermined order on the navigation panel NP so that the user can flip through these images, by using a gesture of moving the position where a finger presses hard. The instruction can be supplied by using the finger press pressure. Consequently, the user can turn the pages of an e-book reader like flipping through the pages of a paper book. The user can search a certain page with the aid of the thumbnail images TN or the index images IND.

<<Structure Example of Sensing Portion 250>>

The sensing portion **250** generates the sensing data DS. The sensing portion **250** has a function of sensing the illuminance of the environment where the data processing device **200** is used and a function of supplying illuminance data, for example.

The sensing portion **250** has a function of sensing the ambient conditions and supplying the sensing data. Specifically, illuminance data, attitude data, acceleration data, direction data, pressure data, temperature data, humidity data, or the like can be supplied.

For example, a photosensor, an attitude sensor, an acceleration sensor, a direction sensor, a GPS (Global positioning system) signal receiving circuit, a pressure-sensitive switch, a pressure sensor, a temperature sensor, a humidity sensor, a camera, or the like can be used as the sensing portion **250**.

<<Communication Portion 290>>

The communication portion **290** has a function of supplying data to a network and obtaining data from the network.

<<Housing>>

Note that the housing has a function of storing the input/output device **220** or the arithmetic device **210**. The housing has a function of supporting the display portion **230** or the arithmetic device **210**.

Thus, the control data CI can be generated on the basis of the input data II or the sensing data DS. The image data VI can be displayed on the basis of the input data II or the sensing data DS. The data processing device is capable of operating with knowledge of the intensity of light that the housing of the data processing device receives in the environment where the data processing device is used. The user of the data processing device can select a display method. Thus, a novel data processing device that is highly convenient, useful, or reliable can be provided.

Note that in some cases, these components cannot be clearly distinguished from each other and one component may also serve as another component or may include part of another component. For example, a touch panel in which a touch sensor overlaps with a display panel is an input portion as well as a display portion.

<<Structure Example 2 of Arithmetic Device **210**>>

The arithmetic device **210** includes an artificial intelligence (AI) portion **213** (see FIG. **16A**).

The artificial intelligence portion **213** is supplied with the input data II or the sensing data DS, and the artificial intelligence portion **213** infers the control data CI on the basis of the input data II or the sensing data DS. Moreover, the artificial intelligence portion **213** supplies the control data CI.

In this manner, the control data CI for display that can be felt suitable can be generated. Display that can be felt suitable is possible. The control data CI for display that can be felt comfortable can be generated. Display that can be felt comfortable is possible. Thus, a novel data processing device that is highly convenient, useful, or reliable can be provided.

[Natural Language Processing on Input Data II]

Specifically, the artificial intelligence portion **213** can perform natural language processing on the input data II to extract one feature from the whole input data II. For example, the artificial intelligence portion **213** can infer emotion or the like put in the input data II, which can be a feature. The artificial intelligence portion **213** can infer the color, design, font, or the like empirically felt suitable for the feature. The artificial intelligence portion **213** can generate data specifying the color, design, or font of a letter or data specifying the color or design of the background, and the data can be used as the control data CI.

Specifically, the artificial intelligence portion **213** can perform natural language processing on the input data II to extract some words included in the input data II. For example, the artificial intelligence portion **213** can extract expressions including a grammatical error, a factual error, emotion, and the like. The artificial intelligence portion **213** can generate the control data CI for displaying extracted part in the color, design, font, or the like different from those of another part, and the data.

[Image Processing on Input Data II]

Specifically, the artificial intelligence portion **213** can perform image processing on the input data II to extract one feature from the input data II. For example, the artificial intelligence portion **213** can infer the age where an image of the input data II is taken, whether the image is taken indoors or outdoors, whether the image is taken in the daytime or at night, or the like, which can be a feature. The artificial intelligence portion **213** can infer the color tone empirically

felt suitable for the feature and generate the control data CI for use of the color tone for display. Specifically, data specifying color (e.g., full color, monochrome, or sepia) used for expression of a gradation can be used as the control data CI.

Specifically, the artificial intelligence portion **213** can perform image processing on the input data II to extract some images included in the input data II. For example, the artificial intelligence portion **213** can generate the control data CI for displaying a boundary between extracted part of the image and another part. Specifically, the artificial intelligence portion **213** can generate the control data CI for displaying a rectangle surrounding the extracted part of the image.

[Inference Using Sensing Data DS]

Specifically, the artificial intelligence portion **213** can make an inference using the sensing data DS. The artificial intelligence portion **213** can generate the control data CI on the basis of the inference RI so that the user of the data processing device **200** can feel comfortable.

Specifically, the artificial intelligence portion **213** can generate the control data CI for adjustment of display brightness on the basis of the ambient illuminance or the like so that the display brightness can be felt comfortable. The artificial intelligence portion **213** can generate the control data CI for adjustment of volume on the basis of the ambient noise or the like so that the volume can be felt comfortable.

As the control data CI, a clock signal, a timing signal, or the like that is supplied to the control portion **238** included in the display portion **230** can be used. A clock signal, a timing signal, or the like that is supplied to a control portion included in the input portion **240** can be used as the control data CI.

<Structure Example 2 of Data Processing Device>

Another structure of the data processing device of one embodiment of the present invention will be described with reference to FIG. **17A** and FIG. **17B**.

<<Program>>

A program of one embodiment of the present invention has the following steps (see FIG. **17A**).

[First Step]

In a first step, settings are initialized (see (S1) in FIG. **17A**).

For example, predetermined image data that is to be displayed on start-up, a predetermined mode displaying the image data, and data determining a predetermined display method for displaying the image data are obtained from the memory portion **212**. Specifically, one still image data or another moving image data can be used as the predetermined image data. Furthermore, a first mode or a second mode can be used as the predetermined mode.

[Second Step]

In a second step, interrupt processing is allowed (see (S2) in FIG. **17A**). Note that an arithmetic device allowed to execute the interrupt processing can perform the interrupt processing in parallel with the main processing. The arithmetic device that has returned from the interrupt processing to the main processing can reflect the results obtained through the interrupt processing in the main processing.

The arithmetic device may execute the interrupt processing when a counter has an initial value, and the counter may be set at a value other than the initial value when the arithmetic device returns from the interrupt processing. Thus, the interrupt processing can be executed any time after the program is started up.

[Third Step]

In a third step, image data is displayed by a predetermined mode or a predetermined display method selected in the first step or the interrupt processing (see (S3) in FIG. 17A). Note that the predetermined mode determines a mode for displaying the data, and the predetermined display method determines a method for displaying the image data. For example, the image data VI can be used as data to be displayed.

One method for displaying the image data VI can be associated with the first mode, for example. Another method for displaying the image data VI can be associated with the second mode. Thus, a display method can be selected on the basis of the selected mode.

<<First Mode>>

Specifically, a method for supplying selection signals to a scan line at a frequency of 30 Hz or higher, preferably 60 Hz or higher, to perform display on the basis of the selection signals can be associated with the first mode.

For example, when selection signals are supplied at a frequency of 30 Hz or higher, preferably 60 Hz or higher, the movement of a displayed moving image can be smooth.

For example, when an image is refreshed at a frequency of 30 Hz or higher, preferably 60 Hz or higher, an image that changes so as to smoothly follow the user's operation can be displayed on the data processing device 200 which is being operated by the user.

<<Second Mode>>

Specifically, a method for supplying selection signals to a scan line at a frequency lower than 30 Hz, preferably lower than 1 Hz, further preferably less than once a minute, to perform display on the basis of the selection signals can be associated with the second mode.

The supply of selection signals at a frequency lower than 30 Hz, preferably lower than 1 Hz, further preferably less than once a minute enables display with a flicker or flickering inhibited. Furthermore, the power consumption can be reduced.

For example, in the case where the data processing device 200 is used for a clock, the display can be refreshed at a frequency of once a second, once a minute, or the like.

In the case where a light-emitting device is used as a display element, for example, the light-emitting device can be made to emit light in a pulsed manner so that the image data is displayed. Specifically, an organic EL element can be made to emit light in a pulsed manner, and its afterglow can be used for display. The organic EL element has excellent frequency characteristics; thus, time for driving the light-emitting device can be shortened and the power consumption can be reduced in some cases. Heat generation is inhibited; thus, the deterioration of the light-emitting device can be suppressed in some cases. When the duty ratio is set to be lower than or equal to 20%, afterimages included in the display can be reduced.

[Fourth Step]

In a fourth step, selection is performed such that the program proceeds to a fifth step when an end instruction has been supplied (Yes), whereas the program proceeds to the third step when the end instruction has not been supplied (No) (see (S4) in FIG. 17A).

For example, the end instruction supplied in the interrupt processing may be used for the determination.

[Fifth Step]

In the fifth step, the program terminates (see (S5) in FIG. 17A).

<<Interrupt Processing>>

The interrupt processing includes a sixth step to an eighth step described below (see FIG. 17B).

[Sixth Step]

In the sixth step, the illuminance of the environment where the data processing device 200 is used is sensed using the sensing portion 250, for example (see (S6) in FIG. 17B). Note that color temperature or chromaticity of ambient light may be sensed instead of the illuminance of the environment.

[Seventh Step]

In the seventh step, a display method is determined on the basis of the sensed illuminance data (see (S7) in FIG. 17B). For example, a display method is determined such that the display brightness is not too dark or too bright.

Note that in the case where the color temperature of the ambient light or the chromaticity of the ambient light is sensed in the sixth step, the color of display may be adjusted.

[Eighth Step]

In the eighth step, the interrupt processing terminates (see (S8) in FIG. 17B).

<Structure Example 3 of Data Processing Device>

Another structure of the data processing device of one embodiment of the present invention will be described with reference to FIG. 18.

FIG. 18A is a flow chart showing a program of one embodiment of the present invention. FIG. 18A is a flow chart showing interrupt processing different from the interrupt processing shown in FIG. 17B.

Note that the structure example 3 of the data processing device is different from the interrupt processing described with reference to FIG. 17B in that the interrupt processing includes a step of changing a mode on the basis of a supplied predetermined event. Different portions are described in detail here, and the above description is referred to for portions that can use similar structures.

<<Interrupt Processing>>

The interrupt processing includes a sixth step to an eighth step described below (see FIG. 18A).

[Sixth Step]

In the sixth step, the program proceeds to the seventh step when a predetermined event has been supplied (Yes), whereas the program proceeds to the eighth step when the predetermined event has not been supplied (No) (see (U6) in FIG. 18A). For example, whether the predetermined event is supplied in a predetermined period or not can be used as a condition. Specifically, the predetermined period can be longer than 0 seconds, and shorter than or equal to 5 seconds, shorter than or equal to 1 second, or shorter than or equal to 0.5 seconds, preferably shorter than or equal to 0.1 seconds.

[Seventh Step]

In the seventh step, the mode is changed (see (U7) in FIG. 18A). Specifically, the second mode is selected in the case where the first mode has been selected, and the first mode is selected in the case where the second mode has been selected.

For example, it is possible to change the display mode of a region that is part of the display portion 230. Specifically, the display mode of a region where one driver circuit in the display portion 230 including the driver circuit GDA, the driver circuit GDB, and a driver circuit GDC supplies a selection signal can be changed (see FIG. 18B).

For example, the display mode of the region where a selection signal is supplied from the driver circuit GDB can be changed when a predetermined event is supplied to the input portion 240 in a region overlapping with the region where a selection signal is supplied from the driver circuit GDB (see FIG. 18B and FIG. 18C). Specifically, the frequency of supply of the selection signal from the driver

circuit GDB can be changed in accordance with a “tap” event supplied to a touch panel with a finger or the like.

A signal GCLK is a clock signal controlling the operation of the driver circuit GDB, and a signal PWC1 and a signal PWC2 are pulse width control signals controlling the operation of the driver circuit GDB. The driver circuit GDB supplies selection signals to a conductive film G2(m+1) to a conductive film G2(2m) on the basis of the signal GCLK, the signal PWC1, the signal PWC2, and the like.

Thus, for example, the driver circuit GDB can supply a selection signal without supply of selection signals from the driver circuit GDA and the driver circuit GDC. The display of the region where a selection signal is supplied from the driver circuit GDB can be refreshed without any change in the display of regions where selection signals are supplied from the driver circuit GDA and the driver circuit GDC. Power consumed by the driver circuits can be reduced. [Eighth Step]

In the eighth step, the interrupt processing terminates (see (U8) in FIG. 18A). Note that in a period during which the main processing is executed, the interrupt processing may be repeatedly executed.

<<Predetermined Event>>

For example, it is possible to use events supplied using a pointing device such as a mouse, such as “click” and “drag”, and events supplied to a touch panel with a finger or the like used as a pointer, such as “tap”, “drag”, and “swipe”.

For example, the position of a slide bar pointed by a pointer, the swipe speed, and the drag speed can be used to assign arguments to an instruction associated with a predetermined event.

For example, data sensed by the sensing portion 250 is compared with a predetermined threshold value, and the compared results can be used for the event.

Specifically, a pressure sensor or the like in contact with a button or the like that is arranged so as to be pushed in a housing can be used for the sensing portion 250.

<<Instruction Associated with Predetermined Event>>

For example, the end instruction can be associated with a predetermined event.

For example, “page-turning instruction” for switching display from one displayed image data to another image data can be associated with a predetermined event. Note that an argument determining the page-turning speed or the like, which is used when the “page-turning instruction” is executed, can be supplied using the predetermined event.

For example, “scroll instruction” for moving the display position of displayed part of image data and displaying another part continuing from that part, or the like can be associated with a predetermined event. Note that an argument determining the moving speed of display, or the like, which is used when the “scroll instruction” is executed, can be supplied using the predetermined event.

For example, an instruction for setting the display method, an instruction for generating image data, or the like can be associated with a predetermined event. Note that an argument determining the brightness of a generated image can be associated with a predetermined event. An argument determining the brightness of a generated image may be determined on the basis of ambient brightness sensed by the sensing portion 250.

For example, an instruction for acquiring data distributed via a push service using the communication portion 290 or the like can be associated with a predetermined event.

Note that position data sensed by the sensing portion 250 may be used for the determination of the presence or absence of a qualification for acquiring data. Specifically, it may be

determined that there is a qualification for acquiring data in the case of presence in a predetermined class room, school, conference room, company, building, or the like or in a predetermined region. Thus, for example, educational materials distributed in a classroom of a school, a university, or the like can be received, so that the data processing device 200 can be used as a schoolbook or the like (see FIG. 16C). Materials distributed in a conference room in, for example, a company can be received and used for a conference material.

<Structure Example 4 of Data Processing Device>

Another structure of the data processing device of one embodiment of the present invention will be described with reference to FIG. 19.

Note that the structure example 4 of the data processing device described with reference to FIG. 19A is different from the structure example described with reference to FIG. 17B in the interrupt processing. Specifically, the interrupt processing includes the step of determining a region, the step of generating an image, the step of displaying the image, and the step of imaging on the basis of a supplied predetermined event. Different portions are described in detail here, and the above description is referred to for portions that can use similar structures.

<<Interrupt Processing>>

The interrupt processing includes a sixth step to an eleventh step (see FIG. 19A).

[Sixth Step]

In the sixth step, the program proceeds to the seventh step when a predetermined event has been supplied (Yes), whereas the program proceeds to the eleventh step when the predetermined event has not been supplied (No) (see (V6) in FIG. 19A).

The predetermined event can be supplied using the sensing portion 250, for example. Specifically, a motion such as lifting of the data processing device can be used as the predetermined event. For example, a motion of the data processing device can be sensed using an angular sensor or an acceleration sensor. Touch of a finger or approach of an object can be sensed using a touch sensor.

[Seventh Step]

In the seventh step, a first region SH is determined (see (V7) in FIG. 19A).

For example, a region where an object such as a finger touches or approaches the input/output device 220 of one embodiment of the present invention can be the first region SH. A region that is set in advance by the user or the like can be used as the first region SH.

Specifically, an image of a finger THM or the like that touches or approaches the functional panel of one embodiment of the present invention is taken using the pixel 703(i, j) and subjected to image processing, whereby the first region SH can be determined (see FIG. 19B).

For example, an image of a shadow caused when external light is blocked by touch or approach of an object such as the finger THM is taken using the pixel 703(i, j) in the functional panel of one embodiment of the present invention and subjected to image processing, whereby the first region SH can be determined.

With the use of the pixel 703(i, j) in the functional panel of one embodiment of the present invention, an object such as the finger THM that touches or approaches the functional panel is irradiated with light, and an image of light reflected by the object is taken using the pixel 703(i, j) and subjected to image processing, whereby the first region SH can be determined.

A region where an object such as the finger THM touches can be determined as the first region SH by a touch sensor. [Eighth Step]

In the eighth step, an image FI including a second region and a third region is generated on the basis of the first region SH (see (V8) in FIG. 19A and FIG. 19B). For example, the shape of the first region SH is used as the shape of the second region, and a region excluding the first region SH is used as the third region.

[Ninth Step]

In the ninth step, the image FI is displayed such that the second region overlaps with the first region SH (see (V9) in FIG. 19A and FIG. 19B).

For example, an image signal is generated from the image FI and supplied to the region 231, and light is emitted from the pixel 703(i,j). In a period during which the first selection signal is supplied to the conductive film G1(i), the generated image signal is supplied to the conductive film S1g(j), and the image signal can be written to the pixel 703(i, j). The generated image signal is supplied to the conductive film S1g(j) and the conductive film S2g(j), and an enhanced image signal can be written to the pixel 703(0). The use of an enhanced image signal enables display with higher luminance.

Thus, the image FI can be displayed to overlap with the first region SH where the object such as a finger touches or approaches the region 231. The region where the object such as a finger touches can be irradiated with light using the pixel 703(0). The touching or approaching object such as the finger THM can be illuminated with a light. The object such as a finger can be led to touch or approach the region that is determined in advance by the user or the like.

[Tenth Step]

In the tenth step, the object that touches or approaches the first region SH is imaged while the image FI is displayed (see (V10) in FIG. 19A and FIG. 19B).

For example, an image of the finger THM or the like approaching the region 231 is taken while the finger or the like is irradiated with light. Specifically, an image of a fingerprint FP of the finger THM touching the region 231 can be taken (see FIG. 19C).

For example, the supply of the first selection signal can be stopped while an image is displayed with the pixel 703(i, j). For example, imaging can be performed using the pixel 703(i, j) while the supply of the first selection signal to the pixel circuit 530G(i, j) is stopped.

Accordingly, the touching or approaching object such as a finger can be imaged while the object is illuminated. Imaging can be performed in a period during which the first selection signal is not supplied. Noise in imaging can be inhibited. A clear image of a fingerprint can be obtained. An image that can be used for the authentication of the user can be obtained. In any area of the region 231, an image of the fingerprint of the finger touching the region 231 can be taken clearly. Thus, a novel data processing device that is highly convenient, useful, or reliable can be provided.

[Eleventh Step]

In the eleventh step, the interrupt processing ends (see (V11) in FIG. 19A).

<Structure Example 5 of Data Processing Device>

Another structure of the data processing device of one embodiment of the present invention is described with reference to FIG. 20.

<<Interrupt Processing>>

The interrupt processing includes a sixth step to a ninth step (see FIG. 20A).

[Sixth Step]

In the sixth step, the program proceeds to the seventh step when a predetermined event has been supplied (Yes), whereas the program proceeds to the ninth step when the predetermined event has not been supplied (No) (see (W6) in FIG. 20A).

For example, an object 30 is placed in a certain position in the data processing device 200, and a predetermined event can be supplied with use of the input portion 240 (see FIG. 20B). Specifically, the contact or approach of a finger or the like is sensed with use of a touch sensor in a region 231(1) and can be used for the predetermined event. For example, a touch sensor placed to overlap with a place where an image associated with the interrupt processing is displayed can be used. Specifically, an image associated with the interrupt processing is displayed in the region 231(1) and the input portion 240 placed to overlap with the region 231(1) can be used.

[Seventh Step]

In the seventh step, imaging is performed using the region 231(1) (see (W7) in FIG. 20A).

For example, when the object 30 approaches or adheres closely to the region 231, a still image is captured (see FIG. 20C). Specifically, a still image is captured when the intensity of external light entering the region 231 becomes smaller than a predetermined value. Alternatively, a still image is captured when an image taken in the region 231 does not show a change exceeding a predetermined level for a predetermined period. Alternatively, a still image is captured after the housing of the data processing device 200 is closed.

[Eighth Step]

In the eighth step, display is performed using the region 231(1) (see (W8) in FIG. 20A).

For example, the still image captured in the seventh step is displayed in the region 231 (see FIG. 20D).

[Ninth Step]

In the ninth step, the interrupt processing ends (see (W9) in FIG. 20A).

Accordingly, the touching or approaching object such as a finger can be imaged while the object is illuminated. A clear image with reduced distortion can be obtained. Information printed on a print or the like can be duplicated to electronic data. Thus, a novel data processing device that is highly convenient, useful, or reliable can be provided.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

Embodiment 8

In this embodiment, structures of a data processing device of one embodiment of the present invention will be described with reference to drawings.

FIG. 21A is a block diagram of the data processing device, and FIG. 21B to FIG. 21E are perspective views illustrating structures of the data processing device.

FIG. 22A to FIG. 22E are perspective views illustrating structures of the data processing device.

FIG. 23A and FIG. 23B are perspective views illustrating structures of the data processing device.

<Data Processing Device>

A data processing device 5200B described in this embodiment includes an arithmetic device 5210 and an input/output device 5220 (see FIG. 21A).

The arithmetic device 5210 has a function of being supplied with operation data and has a function of supplying image data on the basis of the operation data.

The input/output device **5220** includes a display portion **5230**, an input portion **5240**, a sensing portion **5250**, and a communication portion **5290** and has a function of supplying operation data and a function of being supplied with image data. The input/output device **5220** also has a function of supplying sensing data, a function of supplying communication data, and a function of being supplied with communication data.

The input portion **5240** has a function of supplying operation data. For example, the input portion **5240** supplies operation data on the basis of operation by a user of the data processing device **5200B**.

Specifically, a keyboard, a hardware button, a pointing device, a touch sensor, an illuminance sensor, an imaging device, an audio input device, an eye-gaze input device, an attitude detection device, or the like can be used as the input portion **5240**.

The display portion **5230** includes a display panel and has a function of displaying image data. For example, the display panel described in any one of Embodiment 2 to Embodiment 4 can be used for the display portion **5230**.

The sensing portion **5250** has a function of supplying sensing data. For example, the sensing portion **5250** has a function of sensing a surrounding environment where the data processing device is used and supplying sensing data.

Specifically, an illuminance sensor, an imaging device, an attitude detection device, a pressure sensor, a human motion sensor, or the like can be used as the sensing portion **5250**.

The communication portion **5290** has a function of being supplied with communication data and a function of supplying communication data. For example, the communication portion **5290** has a function of being connected to another electronic device or a communication network through wireless communication or wired communication. Specifically, the communication portion **5290** has a function of wireless local area network communication, telephone communication, near field communication, or the like.

<<Structure Example 1 of Data Processing Device>>

For example, the display portion **5230** can have an outer shape along a cylindrical column or the like (see FIG. 21B). In addition, the data processing device has a function of changing its display method in accordance with the illuminance of a usage environment. Furthermore, the data processing device has a function of changing displayed content in response to sensed existence of a person. This allows the data processing device to be provided on a column of a building, for example. The data processing device can display advertising, guidance, or the like. The data processing device can be used for digital signage or the like.

<<Structure Example 2 of Data Processing Device>>

For example, the data processing device has a function of generating image data on the basis of the path of a pointer used by a user (see FIG. 21C). Specifically, the display panel with a diagonal size of 20 inches or longer, preferably 40 inches or longer, and further preferably 55 inches or longer can be used. Alternatively, a plurality of display panels can be arranged and used as one display region. Alternatively, a plurality of display panels can be arranged and used as a multiscreen. Thus, the data processing device can be used for an electronic blackboard, an electronic bulletin board, digital signage, or the like.

<<Structure Example 3 of Data Processing Device>>

The data processing device can receive data from another device and display the data on the display portion **5230** (see FIG. 21D). Several options can be displayed. The user can choose some from the options and send a reply to a transmitter of the data. For example, the data processing device

has a function of changing its display method in accordance with the illuminance of a usage environment. Thus, the power consumption of a smartwatch can be reduced, for example. A smartwatch can display an image to be suitably used even in an environment under strong external light, e.g., outdoors in fine weather, for example.

<<Structure Example 4 of Data Processing Device>>

For example, the display portion **5230** has a surface gently curved along a side surface of a housing (see FIG. 21E). The display portion **5230** includes a display panel, and the display panel has a function of performing display on the front surface, the side surfaces, the top surface, and the rear surface, for example. Thus, for example, a mobile phone can display data not only on its front surface but also on its side surfaces, its top surface, and its rear surface.

<<Structure Example 5 of Data Processing Device>>

For example, the data processing device can receive data via the Internet and display the data on the display portion **5230** (see FIG. 22A). A created message can be checked on the display portion **5230**. The created message can be sent to another device. The data processing device has a function of changing its display method in accordance with the illuminance of a usage environment, for example. Thus, the power consumption of a smartphone can be reduced. A smartphone can display an image to be suitably used even in an environment under strong external light, e.g., outdoors in fine weather, for example.

<<Structure Example 6 of Data Processing Device>>

A remote controller can be used as the input portion **5240** (see FIG. 22B). For example, the data processing device can receive data from a broadcast station or via the Internet and display the data on the display portion **5230**. An image of a user can be taken using the sensing portion **5250**. The image of the user can be transmitted. The data processing device can acquire a viewing history of the user and provide it to a cloud service. The data processing device can acquire recommendation data from a cloud service and display the data on the display portion **5230**. A program or a moving image can be displayed on the basis of the recommendation data. The data processing device has a function of changing its display method in accordance with the illuminance of a usage environment, for example. Accordingly, for example, a television system can display an image to be suitably used even when irradiated with strong external light that enters a room in fine weather.

<<Structure Example 7 of Data Processing Device>>

For example, the data processing device can receive educational materials via the Internet and display them on the display portion **5230** (see FIG. 22C). An assignment can be input with the input portion **5240** and sent via the Internet. A corrected assignment or the evaluation of the assignment can be obtained from a cloud service and displayed on the display portion **5230**. Suitable educational materials can be selected on the basis of the evaluation and displayed.

For example, the display portion **5230** can perform display using an image signal received from another data processing device. When the data processing device is placed on a stand or the like, the display portion **5230** can be used as a sub-display. Thus, for example, a tablet computer can display an image to be suitably used even in an environment under strong external light, e.g., outdoors in fine weather.

<<Structure Example 8 of Data Processing Device>>

The data processing device includes, for example, a plurality of display portions **5230** (see FIG. 22D). For example, the display portion **5230** can display an image that the sensing portion **5250** is capturing. A captured image can

be displayed on the sensing portion. A captured image can be decorated using the input portion 5240. A message can be attached to a captured image. A captured image can be transmitted via the Internet. The data processing device has a function of changing its shooting conditions in accordance with the illuminance of a usage environment. Accordingly, for example, a digital camera can display an object in such a manner that an image is favorably viewed even in an environment under strong external light, e.g., outdoors in fine weather.

<<Structure Example 9 of Data Processing Device>>

For example, the data processing device of this embodiment is used as a master and another data processing device is used as a slave, whereby the other data processing device can be controlled (see FIG. 22E). As another example, part of image data can be displayed on the display portion 5230 and another part of the image data can be displayed on a display portion of another data processing device. Image signals can be supplied to another data processing device. With the communication portion 5290, data to be written can be obtained from an input portion of another data processing device. Thus, a large display region can be utilized by using a portable personal computer, for example.

<<Structure Example 10 of Data Processing Device>>

The data processing device includes, for example, the sensing portion 5250 that senses an acceleration or a direction (see FIG. 23A). The sensing portion 5250 can supply data on the position of the user or the direction in which the user faces. The data processing device can generate image data for the right eye and image data for the left eye in accordance with the position of the user or the direction in which the user faces. The display portion 5230 includes a display region for the right eye and a display region for the left eye. Thus, a virtual reality image that gives the user a sense of immersion can be displayed on a goggles-type data processing device, for example.

<<Structure Example 11 of Data Processing Device>>

The data processing device includes, for example, an imaging device and the sensing portion 5250 that senses an acceleration or a direction (see FIG. 23B). The sensing portion 5250 can supply data on the position of the user or the direction in which the user faces. The data processing device can generate image data in accordance with the position of the user or the direction in which the user faces. Accordingly, the data can be shown together with a real-world scene, for example. An augmented reality image can be displayed on a glasses-type data processing device.

Note that this embodiment can be combined with any of the other embodiments in this specification as appropriate.

In the case where there is an explicit description, X and Y are connected, in this specification and the like, for example, the case where X and Y are electrically connected, the case where X and Y are functionally connected, and the case where X and Y are directly connected are disclosed in this specification and the like. Accordingly, without being limited to a predetermined connection relationship, for example, a connection relationship shown in drawings or texts, a connection relationship other than one shown in drawings or texts is regarded as being disclosed in the drawings or the texts.

Here, X and Y each denote an object (e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, or a layer).

Examples of the case where X and Y are directly connected include the case where an element that allows an electrical connection between X and Y (e.g., a switch, a transistor, a capacitor, an inductor, a resistor, a diode, a

display element, a light-emitting element, and a load) is not connected between X and Y, and the case where X and Y are connected without the element that allows the electrical connection between X and Y (e.g., a switch, a transistor, a capacitor, an inductor, a resistor, a diode, a display element, a light-emitting element, and a load) provided therebetween.

For example, in the case where X and Y are electrically connected, one or more elements that allow an electrical connection between X and Y (e.g., a switch, a transistor, a capacitor, an inductor, a resistor, a diode, a display element, a light-emitting element, and a load) can be connected between X and Y. Note that a switch has a function of being controlled to be turned on or off. That is, a switch has a function of being in a conduction state (on state) or a non-conduction state (off state) to control whether or not current flows. Alternatively, the switch has a function of selecting and changing a current path. Note that the case where X and Y are electrically connected includes the case where X and Y are directly connected.

An example of the case where X and Y are functionally connected is the case where one or more circuits that allow functional connection between X and Y (for example, a logic circuit (an inverter, a NAND circuit, a NOR circuit, or the like), a signal converter circuit (a DA converter circuit, an AD converter circuit, a gamma correction circuit, or the like), a potential level converter circuit (a power supply circuit (for example, a step-up circuit, a step-down circuit, or the like), a level shifter circuit for changing the potential level of a signal, or the like), a voltage source, a current source, a switching circuit, an amplifier circuit (a circuit capable of increasing signal amplitude, the amount of current, or the like, an operational amplifier, a differential amplifier circuit, a source follower circuit, a buffer circuit, or the like), a signal generator circuit, a memory circuit, a control circuit, or the like) can be connected between X and Y. For example, even when another circuit is interposed between X and Y, X and Y are functionally connected when a signal output from X is transmitted to Y. Note that the case where X and Y are functionally connected includes the case where X and Y are directly connected and the case where X and Y are electrically connected.

Note that in the case where there is an explicit description, X and Y are electrically connected, the case where X and Y are electrically connected (i.e., the case where X and Y are connected with another element or another circuit provided therebetween), the case where X and Y are functionally connected (i.e., the case where X and Y are functionally connected with another circuit provided therebetween), and the case where X and Y are directly connected (i.e., the case where X and Y are connected without another element or another circuit provided therebetween) are disclosed in this specification and the like. That is, in the case where there is an explicit description, being electrically connected, the same contents as the case where there is only an explicit description, being connected, are disclosed in this specification and the like.

Note that, for example, the case where a source (or a first terminal or the like) of a transistor is electrically connected to X through (or not through) Z1 and a drain (or a second terminal or the like) of the transistor is electrically connected to Y through (or not through) Z2, or the case where a source (or a first terminal or the like) of a transistor is directly connected to one part of Z1 and another part of Z1 is directly connected to X while a drain (or a second terminal or the like) of the transistor is directly connected to one part of Z2 and another part of Z2 is directly connected to Y can be expressed as follows.

It can be expressed as, for example, “X Y, a source (or a first terminal or the like) of a transistor, and a drain (or a second terminal or the like) of the transistor are electrically connected to each other, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are electrically connected to each other in this order”. Alternatively, the expression “a source (or a first terminal or the like) of a transistor is electrically connected to X, a drain (or a second terminal or the like) of the transistor is electrically connected to Y, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are electrically connected to each other in this order” can be used. Alternatively, the expression “X is electrically connected to Y through a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are provided in this connection order” can be used. When the connection order in a circuit structure is defined by an expression similar to the above examples, a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor can be distinguished from each other to specify the technical scope.

Other examples of the expressions include, “a source (or a first terminal or the like) of a transistor is electrically connected to X through at least a first connection path, the first connection path does not include a second connection path, the second connection path is a path through the transistor and between the source (or the first terminal or the like) of the transistor and a drain (or a second terminal or the like) of the transistor, the first connection path is a path through Z1, the drain (or the second terminal or the like) of the transistor is electrically connected to Y through at least a third connection path, the third connection path does not include the second connection path, and the third connection path is a path through Z2” and “a source (or a first terminal or the like) of a transistor is electrically connected to X through Z1 by at least a first connection path, the first connection path does not include a second connection path, the second connection path includes a connection path through the transistor, a drain (or a second terminal or the like) of the transistor is electrically connected to Y through Z2 by at least a third connection path, and the third connection path does not include the second connection path”. Alternatively, the expression “a source (or a first terminal or the like) of a transistor is electrically connected to X by at least a first electrical path through Z1, the first electrical path does not include a second electrical path, the second electrical path is an electrical path from the source (or the first terminal or the like) of the transistor to a drain (or a second terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor is electrically connected to Y by at least a third electrical path through Z2, the third electrical path does not include a fourth electrical path, and the fourth electrical path is an electrical path from the drain (or the second terminal or the like) of the transistor to the source (or the first terminal or the like) of the transistor” can be used. When the connection path in a circuit structure is defined by an expression similar to the above examples, a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor can be distinguished from each other to specify the technical scope.

Note that these expressions are examples and the expression is not limited to these expressions. Here, X, Y, Z1, and

Z2 denote an object (e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, and a layer).

Even when independent components are electrically connected to each other in a circuit diagram, one component has functions of a plurality of components in some cases. For example, when part of a wiring also functions as an electrode, one conductive film has functions of both of the components that are a wiring and an electrode. Thus, electrical connection in this specification includes, in its category, such a case where one conductive film has functions of a plurality of components.

REFERENCE NUMERALS

ABS: absorption spectrum, ANO: conductive film, CAPSEL: conductive film, CDSBIAS: conductive film, CDSVDD: conductive film, CDSVSS: conductive film, C21: capacitor, C22: capacitor, C31: capacitor, CL: conductive film, FD: node, FPC1: flexible printed circuit, GCLK: signal, G1: conductive film, G2: conductive film, IN: terminal, MD: transistor, M21: transistor, M31: transistor, M32: transistor, N21: node, N22: node, OUT: terminal, P1: position data, PWC1: signal, PWC2: signal, RS: conductive film, S1g: conductive film, S1r: conductive film, S2g: conductive film, SE: conductive film, SH: region, SP11: spectrum, SP12: spectrum, SW21: switch, SW22: switch, SW23: switch, SW31: switch, SW32: switch, SW33: switch, TX: conductive film, VCL: conductive film, VCOM2: conductive film, VCP: conductive film, VIV: conductive film, VLEN: conductive film, VPI: conductive film, VR: conductive film, WX: conductive film, 30: object, 200: data processing device, 210: arithmetic device, 211: arithmetic portion, 212: memory portion, 213: artificial intelligence portion, 214: transmission path, 215: input/output interface, 220: input/output device, 230: display portion, 231: region, 233: timing controller, 234: decompression circuit, 235: image processing circuit, 238: control portion, 240: input portion, 241: sensing region, 243: control circuit, 250: sensing portion, 290: communication portion, 501C: insulating film, 501D: insulating film, 504: conductive film, 506: insulating film, 508: semiconductor film, 508A: region, 508B: region, 508C: region, 510: base material, 512A: conductive film, 512B: conductive film, 512C: conductive film, 512D: conductive film, 512E: conductive film, 512F: conductive film, 516: insulating film, 516A: insulating film, 516B: insulating film, 518: insulating film, 519B: terminal, 520: functional layer, 521: insulating film, 521A: insulating film, 521B: insulating film, 524: conductive film, 528: insulating film, 530G: pixel circuit, 53016: pixel circuit, 550G: device, 55016: optical functional device, 551G: electrode, 551RS: electrode, 552: electrode, 553G: layer, 553R: layer, 553RS: optical functional layer, 553S: layer, 573: insulating film, 573A: insulating film, 573B: insulating film, 591G: opening portion, 591RS: opening portion, 700: functional panel, 700TP: functional panel, 702B: pixel, 702G: pixel, 702RS: pixel, 703: pixel, 705: sealant, 720: functional layer, 770: base material, 770P: functional film, 771: insulating film, 802: sensor, 5200B: data processing device, 5210: arithmetic device, 5220: input/output device, 5230: display portion, 5240: input portion, 5250: sensing portion, 5290: communication portion

49

The invention claimed is:

1. An optical functional device comprising:

a light-emitting function;

a photoelectric conversion function;

a first electrode;

a second electrode; and

an optical functional layer,

wherein the light-emitting function converts electrical energy into first light,

wherein the first light has a first emission spectrum,

wherein the first emission spectrum exhibits a maximum peak at a first wavelength,

wherein the maximum peak exhibits a first value,

wherein at a second wavelength, intensity of the first emission spectrum is 80% of the first value,

wherein the photoelectric conversion function has a spectral sensitivity characteristic,

wherein at a third wavelength, the spectral sensitivity characteristic has a maximum sensitivity within a range of 420 to 720 nm inclusive,

wherein at a fourth wavelength, sensitivity of the spectral sensitivity characteristic is 80% of the maximum sensitivity,

wherein the third wavelength is positioned closer to the second wavelength than to the first wavelength,

wherein the fourth wavelength is positioned closer to the first wavelength than to the third wavelength,

wherein the optical functional layer comprises a region interposed between the first electrode and the second electrode,

wherein the optical functional layer comprises a first layer and a second layer,

wherein the first layer emits light comprising the first light,

wherein the second layer comprises a region overlapping with the first layer,

wherein the second layer comprises a light-absorbing material,

wherein the light-absorbing material has a first absorption spectrum, and

wherein the first absorption spectrum has a region overlapping with the spectral sensitivity characteristic.

2. The optical functional device according to claim 1, wherein the third wavelength is shorter than the first wavelength.

3. The optical functional device according to claim 2, wherein the second wavelength is shorter than the first wavelength,

wherein the fourth wavelength is longer than the third wavelength, and

wherein the fourth wavelength is shorter than the second wavelength.

4. A functional panel comprising:

a first pixel,

a first conductive film,

a second conductive film,

a third conductive film,

a fourth conductive film,

a fifth conductive film,

a sixth conductive film, and

a seventh conductive film,

wherein the first pixel comprises the optical functional device according to claim 1 and a first pixel circuit,

wherein the first pixel circuit is electrically connected to the optical functional device, and

wherein the first pixel circuit is electrically connected to the first conductive film, the second conductive film,

50

the third conductive film, the fourth conductive film, the fifth conductive film, the sixth conductive film, and the seventh conductive film.

5. The functional panel according to claim 4, comprising a pixel set,

wherein the pixel set comprises the first pixel and a second pixel,

wherein the second pixel comprises a light-emitting device and a second pixel circuit,

wherein the light-emitting device is electrically connected to the second pixel circuit,

wherein the light-emitting device emits third light, and wherein the optical functional device is capable of photoelectric conversion of the third light.

6. The functional panel according to claim 4, comprising a functional layer,

wherein the functional layer comprises the first pixel circuit,

wherein the first pixel circuit comprises a first transistor and a second transistor,

wherein the functional layer comprises a driver circuit, wherein the driver circuit comprises a third transistor,

wherein the first transistor comprises a semiconductor film,

wherein the second transistor comprises a semiconductor film that can be formed in a step of forming the semiconductor film, and

wherein the third transistor comprises a semiconductor film that can be formed in the step of forming the semiconductor film.

7. The functional panel according to claim 4, comprising a region,

wherein the region comprises a group of pixel sets and a different group of pixel sets,

wherein the group of pixel sets is arranged in a row direction,

wherein the group of pixel sets comprises the pixel set, wherein the group of pixel sets is electrically connected to the first conductive film,

wherein the group of pixel sets is electrically connected to the fifth conductive film,

wherein the different group of pixel sets is arranged in a column direction intersecting with the row direction,

wherein the different group of pixel sets comprises the pixel set,

wherein the different group of pixel sets is electrically connected to the third conductive film, and

wherein the different group of pixel sets is electrically connected to the seventh conductive film.

8. The functional panel according to claim 4, comprising: a group of sampling circuits;

a multiplexer;

an amplifier circuit; and

an analog-digital converter circuit,

wherein the multiplexer selects one from the group of sampling circuits and obtains an imaging signal,

wherein the multiplexer is configured to supply the imaging signal to the amplifier circuit, and

wherein the amplifier circuit is configured to supply the imaging signal to the analog-digital converter circuit.

9. A display device comprising:

a control portion; and

the functional panel according to claim 4,

wherein the control portion is supplied with image data and control data,

wherein the control portion generates data on the basis of the image data,

51

wherein the control portion generates a control signal on the basis of the control data,
 wherein the control portion supplies the data and the control signal,
 wherein the functional panel is supplied with the data and the control signal, and
 wherein the pixel set performs display on the basis of the data.

10. An input/output device comprising:
 an input portion; and
 a display portion,
 wherein the display portion comprises the functional panel according to claim 4,
 wherein the input portion comprises a sensing region,
 wherein the input portion senses an object approaching the sensing region, and
 wherein the sensing region comprises a region overlapping with the pixel.

11. A data processing device comprising:
 an arithmetic device; and
 an input/output device,
 wherein the arithmetic device is supplied with input data or sensing data,
 wherein the arithmetic device generates control data and image data on the basis of the input data or the sensing data,
 wherein the arithmetic device supplies the control data and the image data,
 wherein the input/output device supplies the input data and the sensing data,
 wherein the input/output device is supplied with the control data and the image data,
 wherein the input/output device comprises a display portion, an input portion, and a sensing portion,
 wherein the display portion comprises the functional panel according to claim 4,
 wherein the display portion displays the image data on the basis of the control data,
 wherein the input portion generates the input data, and
 wherein the sensing portion generates the sensing data.

12. A data processing device comprising:
 one or more of a keyboard, a hardware button, a pointing device, a touch sensor, an illuminance sensor, an imaging device, an audio input device, an eye-gaze input device, and an attitude detection device; and
 the functional panel according to claim 4.

13. The optical functional device according to claim 2,
 wherein the optical functional device is configured to:
 emit red light; and
 perform photoelectric conversion of green light.

14. An optical functional device comprising:
 a light-emitting function;
 a photoelectric conversion function;
 a first electrode;
 a second electrode; and
 an optical functional layer,
 wherein the light-emitting function converts electrical energy into first light,
 wherein the first light has a first emission spectrum,
 wherein the first emission spectrum exhibits a maximum peak at a first wavelength,

52

wherein the photoelectric conversion function has a spectral sensitivity characteristic,
 wherein at a third wavelength, the spectral sensitivity characteristic has a maximum sensitivity within a range of 420 to 720 nm inclusive,
 wherein the third wavelength is positioned in a range of wavelength longer than or equal to 420 nm and shorter than the first wavelength,
 wherein the optical functional layer comprises a region interposed between the first electrode and the second electrode,
 wherein the optical functional layer comprises a first layer and a second layer,
 wherein the first layer emits light comprising the first light,
 wherein the second layer comprises a light-absorbing material,
 wherein the light-absorbing material has a first absorption spectrum, and
 wherein the first absorption spectrum has a region overlapping with the spectral sensitivity characteristic.

15. The optical functional device according to claim 14,
 wherein the optical functional device is configured to:
 emit red light; and
 perform photoelectric conversion of green light.

16. An optical functional device comprising:
 a first electrode;
 a second electrode; and
 an optical functional layer,
 wherein the optical functional layer comprises a region interposed between the first electrode and the second electrode,
 wherein the optical functional layer comprises a first layer and a second layer,
 wherein the first layer comprises a light-emitting material,
 wherein the light-emitting material is configured to emit second light,
 wherein the second light has a second spectrum,
 wherein the second spectrum exhibits a maximum peak at a fifth wavelength,
 wherein the second layer comprises a light-absorbing material,
 wherein the light-absorbing material has a first absorption spectrum,
 wherein at a sixth wavelength, the first absorption spectrum exhibits maximum absorption within a range of 420 to 720 nm inclusive, and
 wherein the sixth wavelength is positioned in a range of wavelength longer than or equal to 420 nm and shorter than the fifth wavelength.

17. The optical functional device according to claim 16,
 wherein the first absorption spectrum comprises an absorption edge at a seventh wavelength, and
 wherein the seventh wavelength is shorter than the fifth wavelength.

18. The optical functional device according to claim 16,
 wherein the second spectrum comprises an emission edge at an eighth wavelength, and
 wherein the eighth wavelength is shorter than the fifth wavelength and longer than the sixth wavelength.

* * * * *